

Zeroizing Attacks against Evasive and Circular Evasive LWE

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Abstract

We develop new attacks against the Evasive LWE family of assumptions, in both the public and private-coin regime. To the best of our knowledge, ours are the *first* attacks against Evasive LWE in the *public-coin* regime, for any instantiation from the family. Our attacks are summarized below.

- **Public-Coin Attacks.**

1. The recent work by Hsieh, Lin and Luo [HLL23] constructed the first Attribute Based Encryption (ABE) for unbounded depth circuits by relying on the “circular” evasive LWE assumption. This assumption has been popularly considered as a safe, public-coin instance of Evasive LWE in contrast to its “private-coin” cousins (for instance, see [CW25, DJM⁺25a]).

We provide the first attack against this assumption, challenging the widely held belief that this is a public-coin assumption.

2. We demonstrate a counter-example against vanilla public-coin evasive LWE by Wee [Wee22] in an unnatural parameter regime. Our attack crucially relies on the error in the pre-condition being larger than the error in the post-condition, necessitating a refinement of the assumption.

- **Private-Coin Attacks.**

1. The recent work by Agrawal, Kumari and Yamada [AKY24a] constructed the first functional encryption scheme for pseudorandom functionalities (PRFE) and extended this to obfuscation for pseudorandom functionalities (PRIO) [AKY24c] by relying on private-coin evasive LWE. We provide a new attack against the assumption stated in the first posting of their work (subsequently refined to avoid these attacks).
2. The recent work by Branco et al. [BDJ⁺25] (concurrently to [AKY24c]) provides a construction of obfuscation for pseudorandom functionalities by relying on private-coin evasive LWE. We provide a new attack against their stated assumption.
3. Branco et al. [BDJ⁺25] showed that there exist contrived, “self-referential” classes of pseudorandom functionalities for which pseudorandom obfuscation cannot exist. We extend their techniques to develop an analogous result for pseudorandom functional encryption.

While Evasive LWE was developed to specifically avoid “zeroizing attacks”, our work shows that in certain settings, such attacks can still apply.

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1 Introduction

The Evasive LWE [Wee22, Tsa22] assumption was introduced independently by Wee [Wee22] and Tsabary [Tsa22] to give the first constructions to the long-standing problems of lattice-based broadcast encryption [Wee22] and witness encryption [Tsa22, VWV22]. In recent years, evasive LWE has been instrumental in making progress in several other challenging questions in lattice based cryptography, for instance [HLL23, AKY24b, WW24, WWW22a, MPV24a].

Evading the Zeroizing Regime. The primary motivation for introducing the evasive LWE assumption was to interpolate assumptions underlying lattice based iO on the one hand (which are considered unstable) and solid assumptions like LWE on the other (which are very stable but afford limited power). At a very high level, known attacks on lattice based iO pertained to the so-called “zeroizing regime” where the adversary may obtain a large number of equations in low norm secret values over the integers, which can then be somehow solved to recover the secrets. Evasive LWE was defined to carefully sidestep the zeroizing regime in its entirety – this was done by ensuring that the attacker *only obtains large norm values* which wraparound the modulus. The hope was that this assumption would allow to make progress on some long-standing, presumably “intermediate” level hard problems such as broadcast encryption from lattices without going all the way to FE/iO.

In more detail, the evasive LWE assumption roughly says that if

$$(\mathbf{B}, \mathbf{P}, \mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{s}^\top \mathbf{P} + \mathbf{e}_\mathbf{P}^\top, \text{aux}) \approx_c (\mathbf{B}, \mathbf{P}, \$, \$, \text{aux})$$

where \$ represents random, then

$$(\mathbf{B}, \mathbf{P}, \mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{B}^{-1}(\mathbf{P}), \text{aux}) \approx_c (\mathbf{B}, \mathbf{P}, \$, \mathbf{B}^{-1}(\mathbf{P}), \text{aux})$$

Above $\mathbf{B}^{-1}(\mathbf{P})$ refers to a low norm matrix, say $\mathbf{K}$, such that $\mathbf{BK} = \mathbf{P} \bmod q$. Evidently, given $\mathbf{K} = \mathbf{B}^{-1}(\mathbf{P})$ and $\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top$, the adversary can compute $\mathbf{s}^\top \mathbf{P} + \mathbf{e}_\mathbf{P}^\top \mathbf{K}$. Here, the error $\mathbf{e}_\mathbf{B}^\top \mathbf{K}$ is not i.i.d, unlike its counterpart $\mathbf{e}_\mathbf{P}^\top$ in the pre-condition. As discussed by Wee [Wee22], irregularities in the error distribution can be very dangerous for LWE security, but the hope here is that the large masking term $\mathbf{s}^\top \mathbf{P}$ prevents any exploitation of this correlation.

Public versus Private Coin. An important distinction between variants of Evasive LWE are “public-coin” and “private-coin”, where the former means that the randomness used by the sampler, who outputs $\mathbf{P}$ and potentially other information, is made available to the adversary, and the latter means that the sampler’s random coins need to be hidden from the adversary. So far, counter-examples have been developed in the stronger private-coin setting but we did not know of any attacks in the public-coin setting.

Prior Work. The original work of [VWV22] defined private coin evasive LWE and showed contrived counterexamples against the assumption – these counterexamples are later formalized and improved in [BUW24, HHY25]. However, this was not considered too problematic as it relied on highly unnatural auxiliary information which contained obfuscations that would output secrets given $\mathbf{B}^{-1}(\mathbf{P})$ but not otherwise. No attacks were known in the public-coin setting used by Wee’s original formulation or its extensions, such as the circular evasive LWE by Hsieh, Lin and Luo [HLL23]. The recent work of [BUW24] also provided attacks against classes of evasive LWE where either $\mathbf{B}$ or $\mathbf{P}$ are not known to the adversary.

1.1 Our Results

We develop new counterexamples against the Evasive LWE family of assumptions, in both the public and private-coin regime. To the best of our knowledge, ours are the first attacks against Evasive LWE in the *public-coin* regime, for any instantiation from the family. Our attacks are summarized below.

- **Public-Coin Attacks.**

1. The recent work by Hsieh, Lin and Luo [HLL23] constructed the first Attribute Based Encryption (ABE) for unbounded depth circuits by relying on the “circular” evasive LWE assumption. This assumption has been popularly considered as a safe, public-coin instance of Evasive LWE in contrast to its more dangerous “private-coin” cousins (for instance, see [CW25, DJM⁺25a]).

We provide the first attack against this assumption, challenging the widely held belief that this is a public-coin assumption (see for instance [DJM⁺25b]).

2. We demonstrate a counter-example against vanilla public-coin evasive LWE by Wee [Wee22] in an unnatural parameter regime. Our attack crucially relies on the error in the pre-condition being larger than the error in the post-condition, necessitating a refinement of the assumption.

- **Private-Coin Attacks.**

1. The recent work by Agrawal, Kumari and Yamada [AKY24a] constructed the first functional encryption scheme for pseudorandom functionalities (PRFE) and extended this to obfuscation for pseudorandom functionalities (PRIO) [AKY24c] by relying on private-coin evasive LWE. We provide a new attack against the assumption stated in the first posting of their work (subsequently refined to avoid these attacks).
2. The recent work by Branco et al. [BDJ⁺25] (concurrently to [AKY24c]) provides a construction of obfuscation for pseudorandom functionalities by relying on private-coin evasive LWE. We provide a new attack against their stated assumption.
3. Branco et al. [BDJ⁺25] showed that there exist contrived, “self-referential” classes of pseudorandom functionalities for which pseudorandom obfuscation cannot exist. We extend their techniques to develop an analogous result for pseudorandom functional encryption.

While Evasive LWE was developed to specifically avoid “zeroizing attacks”, our work shows that in certain settings, such attacks can still apply.

Repercussions of our attacks. Subsequent to the online posting of this work, there have been the following repercussions:

1. As a result of the attack developed in Section 4.1 (independently developed in [HJL25]), Agrawal, Kumari and Yamada [AKY24a] revised their scheme so that the attacks no longer apply. Additionally, to recover the underlying assumption, they incorporated restrictions to exclude the attacks. We believe that similar restrictions will need to be incorporated into the circular evasive LWE assumption used by [HLL23, CW25] and the private coin evasive LWE assumption of [BDJ⁺25] to recover their validity.
2. As a result of the attacks developed in Section 4.2 (independently developed in [HJL25]) as well attacks developed in [DJM⁺25b, BUW24], Branco et al. [BDJ⁺25] demoted their construction from private-coin Evasive LWE to a heuristic.

3. As a result of the attack developed in Section 3.1, the community is now challenging the “public-coin” status of circular evasive LWE – for instance, the work by Dottling et al. [DJM⁺25b] previously referred to this assumption as “public-coin” [DJM⁺25a] but changed it to “private-coin” in the updated version [DJM⁺25b].

1.2 Concurrent and Independent Work

The concurrent work of [DJM⁺25b] and [HJL25] also provided new counter-examples against evasive LWE in the *private coin* setting. The attacks in the public-coin setting developed in Section 3 are not present in either work.

Comparison with the attacks in [HJL25]. The attack provided by [HJL25] in Section 7 is very similar to the attacks presented in Sections 4.1 and 4.2. The attacks we provide in 3.1, 3.2, and 4.3 are new to our work.

Comparison with the attacks in [DJM⁺25b]. Following the classification in [BUW24], based on whether $\mathbf{B}$ and $\mathbf{P}$ are hidden or provided to the adversary in the pre- and the post-condition, [DJM⁺25b] considers four combinations: (i) $(\mathbf{B}, \mathbf{P})$, (ii) $(\mathbf{B}, \neg\mathbf{P})$, (iii) $(\neg\mathbf{B}, \mathbf{P})$, (iv) $(\neg\mathbf{B}, \neg\mathbf{P})$. They rule out all the four cases by proving the indistinguishability in the pre-condition for the strongest setting where both matrices are known to the adversary and providing a distinguisher in the post condition for the weakest setting where both matrices are hidden.

In our counterexamples in Section 4.1 and Section 4.2, the precondition is indistinguishable in the presence of both $\mathbf{B}$ and $\mathbf{P}$ (as in [DJM⁺25b]). Our distinguishers for post-conditions do not use $\mathbf{B}$ in both Section 4.1 and Section 4.2. Thus, our counterexamples also rule out the cases $(\mathbf{B}, \mathbf{P})$ and $(\neg\mathbf{B}, \mathbf{P})$. Our distinguishers do use $\mathbf{P}$, however, it is easy to hide $\mathbf{P}$ by directly providing the term computed using $\mathbf{P}$ in aux. This will further rule out the remaining cases as well. Please see Remark 4.13 for more details.

1.3 Technical Overview

We proceed to outline the main ideas in our work. Our presentation follows a sequence favoring ease of understanding, rather than importance of results.

1.3.1 Attack when Pre-Condition has Larger Error.

We start with describing an attack against evasive LWE in the public coin regime, since this is the simplest attack we present in this paper. Our attack works in the setting where $\|\mathbf{e}_\mathbf{P}\| \gg \|\mathbf{e}_\mathbf{B}^\top \mathbf{K}\|$. Intuitively, having this condition makes the assumption less reliable, since the post condition adversary can recover $\mathbf{s}^\top \mathbf{P} + \mathbf{e}_\mathbf{B}^\top \mathbf{K}$, which is with much smaller noise than the pre-condition noise $\mathbf{e}_\mathbf{P}$. Our attack here confirms this intuition by showing that there is a concrete choice of parameters for which there is an explicit attack even in the public coin regime.

Our attack works in the setting where the modulus q is set to be $q = pr$ for prime numbers p and r . The sampler chooses random $\mathbf{D} \in [0, r - 1]^{n \times \ell}$ and sets $\mathbf{P} = p\mathbf{D} \bmod q$ and $\text{aux} = \mathbf{D}$. Furthermore, in this example, $\mathbf{s}$ is a short Gaussian vector. How to choose other parameters will become clear through the discussion below. We first show that the precondition holds. Even given $(\mathbf{B}, \mathbf{P}, \text{aux} = \mathbf{D})$, we have

$$(\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{s}^\top \mathbf{P} + \mathbf{e}_\mathbf{P}^\top) \equiv (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{s}^\top (p\mathbf{D}) + \mathbf{e}_\mathbf{P}^\top) \quad (1)$$

$$\approx_s (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{s}^\top (p\mathbf{D}) + \mathbf{s}^\top \mathbf{D}' + \mathbf{e}_\mathbf{P}^\top) \quad \text{where } \mathbf{D}' \leftarrow [0, p - 1]^{n \times \ell} \quad (2)$$

$$\equiv (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{s}^\top \mathbf{F} + \mathbf{e}_\mathbf{P}^\top) \quad \text{where } \mathbf{F} \leftarrow \mathbb{Z}_q^{n \times \ell} \quad (3)$$

$$\approx_c (\mathbf{c}_\mathbf{B} \leftarrow \mathbb{Z}_q^m, \mathbf{c}_\mathbf{P} \leftarrow \mathbb{Z}_q^\ell) \quad (4)$$

In the above, Equation (1) follows by definition, Equation (2) by the smudging argument, where we need $\|\mathbf{e}_P\| \gg \|\mathbf{s}^\top \mathbf{D}'\|$ so that $\mathbf{e}_P^\top$ smudges $\mathbf{s}^\top \mathbf{D}'$, and Equation (3) from the fact that $p\mathbf{D} + \mathbf{D}'$ is distributed uniformly at random over $\mathbb{Z}_q^{n \times \ell}$, and Equation (4) from the LWE assumption. We then observe that the post-condition does not hold. Namely, we have to show that the following distributions are distinguishable:

$$(\mathbf{B}, \mathbf{P}, \mathbf{c}_B^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{K}, \mathbf{D}) \quad \text{and} \quad (\mathbf{B}, \mathbf{P}, \mathbf{c}_B^\top \leftarrow \mathbb{Z}_q^m, \mathbf{K}, \mathbf{D}) \quad \text{where} \quad \mathbf{K} \leftarrow \mathbf{B}^{-1}(\mathbf{P}).$$

In the above, $\mathbf{B}^{-1}(\mathbf{P})$ denotes short Gaussian distribution whose output satisfies $\mathbf{BK} = \mathbf{P}$. To see this, observe

$$\mathbf{c}_B^\top \mathbf{K} = (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top) \mathbf{K} = p\mathbf{s}^\top \mathbf{D} + \mathbf{e}_B^\top \mathbf{K} \bmod q.$$

holds when the given terms are from the LHS distribution. By taking modulo p of the above value, we get $\mathbf{e}_B^\top \mathbf{K} \bmod p$. For the attack to work, we take p large enough so that we have $\|\mathbf{e}_B^\top \mathbf{K}\| < p$. This allows us to recover $\mathbf{e}_B^\top \mathbf{K}$ over the integers, which in turn allows us to recover $\mathbf{e}_B^\top$ by solving linear equations. This completes the description of the attack. Recall that in order for the above attack to work, we need $\|\mathbf{e}_B^\top \mathbf{K}\| < p$ and $\|\mathbf{e}_P\| \gg \|\mathbf{s}^\top \mathbf{D}'\|$. Since each entry of $\mathbf{D}'$ is chosen uniformly at random over $[0, p-1]$, the norm of $\mathbf{e}_P$ should be much larger than p . These imply that we need $\|\mathbf{e}_P\| \gg \|\mathbf{e}_B^\top \mathbf{K}\|$ for the attack. Please see Section 3.2.1 for more details.

Second Attack when Pre-Condition has Larger Error. One may argue that our first attack is contrived because $\mathbf{P}$ falls into the ideal generated by p in $\mathbb{Z}_q$. By modifying the sampler, we also show another example of an attack where $\mathbf{P}$ is (marginally) uniformly random over $\mathbb{Z}_q^{n \times \ell}$. The counter-example works in a similar setting where $q = pr$ for prime p and r and $\mathbf{s}$ is short. In this example, the sampler chooses random $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, $\mathbf{D} \in [0, r-1]^{n \times \ell}$, and $\mathbf{E} \in \{0, 1\}^{m \times \ell}$. Then, it sets $\mathbf{P} = \mathbf{AE} + p\mathbf{D}$ and $\text{aux} = (\mathbf{A}, \mathbf{D}, \mathbf{E})$. This is a public coin sampler, since there is no hidden coin that the sampler uses that is not shown to the adversary. We can see that $\mathbf{AE}$ is distributed uniformly at random by the leftover hash lemma and thus so is $\mathbf{P}$. We remark that we only consider its marginal distribution here.

In this example, pre-condition and post-condition adversary will be given additional term of $\mathbf{c}_A$, which is either $\mathbf{s}^\top \mathbf{A} + \mathbf{e}_A$ or random, similarly to the original formulation of evasive LWE by Wee [Wee22]. We next show the precondition. Even given $(\mathbf{B}, \mathbf{P}, \text{aux} = (\mathbf{A}, \mathbf{D}, \mathbf{E}))$, the following holds:

$$\begin{aligned} & (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top) \\ \equiv & (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{s}^\top (p\mathbf{D} + \mathbf{AE}) + \mathbf{e}_P^\top) \tag{5} \\ \approx_s & (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{s}^\top (p\mathbf{D} + \mathbf{AE}) + \mathbf{e}_A^\top \mathbf{E} + \mathbf{e}_P^\top) \tag{6} \\ = & (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{c}_A^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{s}^\top (p\mathbf{D}) + \mathbf{e}_P^\top + \mathbf{c}_A^\top \mathbf{E}) \\ \approx_s & (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{c}_A^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{s}^\top (p\mathbf{D}) + \mathbf{s}^\top \mathbf{D}' + \mathbf{e}_P^\top + \mathbf{c}_A^\top \mathbf{E}) \quad \text{where } \mathbf{D}' \leftarrow [0, p-1]^{n \times \ell} \\ \equiv & (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{c}_A^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{s}^\top \mathbf{F} + \mathbf{e}_P^\top + \mathbf{c}_A^\top \mathbf{E}) \quad \text{where } \mathbf{F} \leftarrow \mathbb{Z}_q^{n \times \ell} \\ \approx_c & (\mathbf{c}_B \leftarrow \mathbb{Z}_q^m, \mathbf{c}_A^\top \leftarrow \mathbb{Z}_q^m, \mathbf{c}_P \leftarrow \mathbb{Z}_q^\ell) \end{aligned}$$

In the above, Equation (5) follows by definition, Equation (6) by the smudging argument, where we need $\|\mathbf{e}_P^\top\| \gg \|\mathbf{e}_A^\top \mathbf{E}\|$ so that $\mathbf{e}_P^\top$ smudges $\mathbf{e}_A^\top \mathbf{E}$. The rest of the indistinguishability follows similarly to our first counter-example. We then proceed to show that the post-condition distributions are distinguishable. Namely, we show that the following distributions are distinguishable given $(\mathbf{B}, \mathbf{P}, \text{aux} = (\mathbf{A}, \mathbf{D}, \mathbf{E}))$:

$$(\mathbf{c}_B^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{c}_A^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{K}) \quad \text{and} \quad (\mathbf{c}_B^\top \leftarrow \mathbb{Z}_q^m, \mathbf{c}_A^\top \leftarrow \mathbb{Z}_q^m, \mathbf{K}) \quad \text{where} \quad \mathbf{K} \leftarrow \mathbf{B}^{-1}(\mathbf{P}).$$

For the attack, the distinguisher computes $\mathbf{c}_B^\top \mathbf{K} - \mathbf{c}_A^\top \mathbf{E}$. If the terms come from the LHS distribution above, we have

$$\begin{aligned} \mathbf{c}_B^\top \mathbf{K} - \mathbf{c}_A^\top \mathbf{E} &= (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top) \mathbf{K} - (\mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top) \mathbf{E} = \mathbf{s}^\top (p\mathbf{D} + \mathbf{AE}) + \mathbf{e}_B^\top \mathbf{K} - (\mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top) \mathbf{E} \\ &= \mathbf{s}^\top (p\mathbf{D}) - \mathbf{e}_A^\top \mathbf{E} + \mathbf{e}_B^\top \mathbf{K}. \end{aligned}$$

Similarly to the case of our first attack, by taking modulo p , we can separate the error term $-\mathbf{e}_A^\top \mathbf{E} + \mathbf{e}_B^\top \mathbf{K}$ over the integer, if we set the parameters so that $\|-\mathbf{e}_A^\top \mathbf{E} + \mathbf{e}_B^\top \mathbf{K}\| < p$. By solving the linear equation, one can recover $\mathbf{e}_A$ and $\mathbf{e}_B$ if ℓ is sufficiently large. This allows the adversary to distinguish the above distributions. Please see Section 3.2.2 for more details.

1.3.2 Attack against the AKY24 Sampler.

Next we describe the attack against the sampler used by the previous version of [AKY24a]¹, who propose the notion of pseudorandom functional encryption (PRFE) and construct it from (certain variant of) evasive LWE. From here on unless stated otherwise, we will consider arithmetic operation on $\mathbb{Z}_q$ for a prime q . To explain their core idea, we recall a variant of the GSW FHE construction they use. In this variant, there are two types of ciphertexts. The first type of ciphertext encrypts a binary string $\mathbf{x} \in \{0, 1\}^\ell$ and is denoted by $E_{\text{pk}_{\text{fhe}}}(\mathbf{x})$, where pk_{fhe} is the public key of the FHE. The second type of ciphertext encrypts a vector $\mathbf{y}$ in $\mathbb{Z}_q^L$ and is denoted by $\overline{E_{\text{pk}_{\text{fhe}}}(\mathbf{y})}$. Furthermore, the first type ciphertext $E_{\text{pk}_{\text{fhe}}}(\mathbf{x})$ can be converted into the second type ciphertext $\overline{E_{\text{pk}_{\text{fhe}}}(f(\mathbf{x}))}$ of $f(\mathbf{x})$ by a homomorphic computation with respect to a function $f: \{0, 1\}^\ell \rightarrow \mathbb{Z}_q^{1 \times L}$. For the resulting ciphertext (which is represented in a form of a matrix) and a secret key $\mathbf{s}$ corresponding to pk_{fhe} , we have $\mathbf{s}^\top \overline{E_{\text{pk}_{\text{fhe}}}(f(\mathbf{x}))} = \mathbf{e}_{\text{fhe}}^\top + f(\mathbf{x})$, where $\mathbf{e}_{\text{fhe}}^\top$ is some small noise. Note that the decryption is not able to recover lower order bits of $f(\mathbf{x})$ due to the noise, but it can still recover the higher order bit information of $f(\mathbf{x})$.

In their construction, a ciphertext $\text{ct}(\mathbf{x})$ encrypting a message $\mathbf{x}$ consists of

$$\mathbf{c}_B^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \quad \mathbf{c}_A^\top = \mathbf{s}^\top (\mathbf{A} - E_{\text{pk}_{\text{fhe}}}(\mathbf{x}) \otimes \mathbf{G}) + \mathbf{e}_A^\top, \quad X = E_{\text{pk}_{\text{fhe}}}(\mathbf{x}). \quad (7)$$

Notice that here, the secret key $\mathbf{s}$ for the FHE is used also as an LWE secret for other terms like $\mathbf{c}_B$ and $\mathbf{c}_A$. They show that one can compute a short-norm matrix $\mathbf{H}_f$ and $\mathbf{H}_{f,X}$ such that $(\mathbf{A} - E_{\text{pk}_{\text{fhe}}}(\mathbf{x}) \otimes \mathbf{G})\mathbf{H}_{f,X} = \mathbf{A}\mathbf{H}_f - \overline{E_{\text{pk}_{\text{fhe}}}(f(\mathbf{x}))}$ following the idea from [BTW17, HLL23]. This leads to the following equation:

$$\mathbf{c}_A^\top \mathbf{H}_{f,X} = \mathbf{s}^\top (\mathbf{A}\mathbf{H}_f - \overline{E_{\text{pk}_{\text{fhe}}}(f(\mathbf{x}))}) + \mathbf{e}_A^\top \mathbf{H}_{f,X} = \mathbf{s}^\top \mathbf{A}\mathbf{H}_f - \mathbf{e}_{\text{fhe}}^\top - f(\mathbf{x}) + \mathbf{e}_A^\top \mathbf{H}_{f,X}, \quad (8)$$

where the second equality follows from $\mathbf{s}^\top \overline{E_{\text{pk}_{\text{fhe}}}(f(\mathbf{x}))} = \mathbf{e}_{\text{fhe}}^\top + f(\mathbf{x})$. A secret sk_f for a function f consists of $\mathbf{K} \leftarrow \mathbf{B}^{-1}(\mathbf{A}\mathbf{H}_f)$. Given the secret key, one can decrypt the ciphertext by computing

$$\mathbf{c}_B^\top \mathbf{K} - \mathbf{c}_A^\top \mathbf{H}_{f,X} = \mathbf{s}^\top \mathbf{B}\mathbf{K} + \mathbf{e}_B^\top \mathbf{K} - (\mathbf{s}^\top \mathbf{A}\mathbf{H}_f - \mathbf{e}_{\text{fhe}}^\top - f(\mathbf{x}) + \mathbf{e}_A^\top \mathbf{H}_{f,X}) = f(\mathbf{x}) + \underbrace{\mathbf{e}_B^\top \mathbf{K} + \mathbf{e}_{\text{fhe}}^\top - \mathbf{e}_A^\top \mathbf{H}_{f,X}}_{\text{small error}}. \quad (9)$$

Similarly to the case of the FHE decryption, we can recover higher order bits of $f(\mathbf{x})$ by removing the error terms.

We outline the security proof of their construction, since it is relevant to us. For simplicity, we restrict to the case of single ciphertext and single secret key. The security of PRFE stipulates that if $f(\mathbf{x})$ looks pseudorandom, then so is the ciphertext $\text{ct}(\mathbf{x})$ even given $\text{sk} = \mathbf{K}$. To show this, they invoke the evasive LWE assumption with respect to $\mathbf{P} = \mathbf{A}\mathbf{H}_f$. The assumption implies that for proving the security, it suffices to show the precondition, namely, pseudorandomness of $\mathbf{c}_B^\top$, $\mathbf{c}_A^\top$, X , and $\mathbf{c}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top$, where $\mathbf{e}_P^\top$ is a fresh Gaussian noise. This is proven by the following hybrids.

Hyb₀. In this hybrid, the adversary is given $\mathbf{c}_B^\top$, $\mathbf{c}_A^\top$, $X = E_{\text{pk}_{\text{fhe}}}(\mathbf{x})$, and $\mathbf{c}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top$.

Hyb₁. In this hybrid, we change how $\mathbf{c}_P$ is computed. Here, we compute $\mathbf{c}_P$ as $\mathbf{c}_P^\top = \mathbf{c}_A^\top \mathbf{H}_{f,X} + f(\mathbf{x}) + \mathbf{e}_P^\top$.

By Equation (8), we have $\mathbf{c}_A^\top \mathbf{H}_{f,X} + f(\mathbf{x}) + \mathbf{e}_P^\top = \mathbf{s}^\top \mathbf{P} - \mathbf{e}_{\text{fhe}}^\top + \mathbf{e}_A^\top \mathbf{H}_{f,X} + \mathbf{e}_P^\top$. Therefore, the

¹Their construction has since been updated so that the attack no longer applies.

difference from the previous hybrid is in the error term. Since we take $\mathbf{e}_P^\top$ large enough so that it smudges the small error term $-\mathbf{e}_{fhe}^\top + \mathbf{e}_A^\top \mathbf{H}_{f,X}$, this hybrid is statistically close to the previous hybrid.

Hyb₂. In this hybrid, we change $\mathbf{c}_B^\top$, $\mathbf{c}_A^\top$, and pk_{fhe} to be random by using the LWE assumption. Note that the public key pk_{fhe} of GSW is actually an LWE sample with respect to the secret key $\mathbf{s}$ and thus this is possible. This hybrid is indistinguishable from the previous one by the LWE assumption.

Hyb₃. In this hybrid, we change X to be a random string. The GSW encryption is essentially the same as Regev encryption [Reg09] and a random key is lossy [PW11]. Namely, the encryption under a random key is a random string. Therefore, this hybrid is statistically indistinguishable from the previous one.

Hyb₄. In this game, we change $\mathbf{c}_P$ to be a random string. This follows from the pseudorandomness of $f(\mathbf{x})$, which can be invoked since the information of $\mathbf{x}$ is not used anywhere else in the previous hybrid.

We are ready to describe our counter-example. The sampler in our counter-example outputs PRFE ciphertexts of AKY encrypting random secret $\mathbf{x}$ and $\mathbf{P}$ is computed so that $\mathbf{K} = \mathbf{B}^{-1}(\mathbf{P})$ constitutes a secret key for function f defined above. Namely, we set $\mathbf{P} = \mathbf{A}\mathbf{H}_f$. For the homomorphic evaluation of the circuit f , we need some contrived implementation, which we will discuss later. We show that the pre-condition distributions are indistinguishable, yet the post-condition distributions are distinguishable. The precondition with respect to this sampler essentially requires that $\mathbf{c}_B, \mathbf{c}_A$ defined as in Equation (7) along with $\mathbf{c}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top$ is pseudorandom. This is exactly what we showed in the above discussion.

We now provide the description of the distinguisher for the post-condition. We note that the description here is oversimplified but conveys the main intuition. The distinguisher is given $\mathbf{c}_B, \mathbf{c}_A, X$, and $\mathbf{K}$ and tries to check whether $\mathbf{c}_B, \mathbf{c}_A$, and X are structured as in Equation (7) or random. The distinguisher first computes $\mathbf{v} = \mathbf{c}_B^\top \mathbf{K} - \mathbf{c}_A^\top \mathbf{H}_{f,X} \bmod q$. We need special contrived circuit implementation of f for the attack to work, which will be explained soon. The distinguisher interprets $\mathbf{v}$ as a vector in $[-(q-1)/2, (q-1)/2]^L$ by regarding the $\mathbb{Z}_q$ element as an integer in $[-(q-1)/2, (q-1)/2]$. By Equation (9), if the terms are structured, we have $\mathbf{v} = f(\mathbf{x}) + \mathbf{e}_B^\top \mathbf{K} + \mathbf{e}_{fhe}^\top - \mathbf{e}_A^\top \mathbf{H}_{f,X}$. Our first observation is that when $f(\mathbf{x}) \in \mathbb{Z}_q^L$ is pseudorandom, each entry of $f(\mathbf{x})$ is unlikely to fall into $[-(q-1)/2, -(q-1)/2 + B] \cup [(q-1)/2 - B, (q-1)/2]$ for small B when it is represented as a vector in $[-(q-1)/2, (q-1)/2]^L$, where B is set so that we have $\|\mathbf{e}_B^\top \mathbf{K} + \mathbf{e}_{fhe}^\top - \mathbf{e}_A^\top \mathbf{H}_{f,X}\| < B$. Therefore, wraparound does not occur with overwhelming probability in the computation of $f(\mathbf{x}) + (\mathbf{e}_B^\top \mathbf{K} + \mathbf{e}_{fhe}^\top - \mathbf{e}_A^\top \mathbf{H}_{f,X}) \bmod q$ and we are able to recover $f(\mathbf{x}) + \mathbf{e}_B^\top \mathbf{K} + \mathbf{e}_{fhe}^\top - \mathbf{e}_A^\top \mathbf{H}_{f,X}$ over the integers, by representing $\mathbf{v}$ as a vector in $[-(q-1)/2, (q-1)/2]^L$. If we were able to separate $f(\mathbf{x})$ from the error terms as is done in our first counter-example, the attack would have been very simple. However, this is impossible since lower-order bits of $f(\mathbf{x})$ are also pseudorandom and mask the error terms.

To overcome this issue, we borrow the idea from [HJL21] to correlate the least significant bit of $f(\mathbf{x})$ with $\mathbf{e}_{fhe}^\top$. Namely, we show that we can make $f(\mathbf{x}) \equiv \mathbf{e}_{fhe}^\top \bmod 2$ if we choose a contrived circuit implementation for the homomorphic computation of f . They induce such a correlation in the setting of dual GSW FHE, while here, we show that similar correlation can also be constructed for the variant of GSW that AKY use. Therefore, by taking $\mathbf{v} \bmod 2$, we get $\mathbf{e}_B^\top \mathbf{K} - \mathbf{e}_A^\top \mathbf{H}_{f,X} \bmod 2$, since $f(\mathbf{x})$ and $\mathbf{e}_{fhe}^\top$ cancel each other modulo 2. Then, the attacker tries to find $\bar{\mathbf{e}}_A$ and $\bar{\mathbf{e}}_B$ such that $\bar{\mathbf{e}}_B^\top \mathbf{K} - \bar{\mathbf{e}}_A^\top \mathbf{H}_{f,X} = \mathbf{v} \bmod 2$. If such $\bar{\mathbf{e}}_A$ and $\bar{\mathbf{e}}_B$ are found, it guesses that the given terms are structured and otherwise if not. From the above description, in the structured case, such vectors can be always found since $\bar{\mathbf{e}}_B = \mathbf{e}_B$ and $\bar{\mathbf{e}}_A = \mathbf{e}_A$ constitute a solution. We can also show that in the case when the given terms are random, then such vectors cannot be found with overwhelming probability if we take L to be sufficiently large. Therefore, this constitutes a valid attack. Please see Section 4.1 for more details.

1.3.3 Attack against the HLL23 Sampler.

Here, we provide a counter-example against the evasive circular LWE assumption introduced in [HLL23] (HLL). Their assumption is similar to that of AKY, but with the crucial difference that there is no additional secret $\mathbf{x}$. Instead of the encryption X of $\mathbf{x}$, the encryption S of $\mathbf{s}$ is included in the pre- and post-condition distributions. Note that S is a circular ciphertext, since $\mathbf{s}$ is a secret key corresponding to pk_{fhe} . Namely, the evasive circular LWE assumption says that if

$$\mathbf{c}_B^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \quad \mathbf{c}_A^\top = \mathbf{s}^\top (\mathbf{A} - S \otimes \mathbf{G}) + \mathbf{e}_A^\top, \quad S = E_{\text{pk}_{\text{fhe}}}(\mathbf{s}), \quad \mathbf{c}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top$$

are pseudorandom, then so are $\mathbf{c}_B^\top$, $\mathbf{c}_A^\top$, and S even given $\mathbf{K} = \mathbf{B}^{-1}(\mathbf{P})$.

Is Circular Evasive LWE Public or Private Coin? The circular evasive LWE assumption has been widely categorized as a “public-coin” assumption, in the original work by [HLL23] as well as subsequent work, such as [BDJ⁺25, CW25]. Brzuska et. al. noticed that circular evasive LWE does not fall into the regime of public coin in the strict sense.

The essential difference between this assumption and the private-coin version used in AKY is that in the former, the value to be hidden within FHE is the FHE secret key $\mathbf{s}$ (resulting in circularity), while in the latter, it is an arbitrary string $\mathbf{x}$ chosen by the sampler. Since in AKY, $\mathbf{x}$ is chosen by the sampler and must be hidden inside the FHE encoding, it becomes necessary that the FHE encoding is output by the sampler itself – this results in it getting categorized as private coin. On the other hand, in HLL, the FHE secret, which is the same as the LWE secret, is chosen *outside* the sampler and hence the FHE encoding need not part of the sampler’s output – this results in it getting categorized as public-coin. Since it is prevalent in the community to consider the circular evasive LWE assumption as public-coin, we adopt the same terminology here. Indeed, we view the main message of our attack as saying that the circular evasive LWE assumption is *not* a kosher public-coin assumption.

The Attack. In our counter-example here, we regard $(\mathbf{c}_B, \mathbf{c}_A, S)$ as an AKY ciphertext encrypting $\mathbf{s}$ and set $\mathbf{P}$ so that $\mathbf{K}$ constitutes a secret key for a function f . We defer the discussion on how to instantiate f to later. The attack against the post-condition is essentially the same as that for AKY, where the only difference is that $\mathbf{x}$ is replaced with $\mathbf{s}$.

We now argue that the pre-condition holds, which requires some more work. For the precondition to hold, we need a stronger property than just requiring $f(\mathbf{s})$ to be pseudorandom. This is because $\mathbf{s}$ is used also for the computation of other terms, for example in $\mathbf{c}_A$ and S , unlike $\mathbf{x}$ in the case of AKY, where $\mathbf{x}$ is a separate randomness chosen independently from other terms. Hence, we need that $(\mathbf{c}_B, \mathbf{c}_A, S, \text{pk}_{\text{fhe}}, f(\mathbf{s}))$ is jointly pseudorandom. Assuming we have such f , the precondition is proven by the following hybrids. The main difference from the AKY counter-example is that here, we collapse Hyb₂, Hyb₃, and Hyb₄ there into a single hybrid using the property of f .

Hyb₀. In this hybrid, the adversary is given $\mathbf{c}_B^\top$, $\mathbf{c}_A^\top$, $S = E_{\text{pk}_{\text{fhe}}}(\mathbf{s})$, and $\mathbf{c}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top$.

Hyb₁. In this hybrid, we change how $\mathbf{c}_P$ is computed. Here, we compute $\mathbf{c}_P$ as $\mathbf{c}_P^\top = \mathbf{c}_A^\top \mathbf{H}_{f,S} + f(\mathbf{s}) + \mathbf{e}_P^\top$. By Equation (8) (where $\mathbf{x}$ is replaced with $\mathbf{s}$ and X with S), we have $\mathbf{c}_A^\top \mathbf{H}_{f,S} + f(\mathbf{s}) + \mathbf{e}_P^\top = \mathbf{s}^\top \mathbf{P} - \mathbf{e}_{\text{fhe}}^\top + \mathbf{e}_A^\top \mathbf{H}_{f,S} + \mathbf{e}_P^\top$. Therefore, the difference from the previous hybrid is in the error term. Since we take $\mathbf{e}_P^\top$ large enough so that it smudges the small error term $-\mathbf{e}_{\text{fhe}}^\top + \mathbf{e}_A^\top \mathbf{H}_{f,S}$, this hybrid is statistically close to the previous hybrid.

Hyb₂. In this hybrid, we change $\mathbf{c}_B^\top$, $\mathbf{c}_A^\top$, pk_{fhe} , S , $\mathbf{c}_P$ to be random. This follows by the joint pseudorandomness property of f , by observing that $\mathbf{c}_P$ is obtained by adding $f(\mathbf{s})$ to a known term.

To finish the description of the counter-example, it remains to find the implementation of such f . Simply treating $\mathbf{s}$ as a random seed for general PRG will not work, since there is a leakage of $\mathbf{s}$ in the form of LWE samples with respect to the secret $\mathbf{s}$. In this setting, we do not know how to prove the joint pseudorandomness that we require. A more promising approach is to make f randomized and define $f(\mathbf{s})$ to be an LWE instance with respect to some public matrix $\mathbf{F}$, which describes the function f . If f is allowed to be randomized, this works since $(f(\mathbf{s}), \mathbf{c}_B, \mathbf{c}_A, \text{pk}_{\text{fhe}}, S)$ forms LWE samples along with circular encoding and thus the pseudorandomness of them follows from the circular LWE assumption.

Our solution is to define $f(\mathbf{s})$ to be learning with rounding instance [BPR12] with respect to the public matrix $\mathbf{F}$ and secret $\mathbf{s}$. Namely, we define $f(\mathbf{s}) = \lceil (p/q)\mathbf{s}^\top \mathbf{F} \rceil$. The joint pseudorandomness is proven by first arguing that $\lceil (p/q)\mathbf{s}^\top \mathbf{F} \rceil = \lceil (p/q)(\mathbf{s}^\top \mathbf{F} + \mathbf{e}) \rceil$ holds with high probability for small noise $\mathbf{e}$. Then, we replace $\mathbf{s}^\top \mathbf{F} + \mathbf{e}$ along with other LWE terms and the circular encoding $\mathbf{S}$ with random values using the circular LWE assumption. This completes the proof, since $\lceil (p/q)\mathbf{v} \rceil$ for random vector $\mathbf{v}$ will result in a random string in $\mathbb{Z}_p$ for appropriately chosen p and q . Please see Section 3.1 for more details.

1.3.4 Attack against the BDJMPV24 Sampler.

Next, we show a counter-example for the sampler used in [BDJ⁺25]. For the sake of explanation, we describe a simplified version of their sampler in this overview. However, the attack applies to the full-fledged version as well. They construct a primitive they call pseudorandom obfuscation (PRO) and prove the security of the construction using evasive LWE. Roughly speaking, this primitive allows us to obfuscate a PRF. Here, we consider a function f which takes as input a key $\mathbf{x}$ of the PRF and outputs the truth table of the PRF under the key $\mathbf{x}$. We assume that the truth table of the PRF is represented as an element of $\mathbb{Z}_q^{k \times \kappa}$. In our context, we set k and κ to be some huge polynomials and thus the input space of the PRF is of polynomial size. For the primitive to be non-trivial, we want the size of the obfuscation to be much smaller than the size of the truth table (i.e., $O(k\kappa \log q)$). The security of PRO says that if the truth table $f(\mathbf{x}) \in \mathbb{Z}_q^{k \times \kappa}$ is pseudorandom, then so is the description of the obfuscation.

To describe their sampler, we follow their notation that is different from AKY and HLL, where $\mathbf{B}$ is a tall matrix of size $m \times n$ and the LWE sample with respect to it will be $\mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}$, where $\mathbf{S} \in \mathbb{Z}_q^{n \times \kappa}$ is an LWE secret and $\mathbf{E}$ is a noise term. Furthermore, we follow their syntax where the sampler does not choose $\mathbf{P} \in \mathbb{Z}_q^{k \times n}$ by itself. Instead, $\mathbf{P}$ is chosen uniformly at random outside of the control by the sampler. The sampler takes as input $\mathbf{P}$, chooses $\mathbf{S}$ by itself, and computes auxiliary information aux depending on $\mathbf{P}$ and $\mathbf{S}$, where aux is defined as follows:

$$\text{aux} = (\mathbf{A} = \mathbf{P}\mathbf{R} + \bar{\mathbf{E}} + \text{encode}(\mathbf{s}), \quad X = \text{E}_{\text{pk}_{\text{fhe}}}(\mathbf{x}), \quad \mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}).$$

In the above, $\mathbf{x}$ is a secret value, $\mathbf{R}$ is a random matrix, $\bar{\mathbf{E}}$ is a matrix with low norm, and encode is an encoding function with certain property that we will specify soon. The matrix $\mathbf{F}$ is a low norm matrix that is computed by first evaluating the FHE ciphertext X with respect to the function f to obtain $\text{E}_{\text{pk}_{\text{fhe}}}(f(\mathbf{x}))$ and then processing it so that

$$\text{encode}(\mathbf{s})\mathbf{F} = f(\mathbf{x}) + \mathbf{E}_{\text{fhe}}$$

holds for low norm $\mathbf{E}_{\text{fhe}}$. They show that the pre-condition holds for a slight modification of the above sampler. We omit the details here.

We proceed to explain the distinguishing attack against the post-condition distribution. Namely, we show that given $\mathbf{D}$ such that $\mathbf{D}\mathbf{B} = \mathbf{P}$ and aux , the following distribution is not pseudorandom

$$(\mathbf{B}, \mathbf{P}, \mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}, \text{aux}).$$

Before giving an attack, we first observe the following equation:

$$\begin{aligned} \mathbf{AF} + \mathbf{DC} - \mathbf{PH} &= (\mathbf{PR} + \bar{\mathbf{E}} + \text{encode}(\mathbf{s}))\mathbf{F} + \mathbf{D}(\mathbf{BS} + \mathbf{E}) - \mathbf{P}(\mathbf{S} + \mathbf{RF}) \\ &= \underbrace{\bar{\mathbf{E}}\mathbf{F} + \mathbf{DE} + \mathbf{E}_{\text{fhe}}}_{\text{small error}} + f(\mathbf{x}). \end{aligned}$$

One can see that the noisy version of the truth table $f(\mathbf{x})$ is obtained by the above computation. Roughly speaking, their obfuscation consists of $(\mathbf{A}, \mathbf{C}, \mathbf{D}, \mathbf{F}, \mathbf{H}, \mathbf{P})$ and the size of it is much smaller than that of the truth table itself. This is an approximate version of PRO in the sense that the lower-order bits cannot be recovered. However, by embedding the information that we want to recover into the higher-order bits, it can be made precise.

The distinguishing strategy is again similar to those for AKY and HLL, which uses contrived circuit for the homomorphic computation to make a correlation between the encrypted value and the noise appearing when we decrypt. Namely, we create a correlation such that $f(\mathbf{x}) \equiv \mathbf{E}_{\text{fhe}} \pmod{2}$ by a contrived homomorphic operation for the underlying GSW FHE. If given terms are structured, we can recover $\bar{\mathbf{E}}\mathbf{F} + \mathbf{DE} + \mathbf{E}_{\text{fhe}} + f(\mathbf{x})$ over the integer by computing $\mathbf{AF} + \mathbf{DC} - \mathbf{PH}$ and representing it as a matrix in $[-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor]^{k \times \kappa}$, similarly to the case of AKY and HLL. We then take modulus 2 for the resulting term, which gives us $\bar{\mathbf{E}}\mathbf{F} + \mathbf{DE} \pmod{2}$. Since $\mathbf{F}$ and $\mathbf{D}$ are public, we can recover $\bar{\mathbf{E}}$ and $\mathbf{E}$ by solving the linear system of equations. On the other hand, if the given terms are random, we can show that the linear system of equations cannot be solved with high probability. This constitutes a valid distinguishing attack. Please see Section 4.2 for more details.

1.3.5 Impossibility of FE for General Pseudorandom Functionalities

Now we describe our impossibility result for the multi-challenge “pseudorandom CT” (prCT) security for pseudorandom FE (PRFE) as described in AKY. We first recall the definition. A scheme PRFE for a function family $\mathcal{F} = \{f : \mathcal{X} \rightarrow \mathcal{Y}\}$ is multi-challenge prCT secure if for any sampler that outputs $\{f_i\}_{i \in [Q_k]}, \{\mathbf{x}_j\}_{j \in [Q_c]}$ and $\text{aux} \in \{0, 1\}^*$ such that

$$(\{f_i\}_i, \{f_i(\mathbf{x}_j)\}_{i,j}, \text{aux}) \approx (\{f_i\}_i, \{\text{random}_{i,j}\}_{i,j}, \text{aux})$$

the following indistinguishability holds:

$$(\{f_i\}_i, \{\text{ct}_{\mathbf{x}_j}\}_j, \{\text{sk}_{f_i}\}_i, \text{aux}) \approx (\{f_i\}_i, \{\text{random-ct}_j\}_j, \{\text{sk}_{f_i}\}_i, \text{aux})$$

Here, Q_c and Q_k are the number of ciphertext and key queries. We show the impossibility for general case, both with and without aux using subexponentially secure instance hiding witness encryption (WE). Recall that a WE scheme [GGSW13] is defined using a language $\mathcal{L} \in \text{NP}$, where a message is encrypted with respect to a string $\text{stmt} \in \{0, 1\}^*$ and the decryption is possible using a witness for $\text{stmt} \in \mathcal{L}$. If $\text{stmt} \notin \mathcal{L}$, then the message remains hidden. Our techniques are inspired by those of [BDJ⁺25], who showed a similar result for pseudorandom obfuscation.

First, we show that for sufficiently large Q_c and Q_k , for any functionality, there exist a contrived aux for which the pre-condition holds but the post condition is false. The idea, similar to [BDJ⁺25], is to hide $\{f_i(\mathbf{x}_j)\}_{i,j}$ values (by using it as stmt) inside instance hiding WE ciphertexts wct, provided as part of aux, and let the secret keys and ciphertexts of the PRFE, namely $\{\text{sk}_{f_i}\}_i$ and $\{\text{ct}_{\mathbf{x}_j}\}_j$ be the witness, using which wct can be successfully decrypted, by appropriately defining the corresponding language. The distinguisher’s strategy (in the post condition) is then simply to attempt decryption of wct using $\{\text{sk}_{f_i}\}_i$ and $\{\text{ct}_{\mathbf{x}_j}\}_j$ as a witness - if successful, then output 0 (LHS), else 1 (RHS). To prove indistinguishability in the pre-condition, it is important that aux hides $f_i(\mathbf{x}_j)$ values, which can be shown using the instance hiding property of WE and simple counting arguments, again similar to that in [BDJ⁺25].

In the absence of auxiliary input, a natural idea would be to output the WE ciphertexts as FE decryptions. However, since WE ciphertext can be polynomial in its witness size, we cannot perform its computation inside any circuit for which a PRFE key is provided. This is because then the PRFE ciphertext size and key size will be polynomial in Q_k and Q_c and we will not be able to apply incompressibility argument that requires the size of stmt to be sufficiently larger than the witness size and is needed in the proof of pre-condition. Instead, our solution is that the sampler itself computes the WE ciphertext, wct, as before, but now instead of providing it as aux, it divides wct into blocks of fixed lengths and encrypts each block into different PRFE ciphertexts. It also provides an additional PRFE key for a function which simply outputs a part of its input. Now, to be able to argue the indistinguishability in the pre-condition, we need pseudorandomness of WE ciphertexts. This can be achieved by combining (plain) WE with compute-and-compare/lockable obfuscation [WZ17, GKW17]. Please see Section 4.3 for details.

1.3.6 Interpreting the Attacks.

With the development of new attacks, the obvious question that arises is – are applications built from the aforementioned versions of evasive LWE dead? To understand, let us revisit the primary intuition of evasive LWE as formulated by Wee. One way to interpret the original formulation of Wee’s evasive LWE is that it is really two assumptions rolled into one²:

1. **Preimages $\mathbf{B}^{-1}(\mathbf{P})$ can only be used semi-honestly**: Short Gaussian preimages $\mathbf{K} = \mathbf{B}^{-1}(\mathbf{P})$ can only be used semi-honestly to compute samples $\mathbf{sP} + \mathbf{eK}$ from $\mathbf{sB} + \mathbf{e}$ and cannot be exploited in any other way. Here, note that this condition implicitly demands that $\mathbf{P}$ is sufficiently “well-behaved”, and in particular does not contain low norm linear dependencies that would permit “mix and match” attacks of preimages [Agr17]³.
2. **LWE with correlated error is “no worse” than standard LWE**: By using $\mathbf{B}^{-1}(\mathbf{P})$ as above, the adversary can compute an LWE sample $\mathbf{sP} + \mathbf{eK}$ with correlated, overdefined error as against i.i.d. error. However, the intuition here is that this LWE with correlated noise can be conjectured “as good as” LWE since the attacker should never be able to remove the large mask $\mathbf{sP}$ and therefore should never obtain anything in the zeroizing regime.

We now study the two assumptions above in turn, as though they are separate. The first assumption was known to yield attacks when the auxiliary input provided to the adversary explicitly contains contrived information, such an obfuscated program which spits out secrets when given short preimages [VWW22, BUW24, HHY25]. However, for “natural” distributions of auxiliary information, we do not know any attack that exploits $\mathbf{K}$ to compute anything other than $\mathbf{sP} + \mathbf{eK}$, which is explicitly desired. The new problems come from the second assumption, where an i.i.d error would have sufficed for some flooding argument (allowing the pre-condition to be proven), but the correlated, overdefined error does not.

The attacks can be prevented in heuristic ways, as in the revised version of [AKY24a]. However, a clean, principled fix to the assumption to avoid the above attacks has remained elusive so far.

2 Preliminaries

Notations. We use bold capital letters to denote a matrix and bold small letters to represent vectors. By default, a vector $\mathbf{v}$ is a column vector and $\mathbf{v}^\top$ is a row vector. $\mathbf{e}_i$ represents a unit vector with 1 at position i .

²These cannot be separated in general, needless to say.

³As an extreme example of badly behaved $\mathbf{P} = [\mathbf{P}_1, \mathbf{P}_2]$, suppose the adversary could obtain $\mathbf{K}_1, \mathbf{K}_2$ such that $\mathbf{BK}_1 = \mathbf{P}_1 \bmod q$ and $\mathbf{BK}_2 = 2\mathbf{P}_1 \bmod q$. Then, the adversary could compute a short trapdoor to $\mathbf{B}$ simply as $2\mathbf{K}_1 - \mathbf{K}_2$, which is clearly disastrous. However, such “pre-image combination” attacks are prevented by the requirement in the pre-condition, that $\mathbf{sP}_1, \mathbf{sP}_2$ are jointly pseudorandom.

Throughout the paper, we use λ to denote the security parameter. For any $x \in \mathbb{Z}_q$, $\lfloor x \rfloor_p \in \mathbb{Z}_p = \lfloor x \cdot \frac{p}{q} \rfloor$. Sometimes, we regard an element in $x \in \mathbb{Z}_q$ as an integer in $[-q/2, q/2]$. When we take an absolute value $|x|$ for $x \in \mathbb{Z}_q$, this operation means that first regarding x as an integer and then taking the absolute value. For any function f , we write $|f|$ to represent its size. For any set S , $|S|$ denotes the number of elements in set S . For any vector $\mathbf{v}$, we use $\mathbf{v}[\ell; \ell']$ to denote the part of vector $\mathbf{v}$ starting from ℓ -th element and ending at ℓ' -th element where $\ell, \ell' \in \mathbb{Z}_q$. For any vector $\mathbf{v}$, $\mathbf{v}[i, j]$ denotes the j th lowest order bit of i -th element of $\mathbf{v}$. For any matrix $\mathbf{V}$, $\mathbf{V}[i, j]$ denotes the element at i -th row and j -th column, $\mathbf{V}[i, \cdot]$ denotes the i th row of $\mathbf{V}$ and $\mathbf{V}[\cdot, j]$ denotes the j th column of $\mathbf{V}$. We consider the infinity norm:

$$\|\mathbf{v}\| = \max_i |\mathbf{v}[i]|, \quad \|\mathbf{V}\| = \max_i \sum_j |\mathbf{V}[i, j]|$$

With slight overload of notation, we also define $|\mathbf{v}| = \max_i |\mathbf{v}[i]|$ and $|\mathbf{V}| = \max_{i,j} |\mathbf{V}[i, j]|$. We define the most-significant bit operator $\text{MSB} : \mathbb{Z}_q \rightarrow \{0, 1\}$ as

$$\text{MSB}(x) = \begin{cases} 0, & \text{if } x \in [-q/4, q/4) \\ 1, & \text{otherwise} \end{cases}$$

We slightly overload the notation and represent MSB of a vector $\mathbf{v} \in \mathbb{Z}_q^n$ as $\text{MSB}(\mathbf{v}) = (\text{MSB}(\mathbf{v}[1]), \dots, \text{MSB}(\mathbf{v}[n]))$.

2.1 Functional Encryption for Pseudorandom Functionalities

In this section, we give the definition of functional encryption for pseudorandom functionalities PRFE [AKY24a]. Consider a function family $\{\mathcal{F}_{\text{prm}} = \{f : \mathcal{X}_{\text{prm}} \rightarrow \mathcal{Y}_{\text{prm}}\}\}_{\text{prm}}$ for a parameter $\text{prm} = \text{prm}(\lambda)$. Each function $f \in \mathcal{F}_{\text{prm}}$ takes as input a string $x \in \mathcal{X}_{\text{prm}}$ and outputs $f(x) \in \mathcal{Y}_{\text{prm}}$.

Syntax. A functional encryption for pseudorandom functionalities PRFE for function family $\mathcal{F}_{\text{prm}}$ consists of four polynomial time algorithms (Setup, KeyGen, Enc, Dec) defined as follows.

$\text{Setup}(1^\lambda, \text{prm}) \rightarrow (\text{mpk}, \text{msk})$. The setup algorithm takes as input the security parameter λ and a parameter prm and outputs a public key mpk and a master secret key msk .

$\text{KeyGen}(\text{msk}, f) \rightarrow \text{sk}_f$. The key generation algorithm takes as input the master secret key msk and a function $f \in \mathcal{F}_{\text{prm}}$ and it outputs a functional secret key sk_f .

$\text{Enc}(\text{mpk}, x) \rightarrow \text{ct}$. The encryption algorithm takes as input the public key mpk and an input $x \in \mathcal{X}_{\text{prm}}$ and outputs a ciphertext $\text{ct} \in \mathcal{CT}$, where $\mathcal{CT}$ is the ciphertext space.

$\text{Dec}(\text{sk}_f, \text{ct}) \rightarrow y$. The decryption algorithm takes as input a functional secret key sk_f and a ciphertext ct and outputs $y \in \mathcal{Y}_{\text{prm}}$.

Definition 2.1 (Correctness). A PRFE scheme is said to be correct if for all prm , any input $x \in \mathcal{X}_{\text{prm}}$ and function $f \in \mathcal{F}_{\text{prm}}$, we have

$$\Pr \left[(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda, 1^{\text{prm}}), \text{sk}_f \leftarrow \text{KeyGen}(\text{msk}, f), \right. \\ \left. \text{Dec}(\text{sk}_f, \text{Enc}(\text{mpk}, x)) = f(x) \right] = 1.$$

where the probability is taken over the coins of Setup, KeyGen, and Enc.

prCT security. [AKY24a] For a PRFE scheme for function family $\{\mathcal{F}_{\text{prm}} = \{f : \mathcal{X}_{\text{prm}} \rightarrow \mathcal{Y}_{\text{prm}}\}\}_{\text{prm}}$ parameter $\text{prm} = \text{prm}(\lambda)$, let Samp be a PPT algorithm that on input 1^λ , outputs $(f_1, \dots, f_{Q_{\text{key}}}, x_1, \dots, x_{Q_{\text{msg}}}, \text{aux} \in \{0, 1\}^*)$ where Q_{key} is the number of key queries, Q_{msg} is the number of message queries, and $f_i \in \mathcal{F}_{\text{prm}}$, $x_j \in \mathcal{X}_{\text{prm}}$ for all $i \in [Q_{\text{key}}]$, $j \in [Q_{\text{msg}}]$. We define the following advantage functions:

$$\begin{aligned} \text{Adv}_{\mathcal{A}_0}^{\text{pre}}(\lambda) &\stackrel{\text{def}}{=} \Pr[\mathcal{A}_0(\text{aux}, \{f_i, f_i(x_j)\}_{i \in [Q_{\text{key}}], j \in [Q_{\text{msg}}]}) = 1] \\ &\quad - \Pr[\mathcal{A}_0(\text{aux}, \{f_i, \Delta_{i,j} \leftarrow \mathcal{Y}_{\text{prm}}\}_{i \in [Q_{\text{key}}], j \in [Q_{\text{msg}}]}) = 1] \\ \text{Adv}_{\mathcal{A}_1}^{\text{post}}(\lambda) &\stackrel{\text{def}}{=} \Pr[\mathcal{A}_1(\text{mpk}, \text{aux}, \{f_i, \text{Enc}(\text{mpk}, x_j), \text{sk}_{f_i}\}_{i \in [Q_{\text{key}}], j \in [Q_{\text{msg}}]}) = 1] \\ &\quad - \Pr[\mathcal{A}_1(\text{mpk}, \text{aux}, \{f_i, \delta_j \leftarrow \mathcal{CT}, \text{sk}_{f_i}\}_{i \in [Q_{\text{key}}], j \in [Q_{\text{msg}}]}) = 1] \end{aligned}$$

where $(f_1, \dots, f_{Q_{\text{key}}}, x_1, \dots, x_{Q_{\text{msg}}}, \text{aux} \in \{0, 1\}^*) \leftarrow \text{Samp}(1^\lambda)$, $(\text{mpk}, \text{msk}) \leftarrow \text{Setup}(1^\lambda, \text{prm})$ and $\mathcal{CT}$ is the ciphertext space. We say that a PRFE scheme for function family $\mathcal{F}_{\text{prm}}$ satisfies prCT security if for every PPT Samp there exists a polynomial $Q(\cdot)$ such that for every PPT adversary $\mathcal{A}_1$, there exists another PPT $\mathcal{A}_0$ such that

$$\text{Adv}_{\mathcal{A}_0}^{\text{pre}}(\lambda) \geq \text{Adv}_{\mathcal{A}_1}^{\text{post}}(\lambda) / Q(\lambda) - \text{negl}(\lambda)$$

and $\text{time}(\mathcal{A}_0) \leq \text{time}(\mathcal{A}_1) \cdot Q(\lambda)$.

2.2 Instance Hiding Witness Encryption with Pseudorandom Ciphertext

Definition 2.2 (Witness Encryption). [GGSW13, BDJ⁺25] A witness encryption scheme for an NP language $\mathcal{L}$ (with the corresponding witness relation R) consists of the following two polynomial-time algorithms:

$\text{WE.Enc}(1^\lambda, \text{stmt}, \mu) \rightarrow \text{wct}$.⁴ On input security parameter λ , an unbounded-length string stmt and message $\mu \in \{0, 1\}$, outputs ciphertext wct .

$\text{WE.Dec}(\text{wt}, \text{wct}) = \mu / \perp$. On input an unbounded length string wt and ciphertext wct , it either outputs message μ or the symbol $\perp$.

These algorithms satisfy the following conditions:

- **Correctness.** For any security parameter λ , for any $\mu \in \{0, 1\}$, for any $\text{stmt} \in \mathcal{L}$, such that $R(\text{stmt}, \mathcal{L})$ holds i.e. $\text{stmt} \in \mathcal{L}$, we have $\Pr[\text{WE.Dec}(\text{wt}, \text{WE.Enc}(1^\lambda, \text{stmt}, \mu)) = \mu] = 1$.
- **Soundness.** For any PPT adversary $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for any $\text{stmt} \notin \mathcal{L}$, we have: $|\Pr[\mathcal{A}(\text{WE.Enc}(1^\lambda, \text{stmt}, 0)) = 1] - \Pr[\mathcal{A}(\text{WE.Enc}(1^\lambda, \text{stmt}, 1)) = 1]| \leq \text{negl}(\lambda)$.
- **Instance Hiding.** For any PPT adversary $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for any $\mu \in \{0, 1\}$ and for any $\text{stmt}, \text{stmt}' \notin \mathcal{L}$, we have: $|\Pr[\mathcal{A}(\text{WE.Enc}(1^\lambda, \text{stmt}, \mu)) = 1] - \Pr[\mathcal{A}(\text{WE.Enc}(1^\lambda, \text{stmt}', \mu)) = 1]| \leq \text{negl}(\lambda)$.
- **Pseudorandom Ciphertext.** For any PPT adversary $\mathcal{A}$, there exists a negligible function $\text{negl}(\cdot)$ such that for any $\text{stmt} \notin \mathcal{L}$ and any $\mu \in \{0, 1\}$, we have: $|\Pr[\mathcal{A}(\text{WE.Enc}(1^\lambda, \text{stmt}, \mu)) = 1] - \Pr[\mathcal{A}(\mathcal{U}) = 1]| \leq \text{negl}(\lambda)$ where $\mathcal{U}$ is sampled uniformly from the ciphertext space.

⁴In the main body, we drop the first input and implicitly assume 1^λ as an input.

2.3 Lattice Preliminaries

Here, we recall some facts on lattices that are needed for the exposition of our construction. Throughout this section, n , m , and q are integers such that $n = \text{poly}(\lambda)$ and $m \geq n \lceil \log q \rceil$. In the following, we define a function ρ_σ as $\rho_\sigma = \exp(-\pi \|\mathbf{x}/\sigma\|_2^2)$, where $\|\cdot\|_2$ is the Euclidean norm. We will overload the notation to denote by $\rho_\sigma(S)$ the summation of $\rho_\sigma(\mathbf{x})$ over all $\mathbf{x} \in S$ for a countable set X . For a countable set S , $\mathcal{D}_{S,\gamma}$ refers to a distribution that outputs $\mathbf{x} \in S$ with probability $\rho_\sigma(\mathbf{x})/\rho(S)$. In the following, let $\text{SampZ}(\gamma)$ be a sampling algorithm for the truncated discrete Gaussian distribution over $\mathbb{Z}$ with parameter $\gamma > 0$ whose support is restricted to $z \in \mathbb{Z}$ such that $|z| \leq \sqrt{n}\gamma$. Namely, $\text{SampZ}(\gamma)$ is the truncated version of $\mathcal{D}_{\mathbb{Z},\gamma}$.

Let

$$\mathbf{g} = (2^0, 2^1, \dots, 2^{\frac{m}{n+1}-1})^\top, \quad \mathbf{G} = \mathbf{I}_{n+1} \otimes \mathbf{g}^\top$$

be the gadget vector and the gadget matrix. For $\mathbf{p} \in \mathbb{Z}_q^n$, we write $\mathbf{G}^{-1}(\mathbf{p})$ for the m -bit vector $(\text{bits}(\mathbf{p}[1]), \dots, \text{bits}(\mathbf{p}[n+1]))^\top$, where $\text{bits}(\mathbf{p}[i])$ are $m/(n+1)$ bits for each $i \in [n+1]$. The notation extends column-wise to matrices and it holds that $\mathbf{G}\mathbf{G}^{-1}(\mathbf{P}) = \mathbf{P}$.

Trapdoors. Let us consider a matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$. For all $\mathbf{V} \in \mathbb{Z}_q^{n \times m'}$, we let $\mathbf{A}^{-1}(\mathbf{V})$ be an output distribution of $\text{SampZ}(\gamma)^{m \times m'}$ conditioned on $\mathbf{A} \cdot \mathbf{A}^{-1}(\mathbf{V}, \gamma) = \mathbf{V}$. A γ -trapdoor for $\mathbf{A}$ is a trapdoor that enables one to sample from the distribution $\mathbf{A}^{-1}(\mathbf{V}, \gamma)$ in time $\text{poly}(n, m, m', \log q)$ for any $\mathbf{V}$. We slightly overload notation and denote a γ -trapdoor for $\mathbf{A}$ by $\mathbf{A}_\gamma^{-1}$. The following properties had been established in a long sequence of works [GPV08, CHKP10, ABB10a, ABB10b, MP12, BLP⁺13].

Lemma 2.3 (Properties of Trapdoors). Lattice trapdoors exhibit the following properties.

1. Given $\mathbf{A}_\tau^{-1}$, one can obtain $\mathbf{A}_{\tau'}^{-1}$ for any $\tau' \geq \tau$.
2. Given $\mathbf{A}_\tau^{-1}$, one can obtain $[\mathbf{A} \parallel \mathbf{B}]_\tau^{-1}$ and $[\mathbf{B} \parallel \mathbf{A}]_\tau^{-1}$ for any $\mathbf{B}$.
3. There exists an efficient procedure $\text{TrapGen}(1^n, 1^m, q)$ that outputs $(\mathbf{A}, \mathbf{A}_{\tau_0}^{-1})$ where $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ for some $m = O(n \log q)$ and is 2^{-n} -close to uniform, where $\tau_0 = \omega(\sqrt{n \log q \log m})$.

Useful Lemmata.

Lemma 2.4 (tail and truncation of $\mathcal{D}_{\mathbb{Z},\gamma}$). There exists $B_0 \in \Theta(\sqrt{\lambda})$ such that

$$\Pr \left[x \leftarrow \mathcal{D}_{\mathbb{Z},\gamma} : |x| > \gamma B_0(\lambda) \right] \leq 2^{-\lambda} \quad \text{for all } \gamma \geq 1 \text{ and } \lambda \in \mathbb{N}.$$

Let $B \geq 0$, the distribution $\mathcal{D}_{\mathbb{Z},\gamma \leq B}$ is sampled by first sampling $x \leftarrow \mathcal{D}_{\mathbb{Z},\gamma}$, then returning x if $|x| \leq B$, and 0 otherwise. Let $\gamma \geq 1$ and $B = \gamma \Theta(\sqrt{\lambda})$, then $\mathcal{D}_{\mathbb{Z},\gamma \leq B}$ is $2^{-\Omega(\lambda)}$ -close to $\mathcal{D}_{\mathbb{Z},\gamma}$.

Lemma 2.5 ([Lyu12, Lemma 4.4]). The following hold.

1. For any $k > 0$, $\Pr[|z| > k\sigma; z \leftarrow \mathcal{D}_{\mathbb{Z},\sigma}] \leq 2\exp(-k^2/2)$.
2. For any $k > 1$,

$$\Pr[\|\mathbf{z}\| > k\sigma\sqrt{m}; \mathbf{z} \leftarrow \mathcal{D}_{\mathbb{Z}^m,\sigma}] < k^m \exp\left(\frac{m}{2}(1 - k^2)\right).$$

Lemma 2.6. [PR06, Adapted from Lemma 2.11] For all but 2^{-n} fraction of $\mathbf{B} \in \mathbb{Z}_q^{n \times m}$, we have

$$H_\infty(\mathbf{v}) \geq \log(\sigma/\lambda)$$

where $\mathbf{v} \leftarrow \mathbf{B}_\sigma^{-1}(\mathbf{u})$ for any vector $\mathbf{u} \in \mathbb{Z}_q^n$.

Proof. By Lemma 2.10 of [PR06], for any $\mathbf{c} \in \mathbb{R}^m$ and m -dimensional lattice Λ , we have $\rho_\sigma(\Lambda + \mathbf{c}) \in [\sigma^m \det(\Lambda^*)(1 - \epsilon), \sigma^m \det(\Lambda^*)(1 + \epsilon)]$ if $\sigma \geq \eta_\epsilon(\Lambda)$, where $\eta_\epsilon(\Lambda)$ is the smoothing parameter of Λ w.r.t the parameter ϵ and $\det(\Lambda^*)$ is the determinant of the dual lattice Λ^* of Λ .⁵ This also implies $1 \leq \rho_\tau(\Lambda + \mathbf{c}) \leq \tau^m \det(\Lambda^*)(1 + \epsilon)$ for $\tau > \eta_\epsilon(\Lambda)$. These together imply that

$$\mathcal{D}_{\Lambda + \mathbf{c}, \sigma}(\mathbf{v}) = \frac{\rho_\sigma(\mathbf{v})}{\rho_\sigma(\Lambda + \mathbf{c})} \leq \frac{1}{\sigma^m \det(\Lambda^*)(1 - \epsilon)} \leq (\tau/\sigma)^m \cdot \frac{1 + \epsilon}{1 - \epsilon}.$$

We then consider the lattice

$$\Lambda^\perp(\mathbf{B}) = \{\mathbf{v} \in \mathbb{Z}^m : \mathbf{B}\mathbf{v} = \mathbf{u} \bmod q\}$$

and $\mathbf{c} \in \mathbb{Z}^m$ such that $\mathbf{B}\mathbf{c} = \mathbf{u} \bmod q$. We then observe that $\mathbf{v}$ follows the distribution $\mathcal{D}_{\Lambda + \mathbf{c}, \sigma}$. Then, the lemma follows since as shown in [GPV08], for $\tau = \omega(\log m)$, we have $\eta_{2^{-n}}(\Lambda^\perp(\mathbf{B})) < \tau$ for all but 2^{-n} fraction of $\mathbf{B} \in \mathbb{Z}_q^{n \times m}$. Taking $\tau = \lambda/2$, the lemma follows. $\square$

Lemma 2.7 ([ABB10a, Lemma 2.14 (Generalized Leftover Hash Lemma)]). Let $\mathcal{H} = \{h : \mathcal{X} \rightarrow \mathcal{Y}\}$ be a 2-universal hash function family and $f : \mathcal{X} \rightarrow \mathcal{Z}$ be a function. Then for any random variable X taking value in $\mathcal{X}$, the distributions

$$(h, h(X), f(X)) \text{ and } (h, \mathcal{U}(\mathcal{Y}), f(X))$$

are within statistical distance $\sqrt{2^{-H_\infty(X)} \cdot |\mathcal{Y}| |\mathcal{Z}|}$, where $\mathcal{U}(\mathcal{Y})$ represents the uniform distribution over $\mathcal{Y}$. More generally, let $X_1, \dots, X_Q$ be independent random variables taking values in $\mathcal{X}$ and let $\gamma = \min_{i \in [Q]} H_\infty(X_i)$. Then, the distributions

$$(h, h(X_1), \dots, h(X_Q), f(X_1), \dots, f(X_Q)) \text{ and } (h, \mathcal{U}_1(\mathcal{Y}), \dots, \mathcal{U}_Q(\mathcal{Y}), f(X_1), \dots, f(X_Q))$$

are within statistical distance $Q\sqrt{2^{-\gamma} \cdot |\mathcal{Y}| |\mathcal{Z}|}$.

We note that even in the special case where there is no side information of the form $f(X)$, the above lemma can be applied by setting $f(X) = \perp$ for all X and $|\mathcal{Z}| = 1$. In this case, we call the above lemma “leftover hash lemma”. We also note that the family $\{\mathbf{A} \in \mathbb{Z}_q^{n \times m} : \mathbf{r} \mapsto \mathbf{A}\mathbf{r}\}_{\mathbf{A}}$ is 2-universal for any prime q . Furthermore, if we restrict the input space to be $\{0, 1\}^m$, the same family is 2-universal for any integer q .

Lemma 2.8 (Smudging Lemma [WW22b]). Let λ be a security parameter. Take any $a \in \mathbb{Z}$ where $|a| \leq B$. Suppose $\gamma \geq B\lambda^{\omega(1)}$. Then the statistical distance between the distributions $\{z : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ and $\{z + a : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ is $\text{negl}(\lambda)$.

Lemma 2.9. For $\gamma \geq c\lambda^{\omega(1)}$, $\{z \bmod 2 : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ is statistically close to the uniform distribution over $\{0, 1\}$.

Proof. By Lemma 2.8, $\{z : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ and $\{z + 1 : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ are statistically close. Therefore, $\{z \bmod 2 : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ and $\{z + 1 \bmod 2 : z \leftarrow \mathcal{D}_{\mathbb{Z}, \gamma}\}$ are statistically close. This implies what we want to show. $\square$

⁵We refer to the definitions of the smoothing parameter, dual lattice, and its determinant to [PR06], since this is not necessary for our purpose.

2.3.1 Hardness Assumptions

Assumption 2.10 (The LWE Assumption). Let $n = n(\lambda)$, $m = m(\lambda)$, and $q = q(\lambda) > 2$ be integers and $\chi = \chi(\lambda)$ be a distribution over $\mathbb{Z}_q$. We say that the $\text{LWE}(n, m, q, \chi)$ hardness assumption holds if for any PPT adversary $\mathcal{A}$ we have

$$|\Pr[\mathcal{A}(\mathbf{A}, \mathbf{s}^\top \mathbf{A} + \mathbf{e}^\top) \rightarrow 1] - \Pr[\mathcal{A}(\mathbf{A}, \mathbf{v}^\top) \rightarrow 1]| \leq \text{negl}(\lambda)$$

where the probability is taken over the choice of the random coins by the adversary $\mathcal{A}$ and $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{s} \leftarrow \mathbb{Z}_q^n$, $\mathbf{e} \leftarrow \chi^m$, and $\mathbf{v} \leftarrow \mathbb{Z}_q^m$. We also say that $\text{LWE}(n, m, q, \chi)$ problem is subexponentially hard if the above probability is bounded by $2^{-n^\epsilon} \cdot \text{negl}(\lambda)$ for some constant $0 < \epsilon < 1$ for all PPT $\mathcal{A}$.

As shown by previous works [Reg09, BLP⁺13], if we set $\chi = \text{SampZ}(\gamma)$, the $\text{LWE}(n, m, q, \chi)$ problem is as hard as solving worst case lattice problems such as gapSVP and SIVP with approximation factor $\text{poly}(n) \cdot (q/\gamma)$ for some $\text{poly}(n)$. Since the best known algorithms for 2^k -approximation of gapSVP and SIVP run in time $2^{\tilde{O}(n/k)}$, it follows that the above $\text{LWE}(n, m, q, \chi)$ with noise-to-modulus ratio 2^{-n^ϵ} is likely to be (subexponentially) hard for some constant ϵ .

Assumption 2.11 ((Circular) Small Secret LWE Assumption). [HLL23] Let $n, m, m', q, \sigma, \sigma'$ be functions of λ and

$$\begin{aligned} \bar{\mathbf{A}}_{\text{fhe}} &\leftarrow \mathbb{Z}_q^{n \times m}, \quad \bar{\mathbf{A}}_0 \leftarrow \mathbb{Z}_q^{n \times m'}, \quad \mathbf{r} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma \lambda}^n, \quad \mathbf{s} \leftarrow (\mathbf{r}^\top, -1)^\top, \\ \mathbf{e}_{\text{fhe}} &\leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma \lambda}^m, \quad \mathbf{e}_0 \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma', \leq \sigma' \lambda}^{m'}, \quad \mathbf{R} \leftarrow \{0, 1\}^{m \times (n+1) \lceil \log_2 q \rceil m} \end{aligned}$$

The circular small-secret LWE assumption $\text{csLWE}_{n, m, m', q, \sigma, \sigma'}$ states that

$$\begin{aligned} &\left\{ \left(1^\lambda, \quad \mathbf{A}_{\text{fhe}} := \left(\mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \right), \quad \mathbf{C} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \quad \left(\mathbf{c}_0^\top = \mathbf{r}^\top \bar{\mathbf{A}}_0 + \mathbf{e}_0^\top \right) \right) \right\}_{\lambda \in \mathbb{N}} \\ &\approx_c \left\{ \left(1^\lambda, \quad \underbrace{\left(\mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \quad \mathbf{C} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log_2 q \rceil m} \right)}_{\text{circular terms}}, \quad \left(\mathbf{c}_0^\top \leftarrow \mathbb{Z}_q^{1 \times m'} \right) \right) \right\}_{\lambda \in \mathbb{N}} \end{aligned}$$

For the small-secret LWE assumption $\text{sLWE}_{n, m, m', q, \sigma, \sigma'}$, the circular terms are removed from both distributions.

Assumption 2.12 (Evasive LWE). [Wee22, ARYY23] Let $n, m, t, m', q \in \mathbb{N}$ be parameters and λ be a security parameter. Let $\sigma_{\text{pre}}, \sigma'_{\text{pre}}, \sigma_{\text{post}}$ and τ be parameters for Gaussian distributions. Let Samp be a PPT algorithm that outputs

$$\mathbf{S} \in \mathbb{Z}_q^{m' \times n}, \mathbf{P} \in \mathbb{Z}_q^{n \times t}, \text{aux} \in \{0, 1\}^*$$

on input 1^λ . For a PPT adversary $\mathcal{A}_0, \mathcal{A}_1$, we define the following advantage functions:

$$\text{Adv}_{\mathcal{A}_0}^{\text{pre}}(\lambda) \stackrel{\text{def}}{=} \Pr[\mathcal{A}_0(\mathbf{B}, \mathbf{P}, \mathbf{S}\mathbf{B} + \mathbf{E}, \mathbf{S}\mathbf{P} + \mathbf{E}', \text{aux}) = 1] - \Pr[\mathcal{A}_0(\mathbf{B}, \mathbf{P}, \mathbf{C}_0, \mathbf{C}', \text{aux}) = 1]$$

$$\text{Adv}_{\mathcal{A}_1}^{\text{post}}(\lambda) \stackrel{\text{def}}{=} \Pr[\mathcal{A}_1(\mathbf{B}, \mathbf{P}, \mathbf{S}\mathbf{B} + \mathbf{E}'', \mathbf{K}, \text{aux}) = 1] - \Pr[\mathcal{A}_1(\mathbf{B}, \mathbf{P}, \mathbf{C}_0, \mathbf{K}, \text{aux}) = 1]$$

$$\begin{aligned} \text{where} \quad &(\mathbf{S}, \mathbf{P}, \text{aux}) \leftarrow \text{Samp}(1^\lambda), \quad \mathbf{B} \leftarrow \mathbb{Z}_q^{n \times m}, \quad \mathbf{C}_0 \leftarrow \mathbb{Z}_q^{m' \times m}, \quad \mathbf{C}' \leftarrow \mathbb{Z}_q^{m' \times t}, \\ &\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}}^{m' \times m}, \quad \mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'_{\text{pre}}}^{m' \times t}, \quad \mathbf{E}'' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{post}}}^{m' \times t}, \\ &\mathbf{K} \leftarrow \mathbf{B}^{-1}(\mathbf{P}) \text{ with standard deviation } \tau \end{aligned}$$

We say that the evasive LWE (EvLWE) assumption holds if for every PPT Samp and $\mathcal{A}_1$, there exists another PPT $\mathcal{A}_0$ and a polynomial $Q(\cdot)$ such that

$$\text{Adv}_{\mathcal{A}_0}^{\text{pre}}(\lambda) \geq \text{Adv}_{\mathcal{A}_1}^{\text{post}}(\lambda)/Q(\lambda) - \text{negl}(\lambda).$$

We also consider a slight variant of the above, as is introduced by Wee [Wee22]. In this variant, $\text{Samp}(1^\lambda)$ outputs $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ along with $(\mathbf{S}, \mathbf{P}, \text{aux})$. Furthermore, the adversary is given $\mathbf{A}$ and corresponding LWE challenge $\mathbf{c}_\mathbf{A}$ in the pre- and post-condition distributions. We have $\mathbf{c}_\mathbf{A}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_\mathbf{A}^\top$ in the LHS distribution and $\mathbf{c}_\mathbf{A} \leftarrow \mathbb{Z}_q^m$ in the RHS distribution, where $\mathbf{e}_\mathbf{A} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{post}}}^m$ in post-condition and $\mathbf{e}_\mathbf{A} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}}^m$ in pre-condition.

Public versus Private Coin. Evasive LWE has been studied both in the “public-coin” and “private-coin” setting, where the former means that the randomness used by the sampler is made available to the adversary, and the latter means that the sampler’s random coins need to be hidden from the adversary. Here, note that it is necessary that the LWE secret $\mathbf{S}$ and error remain hidden from the adversary, whether in the public or private coin setting. In the public-coin setting, these are generated randomly from the appropriate distribution and are the *only* secrets not known to the adversary, whereas in the private coin setting, there may be additional randomness that is not known to the adversary.

2.4 GSW Homomorphic Encryption and Evaluation

We recall the format of the (leveled fully) homomorphic encryption due to [GSW13] and the correctness property. We adapt the syntax from [HLL23].

Lemma 2.13. The leveled FHE scheme works as follows:

- The keys are

$$(\text{public}) \quad \mathbf{A}_{\text{fhe}} = \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe}} \\ \bar{\mathbf{s}}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix} \in \mathbb{Z}_q^{(n+1) \times m}, \quad (\text{secret}) \quad \mathbf{s}^\top = (\bar{\mathbf{s}}^\top, -1),$$

where $\bar{\mathbf{s}} \in \mathbb{Z}^n$, $\bar{\mathbf{A}}_{\text{fhe}} \in \mathbb{Z}_q^{n \times m}$, and $\mathbf{e}_{\text{fhe}}^\top \in \mathbb{Z}^m$.

- A ciphertext of $x \in \{0, 1\}$ is $\mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - x \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m}$, where $\mathbf{R} \in \mathbb{Z}_q^{m \times m}$ is the encryption randomness. The decryption equation is

$$\mathbf{s}^\top \mathbf{X} = -\mathbf{e}_{\text{fhe}}^\top \mathbf{R} - x \mathbf{s}^\top \mathbf{G} \in \mathbb{Z}_q^m,$$

which can be used to extract x via multiplication by $\mathbf{G}^{-1}(\lfloor q/2 \rfloor \iota_{n+1})$.

Lemma 2.14. (homomorphic evaluation for vector-valued functions [HLL23]) For the scheme in Lemma 4, there is an efficient algorithm

$$\text{MakeVEvalCkt}(n, m, q, C) = \text{VEval}_C$$

that takes as input n, m, q and a vector-valued circuit $C : \{0, 1\}^L \rightarrow \mathbb{Z}_q^{1 \times m'}$ and outputs a circuit

$$\text{VEval}_C(\mathbf{X}_1, \dots, \mathbf{X}_L) = \mathbf{C},$$

taking L ciphertexts as input and outputting a new ciphertext $\mathbf{C}$ of different format.

- The depth of VEval_C is $dO(\log m \log \log q) + O(\log^2 \log q)$ for C of depth d .

- Suppose $\mathbf{X}_\ell = \mathbf{A}_{\text{fhe}} \mathbf{R}_\ell - \mathbf{x}[\ell] \mathbf{G}$ for $\ell \in [L]$ with $\mathbf{x} \in \{0, 1\}^L$, then

$$\mathbf{C} = \mathbf{A}_{\text{fhe}} \mathbf{R}_C - \begin{pmatrix} 0^{n \times m'} \\ C(\mathbf{x}) \end{pmatrix} \in \mathbb{Z}_q^{(n+1) \times m'},$$

where $\|\mathbf{R}_C^\top\| \leq (m+2)^d \lceil \log q \rceil \max_{\ell \in [L]} \|\mathbf{R}_\ell^\top\|$. The new decryption equation is

$$\mathbf{s}^\top \mathbf{C} = -\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_C + C(\mathbf{x}) \in \mathbb{Z}_q^{1 \times m'}.$$

We implement the above homomorphic computation in the following specific manner. Let us represent each entry of $C(\mathbf{x})$ by an element in $[-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor]$. We then let $C(\mathbf{x})[u, v] \in \{-1, 0, 1\}$ be the signed version of the v -th bit of $C(\mathbf{x})[u]$, for $u \in [m']$ and $v \in [0, w)$, $w = \lceil \log_2 q \rceil$. Namely, we have $\sum_{v \in [0, w)} C(\mathbf{x})[u, v] 2^v = C(\mathbf{x})[u]$, where if $C(\mathbf{x})[u] \in [-\lfloor q/2 \rfloor, -1]$, then $C(\mathbf{x})[u, v] \in \{-1, 0\}$ and $C(\mathbf{x})[u, v] \in \{0, 1\}$ otherwise, for all u and v . Note that the parity of $C(\mathbf{x})[u]$ represented as an element in $[-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor]$ equals to that of $C(\mathbf{x})[u, 0]$.⁶

Then VEval_C computation firstly homomorphically computes GSW encryption of $C(\mathbf{x})[u, v]$, which is of the form $\mathbf{C}_{u,v} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C,u,v} - C(\mathbf{x})[u, v] \mathbf{G}$. This is done by first computing the GSW encryption of $C[\mathbf{x}]$ in non-signed binary form (i.e., $C[\mathbf{x}]$ is treated as an element in $[0, q-1]$ and is represented in a binary form) using the GSW homomorphic evaluation and adding the sign by the homomorphic operation. Then for each $u \in [m']$, it linearly aggregates $\{\mathbf{C}_{u,v}\}_{v \in [0, w-1]}$ to get $\mathbf{C}_u = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C,u} - \begin{pmatrix} 0^{n \times 1} \\ C(\mathbf{x})[u] \end{pmatrix}$ as $\mathbf{C}_u = \sum_v \mathbf{C}_{u,v} \mathbf{G}^{-1}(2^v \iota_{n+1})$. In particular,

$$\mathbf{R}_{C,u} = \sum_v \mathbf{R}_{C,u,v} \mathbf{G}^{-1}(2^v \iota_{n+1}), \text{ and } \mathbf{R}_C = (\mathbf{R}_{C,1}, \dots, \mathbf{R}_{C,m'}).$$

2.5 Homomorphic Evaluation Procedures

In this section we describe the properties of the attribute encoding and its homomorphic evaluation. We adapt the syntax from [HLL23].

- For L -bit input, the public parameter is $\mathbf{A}_{\text{att}} \in \mathbb{Z}_q^{(n+1) \times (L+1)m}$.
- The encoding of $\mathbf{x} \in \{0, 1\}^L$ is

$$\mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \mathbf{x}^\top) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top,$$

where $\mathbf{s}^\top = (\bar{\mathbf{s}}^\top, -1)$ with $\bar{\mathbf{s}} \in \mathbb{Z}^n$ and $\mathbf{e}_{\text{att}}^\top \in \mathbb{Z}^{(L+1)m}$.

- There are efficient deterministic algorithms [BTW17]

$$\text{MEvalC}(\mathbf{A}_{\text{att}}, C) = \mathbf{H}_C \quad \text{and} \quad \text{MEvalCX}(\mathbf{A}_{\text{att}}, C, \mathbf{x}) = \mathbf{H}_{C,\mathbf{x}}$$

that take as input $\mathbf{A}_{\text{att}}$, a matrix-valued circuit $C : \{0, 1\}^L \rightarrow \mathbb{Z}_q^{n+1 \times m'}$, and (for MEvalCX) some $\mathbf{x} \in \{0, 1\}^L$, and output some matrix in $\mathbb{Z}^{(L+1)m \times m'}$.

- Suppose C is of depth d , then $\|\mathbf{H}_C^\top\|, \|\mathbf{H}_{C,\mathbf{x}}^\top\| \leq (m+2)^d \lceil \log q \rceil$.
- The matrix encoding homomorphism is $(\mathbf{A}_{\text{att}} - (1, \mathbf{x}^\top) \otimes \mathbf{G}) \mathbf{H}_{C,\mathbf{x}} = \mathbf{A}_{\text{att}} \mathbf{H}_C - C(\mathbf{x})$.

⁶We do not have this equality when we do not represent $C(\mathbf{x})[u]$ in a signed form. This is because for an integer $x \in [-\lfloor q/2 \rfloor, -1]$, $x \bmod 2$ (i.e., the LSB of the signed form) does not equal to $x + q \bmod 2$ (i.e., the LSB of the non-signed form) when q is an odd number.

Dual-Use Technique and Extension. In [BTWV17], the attribute encoded with secret $\mathbf{s}^\top$ is FHE ciphertexts under key $\mathbf{s}^\top$ (the same, "dual-use") and the circuit being MEvalCX'ed is some HEval_C. This leads to automatic decryption. Let C be a vector-valued circuit, with co-domain $\mathbb{Z}_q^{1 \times m'}$, then VEval_C is $\mathbb{Z}_q^{(n+1) \times m'}$ -valued and

$$\begin{aligned} (\mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top) \cdot \mathbf{H}_{\text{VEval}_C, \mathbf{X}} \\ (\text{MEvalCX}) &= \mathbf{s}^\top \mathbf{A}_{\text{att}} \mathbf{H}_{\text{VEval}_C} - \mathbf{s}^\top \text{VEval}_C(\mathbf{X}) + (\mathbf{e}')^\top \\ (\text{VEval decryption}) &= \mathbf{s}^\top \mathbf{A}_{\text{att}} \mathbf{H}_{\text{VEval}_C} - C(\mathbf{x}) + (\mathbf{e}'')^\top. \end{aligned}$$

3 Counter-Examples for Public-Coin Evasive LWE

In this section we show counter-examples against "public-coin" evasive LWE.

3.1 Counter-Example for HLL's Circular Evasive LWE

We first recall the assumption from [HLL23]. We refer to Section 2.3 for basic notations related to lattices.

Assumption 3.1 (Evasive Circular Small-Secret LWE). Let n, m, m', J, q be parameters. Let $\text{Samp}(1^\lambda; \text{coins}_{\text{Samp}}^{\text{pub}})$ be a public coin sampler that, given randomness $\text{coins}_{\text{Samp}}^{\text{pub}}$, outputs

$$\overline{\mathbf{A}}_{\text{circ}} \in \mathbb{Z}_q^{n \times (L_S + 1)m}, \overline{\mathbf{A}}' \in \mathbb{Z}_q^{n \times m'}, \mathbf{P} \in \mathbb{Z}_q^{n \times J}, \sigma, \sigma', \sigma_{-1}, \sigma_{\text{post}}, \sigma_{\text{pre}}, \text{aux}$$

where $m \geq O(n \log q)$, $\sigma_{-1} = \omega(\sqrt{n \log q \log m})$ and $\sigma_{\text{post}} \geq \sigma_{\text{pre}}$. Suppose

$$\begin{aligned} \overline{\mathbf{A}}_{\text{fhe}} &\leftarrow \mathbb{Z}_q^{n \times m}, (\mathbf{B}, \mathbf{B}_{\sigma_{-1}}^{-1}) \leftarrow \text{TrapGen}(1^n, 1^m, q), \mathbf{K} \leftarrow \mathbf{B}_{\sigma_{-1}}^{-1}(\mathbf{P}), \\ \mathbf{e}_{\text{fhe}} &\leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma \sqrt{\lambda}}^m, \mathbf{e}_{\text{circ}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma', \leq \sigma' \sqrt{\lambda}}^{(L_S + 1)m}, \mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma', \leq \sigma' \sqrt{\lambda}}^{m'}, \mathbf{e}_{\mathbf{B}} \in \mathbb{Z}^m, \mathbf{e}_{\mathbf{P}} \in \mathbb{Z}^J, \\ \mathbf{r} &\leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma \sqrt{\lambda}}^n, \mathbf{R} \leftarrow \{0, 1\}^{m \times (n+1) \lceil \log q \rceil m}, \mathbf{A}_{\text{fhe}} = \left(\left(\mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \right) \right) \\ \mathbf{s} &= (\mathbf{r}^\top, -1)^\top, L_S = (n+1)^2 \lceil \log q \rceil^2 m, \end{aligned}$$

where L_S is the length of the binary representation of $\mathbf{S}$, which is defined below. In the precondition, the entries of $\mathbf{e}_{\mathbf{B}}, \mathbf{e}_{\mathbf{P}}$ are independent and follow $\mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}, \leq \sigma_{\text{pre}} \sqrt{\lambda}}$ and $\text{evcsLWE}_{\text{pre}}^{\text{Samp}}$ states that

$$\begin{aligned} D_0^{\text{pre}} &:= \left\{ \left(\begin{array}{l} 1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}}, \text{aux}, \overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top, \\ \mathbf{S} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \overline{\mathbf{c}}_{\text{circ}}^\top = \mathbf{r}^\top (\overline{\mathbf{A}}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \overline{\mathbf{G}}) + \mathbf{e}_{\text{circ}}^\top, \\ (\mathbf{c}')^\top = \mathbf{r}^\top \overline{\mathbf{A}}' + (\mathbf{e}')^\top, \mathbf{c}_{\mathbf{B}}^\top = \mathbf{r}^\top \mathbf{B} + \mathbf{e}_{\mathbf{B}}^\top, \\ \mathbf{c}_{\mathbf{P}}^\top = \mathbf{r}^\top \mathbf{P} + \mathbf{e}_{\mathbf{P}}^\top \end{array} \right) \right\}_{\lambda \in \mathbb{N}} \\ \approx_c D_1^{\text{pre}} &:= \left\{ \left(\begin{array}{l} 1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}}, \text{aux}, \overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \\ \mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1)m \lceil \log q \rceil}, \\ \overline{\mathbf{c}}_{\text{circ}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_S + 1)m}, \\ (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times m'}, \mathbf{c}_{\mathbf{B}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \\ \mathbf{c}_{\mathbf{P}}^\top \leftarrow \mathbb{Z}_q^{1 \times J} \end{array} \right) \right\}_{\lambda \in \mathbb{N}}, \end{aligned} \quad (10)$$

where $\overline{\mathbf{G}}$ represents the top n rows of $\mathbf{G}$. In the postcondition, the entries of $\mathbf{e}_B$ are independent and follow $\mathcal{D}_{\mathbb{Z}, \sigma_{\text{post}}, \leq \sigma_{\text{post}} \sqrt{\lambda}}$ and $\text{evcsLWE}_{\text{post}}^{\text{Samp}}$ states that

$$D_0^{\text{post}} := \left\{ \left(\begin{array}{l} 1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}}, \text{aux}, \overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top, \\ \mathbf{S} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \overline{\mathbf{c}}_{\text{circ}}^\top = \mathbf{r}^\top (\overline{\mathbf{A}}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \overline{\mathbf{G}}) + \mathbf{e}_{\text{circ}}^\top, \\ (\mathbf{c}')^\top = \mathbf{r}^\top \overline{\mathbf{A}}' + (\mathbf{e}')^\top, \mathbf{c}_B^\top = \mathbf{r}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{K} \end{array} \right) \right\}_{\lambda \in \mathbb{N}}$$

$$\approx_c D_1^{\text{post}} := \left\{ \left(\begin{array}{l} 1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}}, \text{aux}, \overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \\ \mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1)m \lceil \log_2 q \rceil}, \\ \overline{\mathbf{c}}_{\text{circ}}^\top \leftarrow \mathbb{Z}_q^{1 \times (Ls+1)m}, \\ (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times m'}, \mathbf{c}_B^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \mathbf{K} \end{array} \right) \right\}_{\lambda \in \mathbb{N}} \quad (11)$$

The *evasive circular small-secret LWE assumption* states that $\text{evcsLWE}_{\text{pre}}^{\text{Samp}}$ implies $\text{evcsLWE}_{\text{post}}^{\text{Samp}}$ for all efficient sampler Samp .

3.1.1 Preliminaries for the attack

We require some technical preparations before showing the counter-example.

Correlation Inducing Gates for GSW. We will use the following lemmas adapted from [HJL21] to GSW encryption.

Lemma 3.2 (Even-randomness gate for GSW). Let q be an odd number and $\text{ct}_x = \mathbf{A}_{\text{fhe}} \mathbf{R} - x \mathbf{G}$ be a GSW encryption of some $x \in \{0, 1\}$, where $\mathbf{A}_{\text{fhe}}$ is the public key, $|\mathbf{R}[i, j]| \leq B$ for some $B \leq q/(100m)$ and let ct'_x be the result of homomorphically evaluating the following two (arithmetic) gates:

1. $g_1(x) = \frac{1}{2}x$ and $\text{Eval}(g_1, \text{ct}_x) = \text{ct}_x \cdot \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right)$.
2. $g_2(x) = x + x$ and $\text{Eval}(g_2, \text{ct}_x) = \text{ct}_x + \text{ct}_x$,

where ct_x is treated as a matrix in $\mathbb{Z}_q^{(n+1) \times m}$.⁷ That is, $\text{ct}'_x = \text{Eval}(g_2, \text{Eval}(g_1, \text{ct}_x))$. Then $\text{ct}'_x = \mathbf{A}_{\text{fhe}} \mathbf{R}' - x \mathbf{G}$ for some $\mathbf{R}' \in \mathbb{Z}_q^{m \times m}$ (here $m = O(n \log q)$) for which every entry satisfies $\mathbf{R}'[i, j] = 2 \cdot r_{i,j}$ for some $r_{i,j} \in \mathbb{Z}_q$ and $i, j \in [m]$ with $|r_{i,j}| \leq O(mB)$.

Proof. We first expand $\text{Eval}(g_1, \text{ct}_x)$.

$$\begin{aligned} \text{Eval}(g_1, \text{ct}_x) &= (\mathbf{A}_{\text{fhe}} \mathbf{R} - x \mathbf{G}) \cdot \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) = \mathbf{A}_{\text{fhe}} \mathbf{R} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) - x \mathbf{G} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) \\ &= \mathbf{A}_{\text{fhe}} \left(\mathbf{R} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) \right) - \frac{x}{2} \mathbf{G}. \end{aligned}$$

⁷Here, we specify the homomorphic evaluation circuit concretely, since there are multiple different homomorphic evaluation circuits implementing the same operation on the encoded message. For example, we can homomorphically evaluate $x \mapsto x + x$ by $\text{ct} \cdot \mathbf{G}^{-1}(2\mathbf{G})$ alternatively.

$$\begin{aligned}
\text{Then, } \text{Eval}(g_2, \text{Eval}(g_1, \text{ct}_x)) &= \mathbf{A}_{\text{fhe}} \left(\mathbf{R} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) \right) - \frac{x}{2} \mathbf{G} + \mathbf{A}_{\text{fhe}} \left(\mathbf{R} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) \right) - \frac{x}{2} \mathbf{G} \\
&= \mathbf{A}_{\text{fhe}} \left(2 \cdot \mathbf{R} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) \right) - x \mathbf{G}
\end{aligned}$$

Note that $\mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right) \in \{0, 1\}^{m \times m}$, hence $|(\mathbf{R} \mathbf{G}^{-1} \left(\frac{1}{2} \mathbf{G} \right)) [i, j]| \leq O(Bm)$. $\square$

Lemma 3.3 (Multiplication-by-one gate for GSW). Let $\text{ct}^* = \mathbf{A}_{\text{fhe}} \mathbf{R}^* - \mathbf{G}$ be a GSW encryption for constant 1, where $|\mathbf{R}^*[i, j]| \leq q/10$ for all $i, j \in [m]$. Let $x \in \{-1, 0, 1\}$, and let $\text{ct}_x = \mathbf{A}_{\text{fhe}} \mathbf{R} - x \mathbf{G}$ be the GSW encryption of x such that $\mathbf{R}[i, j] = 2 \cdot r_{i,j}$ where $r_{i,j} \in \mathbb{Z}$ and $|r_{i,j}| \leq q/(100m)$, for all $i, j \in [m]$. Let $g(x, y) = x \cdot y$ and the corresponding homomorphic evaluation be defined as $\text{Eval}(g, \text{ct}_1, \text{ct}_2) = -\text{ct}_1 \cdot \mathbf{G}^{-1}(\text{ct}_2)$. Then $\text{Eval}(g, \text{ct}_x, \text{ct}^*) = \mathbf{A}_{\text{fhe}} \mathbf{R}' - x \cdot \mathbf{G}$ where $\mathbf{R}' \in \mathbb{Z}_q^{m \times m}$ and $\mathbf{R}' = x \cdot \mathbf{R}^* \bmod 2$.

Proof. On expanding,

$$\begin{aligned}
\text{Eval}(g, \text{ct}_x, \text{ct}^*) &= -(\mathbf{A}_{\text{fhe}} \mathbf{R} - x \mathbf{G}) \mathbf{G}^{-1} (\mathbf{A}_{\text{fhe}} \mathbf{R}^* - \mathbf{G}) \\
&= \mathbf{A}_{\text{fhe}} (-\mathbf{R} \mathbf{G}^{-1} (\mathbf{A}_{\text{fhe}} \mathbf{R}^* - \mathbf{G}) + x \mathbf{R}^*) - x \cdot \mathbf{G}
\end{aligned}$$

Now, firstly, we note that each entry, $\mathbf{R} \mathbf{G}^{-1} (\mathbf{A}_{\text{fhe}} \mathbf{R}^* - \mathbf{G}) [i, j]$ is small, i.e., $\leq (q/(100m)) * m = q/100$. Further we observe that since each entry $\mathbf{R}[i, j]$ is even, $\mathbf{R} \mathbf{G}^{-1} (\mathbf{A}_{\text{fhe}} \mathbf{R}^* - \mathbf{G}) [i, j] = 0 \bmod 2, \forall i, j \in [m]$. Hence, we have $\mathbf{R}' = x \cdot \mathbf{R}^* \bmod 2$. $\square$

Corollary 3.4 (Correlation inducing gate for GSW). Let $\text{ct}^* = \mathbf{A}_{\text{fhe}} \mathbf{R}^* - \mathbf{G}$ be a GSW encryption of the constant 1, where $|\mathbf{R}^*[i, j]| \leq q/10$ for all $i, j \in [m]$. Let $x \in \{-1, 0, 1\}$ and let $\text{ct}_x = \mathbf{A}_{\text{fhe}} \mathbf{R} - x \mathbf{G}$ be a GSW encryption of x such that $|\mathbf{R}[i, j]| \leq q/\text{poly}(n, \log q)$ for all i, j . Then, for g_1, g_2 as in Lemma 3.2 and g as in Lemma 3.3,

$$\text{Eval}(g, \text{Eval}(g_2, \text{Eval}(g_1, \text{ct}_x)), \text{ct}^*) = \mathbf{A}_{\text{fhe}} \mathbf{R}' - x \mathbf{G},$$

where $\mathbf{R}' = x \mathbf{R}^* \bmod 2$.

Publicly Samplable PRG against Circular-LWE Leakage. Here, we introduce the notion of publicly samplable PRG secure against circular-LWE leakage and show an instantiation.

Definition 3.5 (Publicly Samplable PRG against Circular-LWE leakage). Let $\text{prm} = (n, \ell, q, \sigma, \sigma')$ be parameters. Let $\mathcal{F}_{\text{prm}} = \{f : \mathbb{Z}_q^{n+1} \rightarrow \mathbb{Z}_q^{1 \times \ell}\}$ be a family of functions. We require that uniform sampling from the function family $\mathcal{F}_{\text{prm}}$ can be efficiently done. Furthermore, we identify random coin for sampling f from $\mathcal{F}_{\text{prm}}$ with the description of the function f itself. We say that the family $\mathcal{F}_{\text{prm}}$ is publicly samplable PRG against circular-LWE leakage if the following holds for all polynomials $m' = m'(\lambda)$ and $Q = Q(\lambda)$:

$$\left(\begin{array}{l} 1^\lambda, \mathbf{A}_{\text{fhe}} = \left(\mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \right), \\ \mathbf{C} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \left(\mathbf{c}^\top = \mathbf{r}^\top \overline{\mathbf{A}} + \mathbf{e}^\top \right), \\ \{f_i \leftarrow \mathcal{F}_{\text{prm}}, \mathbf{c}_i^\top = f_i(\mathbf{s})\}_{i \in [Q]} \end{array} \right) \approx_c \left(\begin{array}{l} 1^\lambda, \left(\mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m} \right), \\ \mathbf{C} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1)m \lceil \log q \rceil}, \\ \left(\mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m'} \right), \\ \{f_i \leftarrow \mathcal{F}_{\text{prm}}, \mathbf{c}_i^\top \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]} \end{array} \right) \quad (12)$$

where $\mathbf{r} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma\sqrt{\lambda}}^n$, $\mathbf{s}^\top = (\mathbf{r}^\top, -1)$, $\overline{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\overline{\mathbf{A}} \leftarrow \mathbb{Z}_q^{n \times m'}$, $\mathbf{R} \leftarrow \{0, 1\}^{m \times (n+1) \lceil \log q \rceil m}$, $\mathbf{e}_{\text{fhe}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma\sqrt{\lambda}}^m$, $\mathbf{e} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma', \leq \sigma'\sqrt{\lambda}}^{m'}$.

Remark 3.6. Notice that by our notational convention of identifying the description of the function with the randomness for sampling it, the fact that the distinguisher above is given $\{f_i\}_i$ means that it is given the random coins for sampling the functions $\{f_i\}_i$. We therefore call the function "publicly samplable," meaning that it is secure even if the randomness for sampling it is revealed to the distinguisher.

The following lemma provides a concrete instantiation of the above notion. Before stating the lemma, we prepare several notations. We denote $\mathbf{G}_q = \mathbf{I}_\ell \otimes (1, 2, 2^2, \dots, 2^{\lceil \log_2 q \rceil - 1})$, $\mathbf{G}_p^{-1}(x)$ for any $x \in \mathbb{Z}_p$ represents bit decomposition of x , resulting in a vector in $\{0, 1\}^{\lceil \log p \rceil}$ (similarly for a vector), and $[x]_B$ for any $x \in \mathbb{Z}_q$ represents truncating the range to $[-\lfloor q/2 \rfloor + B, \lfloor q/2 \rfloor - B]$ by mapping the out of range values to 0 (again similarly for a vector).

Lemma 3.7. Let ℓ, q, p be integers and σ and σ' be a parameter satisfying $\ell' = \lceil \ell(\lceil \log q \rceil / \lceil \log p \rceil) \rceil^8$ and $q \geq p\lambda^{\omega(1)}\sigma$. Let B be an integer that is exponentially smaller than q . Then, $\mathcal{F}_{\text{prm}} = \{f_{\mathbf{A}} \in \mathcal{F}_{\text{prm}}\}_{\mathbf{A} \in \mathbb{Z}_q^{n \times \ell'}}$ defined as

$$f_{\mathbf{A}}(\mathbf{s}) = [\mathbf{G}_q \mathbf{G}_p^{-1}(\lfloor (\mathbf{s}^\top \mathbf{A})^\top \rfloor_p)]_B$$

is publicly samplable PRG agaist circular-LWE leakage as per Definition 3.5 with respect to $\text{prm} = (n, \ell, q, \sigma, \sigma')$, assuming circular small secret LWE assumption (Assumption 2.11). The function maps an element in $\mathbb{Z}_q^n$ to that in $[-\lfloor q/2 \rfloor + B, \lfloor q/2 \rfloor - B]^{1 \times \ell}$ and can be implemented by a circuit with depth $d(\lambda) = \text{poly}(\log n, \log \log q)$.

Proof. The last statement about the depth of the circuit follows from the fact that the arithmetic operation can be performed in NC_1 , including the computation of inner-product. We then move to prove the security. We rewrite equation (12) here, with the above definition.

$$D_0 := \left(\begin{array}{l} 1^\lambda, \mathbf{A}_{\text{fhe}} = \left(\mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \right), \\ \mathbf{C} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \left(\mathbf{c}_0^\top = \mathbf{r}^\top \overline{\mathbf{A}}_0 + (\mathbf{e}')^\top \right), \\ \left\{ \left(\overline{\mathbf{A}}_i \right), \mathbf{c}_i^\top = [\mathbf{G}_q \mathbf{G}_p^{-1}(\lfloor (\mathbf{r}^\top \overline{\mathbf{A}}_i - \mathbf{a}_i^\top)^\top \rfloor_p)]_B \right\}_{i \in [Q]} \end{array} \right) \approx_c D_1 := \left(\begin{array}{l} 1^\lambda, \left(\mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m} \right), \\ \mathbf{C} \leftarrow \mathbb{Z}_q^{(n+1) \times ((n+1) \lceil \log_2 q \rceil m)}, \\ \left(\mathbf{c}_0^\top \leftarrow \mathbb{Z}_q^{1 \times m'} \right), \\ \left\{ \left(\overline{\mathbf{A}}_i \right), \mathbf{c}_i^\top \leftarrow \mathbb{Z}_q^{1 \times \ell'} \right\}_{i \in [Q]} \end{array} \right),$$

where $\overline{\mathbf{A}}_i \leftarrow \mathbb{Z}_q^{n \times \ell'}$ and $\mathbf{a}_i \leftarrow \mathbb{Z}_q^{\ell'}$.

Let us define an intermediate distribution D' between D_0 and D_1 , where

$$\mathbf{c}_i^\top = [\mathbf{G}_q \mathbf{G}_p^{-1}(\lfloor \mathbf{r}^\top \overline{\mathbf{A}}_i - \mathbf{a}_i^\top + \mathbf{e}_i^\top \rfloor_p)]_B,$$

for $\mathbf{e}_i \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma\sqrt{\lambda}}^{\ell'}$. Then D_0 and D' are statistically indistinguishable because of the following arguments. First we observe that since $\mathbf{a}_i \leftarrow \mathbb{Z}_q^{\ell'}$, $\mathbf{r}^\top \overline{\mathbf{A}}_i - \mathbf{a}_i^\top$ is also uniformly distributed over $\mathbb{Z}_q^{1 \times \ell'}$. Then we observe that since $\mathbf{e}_i \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma\sqrt{\lambda}}^{\ell'}$, $\|\mathbf{e}_i\| \leq \sigma\sqrt{\lambda}$, and hence,

$$\Pr(\lfloor \mathbf{r}^\top \overline{\mathbf{A}}_i - \mathbf{a}_i^\top \rfloor_p \neq \lfloor \mathbf{r}^\top \overline{\mathbf{A}}_i - \mathbf{a}_i^\top + \mathbf{e}_i^\top \rfloor_p) \leq \sigma\sqrt{\lambda}p/q = \text{negl}(\lambda) \text{ due to the choice of } p \text{ and } q.$$

⁸To keep the notations simple, we set q and p in such a way that $\ell(\lceil \log q \rceil / \lceil \log p \rceil)$ is an integer. This restriction is easy to remove.

Thus, probability that $(\lfloor \mathbf{r}^\top \bar{\mathbf{A}}_i - \mathbf{a}_i^\top \rfloor_p \neq \lfloor \mathbf{r}^\top \bar{\mathbf{A}}_i - \mathbf{a}_i^\top + \mathbf{e}_i^\top \rfloor_p)$, for any $i \in [Q]$ is at most $Q\sigma\sqrt{\lambda}p/q$, which is again negligible.

Next, we show that if there exists an adversary $\mathcal{A}$ who can distinguish between D' and D_1 with non-negligible advantage, then there is a reduction $\mathcal{B}$ that breaks csLWE security with non-negligible advantage. The reduction is as follows.

1. On receiving 1^λ from $\mathcal{A}$, forward it to csLWE challenger.
2. The csLWE challenger samples $\beta \leftarrow \{0, 1\}$, $\bar{\mathbf{A}}_{\text{fhe,csLWE}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\bar{\mathbf{A}}_{0,\text{csLWE}} \leftarrow \mathbb{Z}_q^{n \times (m' + \ell'Q)}$, $\mathbf{r} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma, \leq \sigma\sqrt{\lambda}}^n$, and returns

$$\left(1^\lambda, \mathbf{A}_{\text{fhe,csLWE}} = \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe,csLWE}} \\ \mathbf{c}_{\text{fhe,csLWE}}^\top \end{pmatrix}, \mathbf{C}_{\text{csLWE}}, \begin{pmatrix} \bar{\mathbf{A}}_{0,\text{csLWE}} \\ \mathbf{c}_{0,\text{csLWE}}^\top \end{pmatrix} \right),$$

where $\mathbf{c}_{\text{fhe,csLWE}}^\top = \mathbf{r}^\top \bar{\mathbf{A}}_{\text{fhe,csLWE}} + \mathbf{e}_{\text{fhe,csLWE}}^\top$, $\mathbf{C}_{\text{csLWE}} = \mathbf{A}_{\text{fhe,csLWE}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}$, and $\mathbf{c}_{0,\text{csLWE}}^\top = \mathbf{r}^\top \bar{\mathbf{A}}_{0,\text{csLWE}} + \mathbf{e}_{0,\text{csLWE}}^\top$, if $\beta = 0$, and $\mathbf{c}_{\text{fhe,csLWE}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}$, $\mathbf{c}_{0,\text{csLWE}}^\top \leftarrow \mathbb{Z}_q^{1 \times m' + \ell'Q}$ and $\mathbf{C}_{\text{csLWE}} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log_2 q \rceil^m}$, if $\beta = 1$. Then $\mathcal{B}$ does the following:

- (a) Parse $\bar{\mathbf{A}}_{0,\text{csLWE}} = (\bar{\mathbf{A}}_0, \bar{\mathbf{A}}_1, \dots, \bar{\mathbf{A}}_Q)$ where $\bar{\mathbf{A}}_0 \in \mathbb{Z}_q^{n \times m'}$ and $\bar{\mathbf{A}}_i \in \mathbb{Z}_q^{n \times \ell'}$ for all $i \in [Q]$. Also, parse $\mathbf{c}_{0,\text{csLWE}}^\top = (\mathbf{c}_0^\top, \mathbf{c}_1^\top, \dots, \mathbf{c}_Q^\top)$ where $\mathbf{c}_0^\top \in \mathbb{Z}_q^{1 \times m'}$ and $\mathbf{c}_i^\top \in \mathbb{Z}_q^{1 \times \ell'}$ for all $i \in [Q]$.
- (b) For all $i \in [Q]$, sample $\mathbf{a}_i^\top \leftarrow \mathbb{Z}_q^{1 \times \ell'}$ and set $\tilde{\mathbf{c}}_i^\top = \mathbf{c}_i^\top - \mathbf{a}_i^\top$.
- (c) Compute and set $\mathbf{c}_i^\top = [\mathbf{G}_q \mathbf{G}_p^{-1}(\lfloor \tilde{\mathbf{c}}_i \rfloor_p)]_B \in [-\lfloor q/2 \rfloor + B, \lfloor q/2 \rfloor - B]^{1 \times \ell}$.
- (d) Set $\bar{\mathbf{A}}_{\text{fhe}} = \bar{\mathbf{A}}_{\text{fhe,csLWE}}$, $\mathbf{c}_{\text{fhe}}^\top = \mathbf{c}_{\text{fhe,csLWE}}^\top$, and $\mathbf{C} = \mathbf{C}_{\text{csLWE}}$.
- (e) Forward the following to $\mathcal{A}$.

$$\left(1^\lambda, \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe}} \\ \mathbf{c}_{\text{fhe}}^\top \end{pmatrix}, \mathbf{C}, \begin{pmatrix} \bar{\mathbf{A}}_0 \\ \mathbf{c}_0^\top \end{pmatrix}, \left\{ \begin{pmatrix} \bar{\mathbf{A}}_i \\ \mathbf{a}_i^\top \end{pmatrix}, \mathbf{c}_i^\top \right\}_{i \in [Q]} \right)$$

3. On receiving β' from $\mathcal{A}$, forward β' to csLWE challenger.

We observe that if $\beta = 0$, then $\mathcal{B}$ simulates the distribution D' . When $\beta = 1$, we observe that $\mathcal{B}$ simulates the distribution D_1 except that in D_1 , $\mathbf{c}_i^\top \leftarrow \mathbb{Z}_q^{1 \times \ell}$, while $\mathcal{B}$ returns $\mathbf{c}_i^\top \in [-\lfloor q/2 \rfloor + B, \lfloor q/2 \rfloor - B]^{1 \times \ell}$. However, since B is exponentially smaller than $q/2$, the two distributions are negligibly close. Hence, the advantage of $\mathcal{B}$ is negligibly close to the advantage of $\mathcal{A}$. $\square$

3.1.2 The Attack

Theorem 3.8. There exists an efficient evasive circular small-secret LWE (evcsLWE) sampler Samp as defined in Assumption 3.1 such that the pre-condition holds, but the post-condition does not- i.e., there exists a distinguisher $\mathcal{A}^{\text{post}}$ that breaks post-condition with non-negligible advantage.

Proof. We first describe the *contrived* circuit used by our sampler. Let $q, p, B, \sigma, \sigma', \sigma_{-1}, \sigma_{\text{pre}}, \sigma_{\text{post}}, Q$ be parameters. The parameters will be set after we describe the sampler.

Defining Evasive Circular small-secret LWE Sampler. The public coin sampler Samp of the evcsLWE assumption, on input $(1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}})$, does the following:

1. Let $\mathcal{F}_{\text{prm}} = \{f : \mathbb{Z}_q^n \rightarrow [-\lfloor q/2 \rfloor + B, \lfloor q/2 \rfloor - B]^\ell\}$ be a family of publicly samplable PRG against circular-LWE leakage that is given Lemma 3.7. The function family is fully specified by p and B , which will be set after we describe the sampler. We recall that the function can be implemented by a circuit of depth $d(\lambda) = \text{poly}(\log n, \log \log q)$.
2. Parse the random coin into $\text{coins}_{\text{Samp}}^{\text{pub}} = \left(\{f_i\}_{i \in [Q]}, \mathbf{A}_{\text{circ}} = \begin{pmatrix} \overline{\mathbf{A}}_{\text{circ}} \\ \mathbf{a}_{\text{circ}}^\top \end{pmatrix}, \overline{\mathbf{A}}' \right)$, where $\overline{\mathbf{A}}_{\text{circ}}$ and $\mathbf{a}_{\text{circ}}^\top$ denote the first n rows and $n+1$ -th row of the matrix $\mathbf{A}_{\text{circ}} \in \mathbb{Z}_q^{(n+1) \times (L_S+1)m}$.
3. Define the following circuit C'_{f_i} (with depth at most $d+3$) implementing f_i .
 - (a) Start with any circuit C_{f_i} (of depth at most d) outputting $f_i(\mathbf{s})$ (in binary) on input $\mathbf{s}$.
 - (b) Recall that we have $\mathbf{s}^\top = (\mathbf{r}^\top, -1)$. Therefore, no matter the value of $\mathbf{r}$, there is a fixed position c such that the c -th bit of $\text{bits}(\mathbf{s})$ is 1. We let C_0 be a circuit which outputs the c -th bit of its input.
 - (c) To the output wire of C_{f_i} corresponding to the lowest-order bit of $f_i(\mathbf{s})[1]$, attach a new gate which performs the correlation-inducing transformation described in Corollary 3.4, using the output wire of C_0 as the special “1” input. By the definition, output of C_0 is 1 if the input is $\text{bits}(\mathbf{s})$.
 - (d) To all the other output wires of C_{f_i} , attach a gate performing “even randomness” transformation as in Lemma 3.2.
4. Define the homomorphic evaluation circuit $\text{VEval}_{C'_{f_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{f_i})$, where the homomorphic evaluations of the even randomness gate (Step 3c above) and the correlation inducing gates (Step 3d above) are implemented as specified in Lemma 3.2 and Corollary 3.4, respectively. Compute $\mathbf{H}_{\mathbf{A}_{\text{circ}}}^{f_i} = \text{MEvalC}(\mathbf{A}_{\text{circ}}, \text{VEval}_{C'_{f_i}}) \in \mathbb{Z}_q^{(L_S+1)m \times \ell}$ for all $i \in [Q]$. It then computes $\mathbf{A}_{f_i} = \mathbf{A}_{\text{circ}} \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}}^{f_i} \in \mathbb{Z}_q^{(n+1) \times \ell}$ for all $i \in [Q]$ and sets $\mathbf{P} = [\overline{\mathbf{A}}_{f_1}, \dots, \overline{\mathbf{A}}_{f_Q}]$, where $\overline{\mathbf{A}}_{f_i}$ is the matrix formed by taking upper n rows of $\mathbf{A}_{f_i}$.
5. Output $\overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \mathbf{P} \in \mathbb{Z}_q^{n \times \ell \cdot Q}, \text{aux} = (\mathbf{a}_{\text{circ}}, \{f_i\}_{i \in [Q]}), \sigma, \sigma', \sigma_{-1}, \sigma_{\text{post}}, \sigma_{\text{pre}}$ where $\sigma, \sigma', \sigma_{-1}, \sigma_{\text{post}}, \sigma_{\text{pre}}$ are as defined later in parameter settings.

Parameter Setting. We recall the parameters from [HLL23] except p and B , which we define and set for our purpose.

$$q \in (2^{15\lambda-1}, 2^{15\lambda}], \quad p \in (2^{5\lambda-1}, 2^{5\lambda}], \quad B = 2^{8\lambda}, \quad \sigma = 2^\lambda, \\ \sigma' = 2^{2\lambda}, \quad \sigma_{-1} = 2^\lambda, \quad \sigma_{\text{pre}} = 2^{6\lambda}, \quad \sigma_{\text{post}} = 2^{7\lambda}, \quad Q = \lambda + m + (L_S + 1)m$$

Counter-example Roadmap. Now, for our counter-example, we need to show that for $(\overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \mathbf{P}, \text{aux}, \sigma, \sigma', \sigma_{-1}, \sigma_{\text{post}}, \sigma_{\text{pre}}) \leftarrow \text{Samp}(1^\lambda; \text{coins}_{\text{Samp}}^{\text{pub}} = (\{f_i\}_{i \in [Q]}, \mathbf{A}_{\text{circ}}, \overline{\mathbf{A}}'))$ where $\text{aux} = (\mathbf{a}_{\text{circ}}, \{f_i\}_{i \in [Q]})$,

1. (Attack against postcondition) There exists an adversary $\mathcal{A}^{\text{post}}$ who distinguishes the distributions, D_0^{post} and D_1^{post} as in Equation (11) with non-negligible probability.
2. (Pre-condition) For all PPT adversary $\mathcal{A}^{\text{pre}}$, the two distributions, D_0^{pre} and D_1^{pre} as defined in Equation (10) are indistinguishable. We prove this in Lemma 3.11.

Distinguishing Algorithm/Attack Strategy. The adversary $\mathcal{A}^{\text{post}}$ holds the distribution D_β^{post} for $\beta \leftarrow \{0, 1\}$. Thus, if $\beta = 0$, $\mathbf{c}_\mathbf{B}^\top = \mathbf{r}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top$, $\mathbf{S} = \left(\left(\mathbf{r}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \right) \right) \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}$, $\bar{\mathbf{c}}_{\text{circ}}^\top = \mathbf{r}^\top (\bar{\mathbf{A}}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G}) + \mathbf{e}_{\text{circ}}^\top$ and if $\beta = 1$, $\mathbf{c}_\mathbf{B}^\top \leftarrow \mathbb{Z}_q^m$, $\mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1)m \lceil \log_2 q \rceil}$, $\bar{\mathbf{c}}_{\text{circ}}^\top \leftarrow \mathbb{Z}_q^{(L_S+1)m}$. It does the following:

1. Compute $\mathbf{c}_{\text{circ}}^\top := \bar{\mathbf{c}}_{\text{circ}}^\top - (\mathbf{a}_{\text{circ}}^\top - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G})$. If $\beta = 0$, we have $\mathbf{c}_{\text{circ}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G}) + \mathbf{e}_{\text{circ}}^\top$ and otherwise, it is random.
2. For each $i \in [Q]$, compute $\mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} = \text{MEvalCX}(\mathbf{A}_{\text{circ}}, \text{VEval}_{C'_{f_i}}(\text{bits}(\mathbf{S})))$ and $\mathbf{z}_i = \mathbf{c}_\mathbf{B}^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \bmod q$ where $\mathbf{K}_i \leftarrow \mathbf{B}_{\sigma^{-1}}^{-1}(\bar{\mathbf{A}}_{f_i})$ and $\mathbf{a}_{\text{circ}}^\top$ is the $(n+1)$ th row of $\mathbf{A}_{\text{circ}}$.
3. Let $L = (L_S + 1)m$. $\mathcal{A}^{\text{post}}$ defines the following set of linear equations (Equation (13)) in variables $\mathbf{e}_\mathbf{B}[1], \dots, \mathbf{e}_\mathbf{B}[m]$ and $\mathbf{e}_{\text{circ}}[1], \dots, \mathbf{e}_{\text{circ}}[L]$ and outputs $\beta' = 0$ if Equation (13) is solvable; otherwise, outputs $\beta' = 1$.

$$\begin{aligned}
& \mathbf{e}_\mathbf{B}[1] \mathbf{K}_1[1, 1] + \dots + \mathbf{e}_\mathbf{B}[m] \mathbf{K}_1[m, 1] - \mathbf{e}_{\text{circ}}[1] \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_1}[1, 1] - \dots \\
& \quad - \mathbf{e}_{\text{circ}}[L] \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_1}[L, 1] = \mathbf{z}_1[1] \bmod 2 \\
& \mathbf{e}_\mathbf{B}[1] \mathbf{K}_2[1, 1] + \dots + \mathbf{e}_\mathbf{B}[m] \mathbf{K}_2[m, 1] - \mathbf{e}_{\text{circ}}[1] \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_2}[1, 1] - \dots \\
& \quad - \mathbf{e}_{\text{circ}}[L] \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_2}[L, 1] = \mathbf{z}_2[1] \bmod 2 \quad (13) \\
& \quad \vdots \\
& \mathbf{e}_\mathbf{B}[1] \mathbf{K}_Q[1, 1] + \dots + \mathbf{e}_\mathbf{B}[m] \mathbf{K}_Q[m, 1] - \mathbf{e}_{\text{circ}}[1] \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_Q}[1, 1] - \dots \\
& \quad - \mathbf{e}_{\text{circ}}[L] \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_Q}[L, 1] = \mathbf{z}_Q[1] \bmod 2
\end{aligned}$$

Next, we analyze the success probability of $\mathcal{A}^{\text{post}}$.

Lemma 3.9. $\mathcal{A}^{\text{post}}$ wins (i.e., $\beta = \beta'$) with at least $3/4 - \text{negl}(\lambda)$ probability when $Q \geq \lambda + m + (L_S + 1)m$.

Proof. Let us first analyze the solvability of equations defined in Equation (13) for the two cases: $\beta = 0$ and $\beta = 1$.

When $\beta = 0$, $\forall i \in [Q]$,

$$\begin{aligned}
\mathbf{z}_i &= \mathbf{c}_\mathbf{B}^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \\
&= (\mathbf{r}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top) \cdot \mathbf{K}_i - (\mathbf{s}^\top (\mathbf{A}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G}) + \mathbf{e}_{\text{circ}}^\top) \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \\
&= \mathbf{r}^\top \mathbf{B} \mathbf{K}_i + \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i - \mathbf{s}^\top \mathbf{A}_{\text{circ}} \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} + \mathbf{s}^\top \text{VEval}_{C'_{f_i}}(\text{bits}(\mathbf{S})) - \mathbf{e}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \\
&= \mathbf{r}^\top \mathbf{B} \mathbf{K}_i + \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i - \mathbf{s}^\top \begin{pmatrix} \bar{\mathbf{A}}_{\text{circ}} \\ \mathbf{a}_{\text{circ}}^\top \end{pmatrix} \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} + \mathbf{s}^\top \text{VEval}_{C'_{f_i}}(\text{bits}(\mathbf{S})) - \mathbf{e}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \\
&= \mathbf{r}^\top \bar{\mathbf{A}}_{f_i} + \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i - (\mathbf{r}^\top, -1) \begin{pmatrix} \bar{\mathbf{A}}_{f_i} \\ \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \end{pmatrix} + (\mathbf{r}^\top, -1) \left(\left(\mathbf{r}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \right) \mathbf{R}_{f_i} - \begin{pmatrix} 0^{n \times \ell} \\ f_i(\mathbf{s}) \end{pmatrix} \right) \\
& \quad - \mathbf{e}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \\
&= \mathbf{r}^\top \bar{\mathbf{A}}_{f_i} + \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i - \mathbf{r}^\top \bar{\mathbf{A}}_{f_i} + \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i} + f_i(\mathbf{s}) - \mathbf{e}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} \\
&= \mathbf{e}_\mathbf{B}^\top \cdot \mathbf{K}_i + f_i(\mathbf{s}) - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i} - \mathbf{e}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i}.
\end{aligned} \tag{14}$$

Here, we argue that $\mathbf{z}_i \bmod q \in [-\lfloor q/2 \rfloor, \lfloor q/2 \rfloor]$ gives $\mathbf{e}_B^\top \cdot \mathbf{K}_i + f_i(\mathbf{s}) - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i} - \mathbf{e}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{s}}^{f_i}$ over the integer. This can be seen by observing $f_i(\mathbf{s}) \in [-\lfloor q/2 \rfloor + B, \lfloor q/2 \rfloor - B]$ and $\mathbf{e}_B^\top \cdot \mathbf{K}_i - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i} - \mathbf{e}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{s}}^{f_i} \in [-B, B]$, where the latter follows from $\|\mathbf{e}_B\| \leq \sigma_{\text{pre}} \sqrt{\lambda}$, $\|\mathbf{B}^{-1}(\mathbf{P})\| \leq \sigma_{-1} \sqrt{\lambda}$, $\|\mathbf{e}_{\text{fhe}}\| \leq \sigma \sqrt{\lambda}$, $\|\mathbf{e}_{\text{circ}}\| \leq \sigma' \sqrt{\lambda}$, $\|\mathbf{R}_{f_i}\| \leq (m+2)^{d+3} \lceil \log q \rceil m$, $\|\mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{s}}^{f_i}\| \leq (m+2)^{d+3} \lceil \log q \rceil$ (from Lemma 2.14 and Section 2.5), and our choice of B .

We prove in Claim 3.10 that with probability $1/2 + \text{negl}(\lambda)$, we have $\forall i \in [Q]$, $(f_i(\mathbf{s}) - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i})[1] = 0 \bmod 2$. This gives us that with probability $1/2 + \text{negl}(\lambda)$,

$$\forall i \in [Q], \mathbf{z}_i[1] = \mathbf{e}_B^\top \mathbf{K}_i[1] - \mathbf{e}_{\text{circ}}^\top \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{s}}^{f_i}[1] \bmod 2.$$

Here, $\mathbf{K}_i[1]$ is the first column of $\mathbf{K}_i$. Observe that this is the same as the set of equations defined in Equation (13). Hence, in case of $\beta = 0$, Equation (13) is solvable with probability $1/2 + \text{negl}(\lambda)$. This gives us

$$\Pr(\beta' = 0 \mid \beta = 0) = 1/2 + \text{negl}(\lambda). \quad (15)$$

When $\beta = 1$,

$$\forall i \in [Q], \mathbf{z}_i = \mathbf{c}_B^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{s}}^{f_i} - \mathbf{a}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{s}}^{f_i},$$

where $\mathbf{c}_B \leftarrow \mathbb{Z}_q^m$. Here, we argue that the distribution of $\{\mathbf{c}_B^\top \cdot \mathbf{K}_i \bmod q\}_{i \in [Q]}$ is statistically close to uniform distribution over $(\mathbb{Z}_q^{1 \times \ell})^Q$, even given $\{\mathbf{K}_i \bmod 2\}_{i \in [Q]}$. To see this, we first observe that by Lemma 2.6, each column $\mathbf{K}_i[\cdot, j]$ of $\mathbf{K}_i$ has min-entropy at least $m \log(\sigma_{-1}/\lambda) > 2m$ except for negligible probability over $\mathbf{B}$. Furthermore, $(\mathbf{K}_i[\cdot, j] \bmod 2)$ can be represented by a string of m -bits. We then apply the generalized Leftover Hash Lemma (Lemma 2.7), which implies that the two distributions are within statistical distance $Q\sqrt{2^m \cdot 2^{-2m}q} = \text{negl}(\lambda)$ by our choice of parameters. We therefore have that $\{\mathbf{z}_i \bmod q\}_{i \in [Q]}$ is statistically close to being uniformly distributed at random over $(\mathbb{Z}_q^{1 \times \ell})^Q$, even given $\{\mathbf{K}_i \bmod 2\}_i$. This implies that $\{\mathbf{z}_i \bmod 2\}_{i \in [Q]}$ (which forms the RHS of equations in Equation (13)) is also statistically close to a uniform distribution, independent of $\{\mathbf{K}_i \bmod 2\}_i$ when $\beta = 1$. Hence, Equation (13), having $m + (L_S + 1)m$ number of variables, is unsolvable with overwhelming probability if $Q > \lambda + m + (L_S + 1)m$, since the RHS of the equation falls into the space for which there exists a solution with probability at most $2^{m+(L_S+1)m} / 2^{\lambda+m+(L_S+1)m} = 2^{-\lambda}$.

This gives us

$$\Pr(\beta' = 1 \mid \beta = 1) \geq 1 - \text{negl}(\lambda). \quad (16)$$

Thus, from Equation (15) and Equation (16), we get $\Pr(\beta' = \beta) \geq 3/4 - \text{negl}(\lambda)$. $\square$

Claim 3.10. With probability $1/2 + \text{negl}(\lambda)$, $(f_i(\mathbf{s}) - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i})[1] = 0 \bmod 2$ holds for all $i \in [Q]$.

Proof. Let us start by analyzing the $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i}$ error. Recall that $\mathbf{R}_{f_i}$ is the randomness in the homomorphically evaluated encryption of $f_i(\mathbf{s})$ (of different form: $\mathbf{A}_{\text{fhe}} \mathbf{R}_{f_i} - \begin{pmatrix} 0^{n \times \ell} \\ f_i(\mathbf{s}) \end{pmatrix}$), output by $\text{VEval}_{C_{f_i}'}^{\text{circuit}}$.

We recall from Section 2.4 that for any circuit $C : \{0, 1\}^L \rightarrow \mathbb{Z}_q^{1 \times \ell}$, VEval_C on input $\mathbf{X}$ - a GSW encryption of some $\mathbf{x} \in \{0, 1\}^L$ - outputs $\mathbf{A}_{\text{fhe}} \mathbf{R}_C - \begin{pmatrix} 0^{n \times \ell} \\ C(\mathbf{x}) \end{pmatrix}$. Let $C(\mathbf{x})[u, v] \in \{-1, 0, 1\}$ be (the signed version of) the v -th bit of $C(\mathbf{x})[u]$, for $u \in [\ell]$ and $v \in [0, w)$, $w = \lceil \log_2 q \rceil$, then VEval_C computation firstly homomorphically computes GSW encryption of $C(\mathbf{x})[u, v]$, which is of the form $\mathbf{C}_{u,v} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C,u,v} - C(\mathbf{x})[u, v] \mathbf{G}$. Then for each $u \in [\ell]$, it linearly aggregates $\{\mathbf{C}_{u,v}\}_{v \in [0, w-1]}$ to get $\mathbf{C}_u = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C,u} - \begin{pmatrix} 0^{n \times 1} \\ C(\mathbf{x})[u] \end{pmatrix}$ as $\mathbf{C}_u = \sum_{v=0}^{w-1} \mathbf{C}_{u,v} \mathbf{G}^{-1}(2^v \iota_{n+1})$. In particular,

$$\mathbf{R}_{C,u} = \sum_{v=0}^{w-1} \mathbf{R}_{C,u,v} \mathbf{G}^{-1}(2^v \iota_{n+1}), \text{ and } \mathbf{R}_C = (\mathbf{R}_{C,1}, \dots, \mathbf{R}_{C,\ell}).$$

Coming back to our analysis, $(\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i})([1]) = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_{i,1}} = \mathbf{e}_{\text{fhe}}^\top \sum_{v=0}^{w-1} \mathbf{R}_{f_{i,1,v}} \mathbf{G}^{-1}(2^v \iota_{n+1})$, where $\mathbf{R}_{f_{i,1}}$ is the first column of $\mathbf{R}_{f_i}$ and $\mathbf{R}_{f_{i,1,v}}$ is the randomness in homomorphically computed GSW encryption of the v -th (signed) bit of $f_i(\mathbf{s})[1]$. Here $v \in [0, w)$ where $w = \lceil \log_2 q \rceil$. Since f_i is implemented by C'_{f_i} , in which all the output bits pass through even randomness, except the lowest order bit of the first element, which passes through the correlation inducing gate, we have

$$\begin{aligned}
& f_i(\mathbf{s})[1] - (\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i})([1]) \bmod 2 \\
&= f_i(\mathbf{s})[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_{i,1,0}} \mathbf{G}^{-1}(\iota_{n+1}) - \mathbf{e}_{\text{fhe}}^\top \sum_{v=1}^{w-1} \mathbf{R}_{f_{i,1,v}} \mathbf{G}^{-1}(2^v \iota_{n+1}) \bmod 2 \\
&= f_i(\mathbf{s})[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_{i,1,0}} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \quad (\text{due to Lemma 3.2}) \\
&= f_i(\mathbf{s})[1, 0] + \sum_{v=1}^{w-1} 2^v f_i(\mathbf{s})[1, v] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_{i,1,0}} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\
&= f_i(\mathbf{s})[1, 0] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_{i,1,0}} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\
&= f_i(\mathbf{s})[1, 0] - f_i(\mathbf{s})[1, 0] \mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \quad (\text{due to Corollary 3.4}) \\
&= 0 \bmod 2, \text{ if } \mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \text{ is odd.}
\end{aligned}$$

In the above, $f_i(\mathbf{s})[u, v] \in \{-1, 0, 1\}$, represents the v -th bit in the binary representation of $f_i(\mathbf{s})[u]$ and $\mathbf{R}^*$ is the randomness in the ciphertext of special "1" input used in Corollary 3.4. We then argue that $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1})$ is odd with probability negligibly close to $1/2$. To see this, we first observe that each entry of $\mathbf{R}^*$ is distributed over $\{0, 1\}$ uniformly at random and thus so is each entry of $\mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2$, since the multiplication of $\mathbf{G}^{-1}(\iota_{n+1})$ from the right equals to selecting a column of $\mathbf{R}^*$ in the fixed position. In particular, this implies that $\mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2$ is a non-zero vector with overwhelming probability. We then observe that each entry of $\mathbf{e}_{\text{fhe}} \bmod 2$ is statistically close to the uniform distribution over $\{0, 1\}$ by Lemma 2.9. Combining these facts, it follows that the distribution of $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2$ is statistically close to the uniform distribution over $\{0, 1\}$, as desired. Note that the term $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1})$ does not depend on i and hence, with probability $1/2 + \text{negl}(\lambda)$, we have $f_i(\mathbf{s})[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{f_i}}[1] = 0 \bmod 2$ for all $i \in [Q]$. Hence, the proof. $\square$

Proving the pre-condition. We now show that the pre-condition of evcsLWE holds with respect to the sampler defined in Section 3.1.2.

Lemma 3.11. For $(\overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \mathbf{P}, \text{aux}, \sigma, \sigma', \sigma_{-1}, \sigma_{\text{post}}, \sigma_{\text{pre}}) \leftarrow \text{Samp}(1^\lambda; \text{coins}_{\text{Samp}}^{\text{pub}} = (\mathbf{A}_{\text{circ}}, \overline{\mathbf{A}}', \{f_i\}_{i \in [Q]}))$ where $\text{aux} = (\mathbf{a}_{\text{circ}}, \{f_i\}_{i \in [Q]})$ and for a $\lambda \in \mathbb{N}$,

$$\begin{aligned}
D_0^{\text{pre}} &:= \left(\begin{array}{l} 1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}}, \text{aux}, \overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \mathbf{S} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \overline{\mathbf{c}}_{\text{circ}}^\top = \mathbf{r}^\top (\overline{\mathbf{A}}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \overline{\mathbf{G}}) + \mathbf{e}_{\text{circ}}^\top, \\ (\mathbf{c}')^\top = \mathbf{r}^\top \overline{\mathbf{A}}' + (\mathbf{e}')^\top, \mathbf{c}_{\mathbf{B}}^\top = \mathbf{r}^\top \mathbf{B} + \mathbf{e}_{\mathbf{B}}^\top, \\ \mathbf{c}_{\mathbf{P}}^\top = \mathbf{r}^\top \mathbf{P} + \mathbf{e}_{\mathbf{P}}^\top \end{array} \right) \\
&\approx_c D_1^{\text{pre}} := \left(\begin{array}{l} 1^\lambda, \text{coins}_{\text{Samp}}^{\text{pub}}, \text{aux}, \overline{\mathbf{A}}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log_2 q \rceil m}, \\ \overline{\mathbf{c}}_{\text{circ}}^\top \leftarrow \mathbb{Z}_q^{1 \times (Ls+1)m}, \\ (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times m'}, \mathbf{c}_{\mathbf{B}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \\ \mathbf{c}_{\mathbf{P}}^\top \leftarrow \mathbb{Z}_q^{1 \times J} \end{array} \right) \tag{17}
\end{aligned}$$

where $J = \ell \cdot Q$, sampler Samp is as defined in Section 3.1.2 and the vectors and matrices are sampled as in Assumption 3.1.

Proof. We can write $\mathbf{r}^\top \mathbf{P} + \mathbf{e}_\mathbf{P}^\top = \mathbf{r}^\top [\bar{\mathbf{A}}_{f_1}, \dots, \bar{\mathbf{A}}_{f_Q}] + [\mathbf{e}_{\mathbf{P},1}^\top, \dots, \mathbf{e}_{\mathbf{P},Q}^\top]$ where $\mathbf{e}_{\mathbf{P},i}^\top$ is the i th block of $\mathbf{e}_\mathbf{P}^\top$ with length ℓ . We rearrange the terms in Equation (17) and change it slightly as:

$$\begin{aligned} \hat{D}_0^{\text{pre}} &:= \left(\begin{array}{l} 1^\lambda, \mathbf{A}_{\text{circ}}, \bar{\mathbf{A}}', \bar{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top, \mathbf{S} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{circ}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G}) + \mathbf{e}_{\text{circ}}^\top, \\ (\mathbf{c}')^\top = \mathbf{r}^\top \bar{\mathbf{A}}' + (\mathbf{e}')^\top, \mathbf{c}_\mathbf{B}^\top = \mathbf{r}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \\ \{f_i \leftarrow \mathcal{F}_{\text{prm}}, \mathbf{c}_{\mathbf{P},i}^\top = \mathbf{r}^\top \bar{\mathbf{A}}_{f_i} + \mathbf{e}_{\mathbf{P},i}^\top\}_{i \in [Q]} \end{array} \right) \\ \approx_c \hat{D}_1^{\text{pre}} &:= \left(\begin{array}{l} 1^\lambda, \mathbf{A}_{\text{circ}}, \bar{\mathbf{A}}', \bar{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \\ \mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log_2 q \rceil m}, \\ \mathbf{c}_{\text{circ}}^\top \leftarrow \mathbb{Z}_q^{1 \times (Ls+1)m}, \\ (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times m'}, \mathbf{c}_\mathbf{B}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \\ \{f_i \leftarrow \mathcal{F}_{\text{prm}}, \mathbf{c}_{\mathbf{P},i}^\top \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]} \end{array} \right). \end{aligned} \quad (18)$$

In particular, we replace the term $\bar{\mathbf{c}}_{\text{circ}}^\top$ in Equation (17) with $\mathbf{c}_{\text{circ}}^\top$ in Equation (18). For $\bar{\mathbf{c}}_{\text{circ}}^\top$ in D_0^{pre} and $\mathbf{c}_{\text{circ}}^\top$ in $\hat{D}_0^{\text{pre}}$, $\mathbf{c}_{\text{circ}}^\top = \bar{\mathbf{c}}_{\text{circ}}^\top - (\mathbf{a}_{\text{circ}}^\top - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G})$ holds. Since $-(\mathbf{a}_{\text{circ}}^\top - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G})$ is publicly computable, it is easy to see that Equation (17) and Equation (18) are equivalent.

We now prove Equation (18) by a series of hybrids Hyb_0 to Hyb_2 where Hyb_0 is the $\hat{D}_0^{\text{pre}}$ distribution and Hyb_2 is the $\hat{D}_1^{\text{pre}}$ distribution of Equation (18). We prove that $\text{Hyb}_0 \approx \text{Hyb}_1 \approx \text{Hyb}_2$.

Hyb₁: This hybrid is same as Hyb_0 , except that for all $i \in [Q]$, $\mathbf{c}_{\mathbf{P},i}^\top$ can be computed as:

$$\mathbf{c}_{\mathbf{P},i}^\top = \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} + \mathbf{a}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} + f_i(\mathbf{s}) + \mathbf{e}_{\mathbf{P},i}^\top$$

where $\mathbf{e}_{\mathbf{P},i} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}, \leq \sigma_{\text{pre}} \sqrt{\lambda}}^\ell$. We claim that Hyb_0 and Hyb_1 are statistically indistinguishable. To see this, note that from Equation (14),

$$\mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} = \mathbf{r}^\top \bar{\mathbf{A}}_{f_i} - \mathbf{a}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} - f_i(\mathbf{s}) + \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i} + \mathbf{e}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i}$$

where $\mathbf{R}_{f_i}$ is the randomness in the homomorphically evaluated encryption (of different form) of $f_i(\mathbf{s})$, output by $\text{VEval}_{C'_{f_i}}$. This implies,

$$\mathbf{c}_{\mathbf{P},i}^\top = \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} + f_i(\mathbf{s}) + \mathbf{a}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i} + \mathbf{e}_{\mathbf{P},i}^\top = \mathbf{r}^\top \bar{\mathbf{A}}_{f_i} + \tilde{\mathbf{e}}_i^\top + \mathbf{e}_{\mathbf{P},i}^\top$$

where $\tilde{\mathbf{e}}_i^\top = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{f_i} + \mathbf{e}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i}$.

Let $d' = d_{\text{VEval}_{C'_{f_i}}} \leq d + 3$. Then, we have, $\|\mathbf{e}_{\text{fhe}}\| \leq \sigma \sqrt{\lambda}$, $\|\mathbf{e}_{\text{circ}}\| \leq \sigma' \sqrt{\lambda}$, $\|\mathbf{R}_{f_i}\| \leq (m + 2)^{d+3} \lceil \log q \rceil m$, $\|\mathbf{H}_{\mathbf{A}_{\text{circ}}, \mathbf{S}}^{f_i}\| \leq (m + 2)^{d+3} \lceil \log q \rceil$ (from Lemma 2.14 and Section 2.5). Hence, $\|\tilde{\mathbf{e}}_i\| \leq m^2 \sigma \sqrt{2\lambda} (m + 2)^{d+3} \lceil \log q \rceil + L \sigma' \sqrt{2\lambda} (m + 2)^{d'+3} \lceil \log q \rceil \leq 2^{O(d \log^3 \lambda)} \sigma' \leq 2^{3\lambda}$ (as $d = \text{poly}(\log \lambda)$). Further, since $\mathbf{e}_{\mathbf{P},i} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}, \leq \sigma_{\text{pre}} \sqrt{\lambda}}^\ell$, where $\sigma_{\text{pre}} \leq 2^{6\lambda}$ and hence, by noise

flooding (Lemma 2.8), $\mathbf{e}_{\mathbf{P},i}^\top \approx_s \mathbf{e}_{\mathbf{P},i}^\top + \tilde{\mathbf{e}}_i^\top$ with a statistical distance of $\text{poly}(\lambda)2^{-4\lambda}$. Therefore, the statistical distance Δ between Hyb_0 and Hyb_1 is:

$$\Delta(\text{Hyb}_0, \text{Hyb}_1) = \frac{Q \cdot \text{poly}(\lambda)}{2^{4\lambda}}.$$

We now have the following distribution:

$$\left(\begin{array}{l} 1^\lambda, \mathbf{A}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top, \mathbf{S} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{circ}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{circ}} - (1, \text{bits}(\mathbf{S})) \otimes \mathbf{G}) + \mathbf{e}_{\text{circ}}^\top, (\mathbf{c}')^\top = \mathbf{r}^\top \overline{\mathbf{A}}' + (\mathbf{e}')^\top, \mathbf{c}_{\mathbf{B}}^\top = \mathbf{r}^\top \mathbf{B} + \mathbf{e}_{\mathbf{B}}^\top, \\ \{f_i \leftarrow \mathcal{F}_{\text{prm}}, \mathbf{c}_{\mathbf{P},i}^\top = \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}},\mathbf{S}}^{f_i} + \mathbf{a}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}},\mathbf{S}}^{f_i} + f_i(\mathbf{s}) + \mathbf{e}_{\mathbf{P},i}^\top\}_{i \in [Q]} \end{array} \right)$$

Hyb₂: This hybrid is same as Hyb_1 , except that, we sample $\mathbf{c}_{\text{fhe}} \leftarrow \mathbb{Z}_q^m, \mathbf{c}_{\text{circ}} \leftarrow \mathbb{Z}_q^L, (\mathbf{c}') \leftarrow \mathbb{Z}_q^{m'}, \mathbf{c}_{\mathbf{B}} \leftarrow \mathbb{Z}_q^m, \mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log q \rceil m}$ and $\mathbf{c}_{\mathbf{P},i}^\top \leftarrow \mathbb{Z}_q^{1 \times \ell}$, where $L = (L_{\mathbf{S}} + 1)m$. We prove that $\text{Hyb}_1 \approx \text{Hyb}_2$ via the following claim.

Claim 3.12. $\text{Hyb}_1 \approx_c \text{Hyb}_2$, assuming the indistinguishability in Equation (12).

Proof. We show that if there exists an adversary $\mathcal{A}$ who can distinguish between the two hybrids with non-negligible advantage, then there is a reduction $\mathcal{B}$ that can distinguish L.H.S and R.H.S of Equation (12) with non-negligible advantage. The reduction is as follows.

1. On receiving 1^λ from $\mathcal{A}$, forward it to challenger of Equation (12).
2. Equation (12) challenger samples $\beta \leftarrow \{0, 1\}, \overline{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}, \overline{\mathbf{A}} \leftarrow \mathbb{Z}_q^{n \times (L+m+m')},$ and $\mathbf{r} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\mathbf{r}}, \leq \sigma \sqrt{\lambda}}^n$, and returns

$$\left(1^\lambda, \mathbf{A}_{\text{fhe}} = \begin{pmatrix} \overline{\mathbf{A}}_{\text{fhe}} \\ \mathbf{c}_{\text{fhe}}^\top \end{pmatrix}, \mathbf{C}, \begin{pmatrix} \overline{\mathbf{A}} \\ \mathbf{c}^\top \end{pmatrix}, \{f_i, \mathbf{c}_i^\top\}_{i \in [Q]} \right)$$

where $\mathbf{c}_{\text{fhe}}^\top = \mathbf{r}^\top \overline{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top, \mathbf{c}^\top = \mathbf{r}^\top \overline{\mathbf{A}} + \mathbf{e}^\top, \mathbf{C} = \mathbf{A}_{\text{fhe}} \mathbf{R} - \text{bits}(\mathbf{s}) \otimes \mathbf{G}$ and $\{\mathbf{c}_i^\top = f_i(\mathbf{s})\}_{i \in [Q]}$ if $\beta = 0$; else $\mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times (L+m+m')}, \mathbf{C} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log q \rceil m}$ and $\{\mathbf{c}_i^\top \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]}$, if $\beta = 1$. Then $\mathcal{B}$ does the following.

- (a) Parse $\overline{\mathbf{A}} = (\overline{\mathbf{A}}_{\text{circ}}, \mathbf{B}, \overline{\mathbf{A}}')$ where $\overline{\mathbf{A}}_{\text{circ}} \in \mathbb{Z}_q^{n \times L}, \mathbf{B} \in \mathbb{Z}_q^{n \times m}, \overline{\mathbf{A}}' \in \mathbb{Z}_q^{n \times m'}$ and $\mathbf{c}^\top = (\tilde{\mathbf{c}}_{\text{circ}}^\top, \mathbf{c}_{\mathbf{B}}^\top, (\mathbf{c}')^\top)$ where $\tilde{\mathbf{c}}_{\text{circ}}^\top \in \mathbb{Z}_q^{1 \times L}, \mathbf{c}_{\mathbf{B}}^\top \in \mathbb{Z}_q^{1 \times m}, (\mathbf{c}')^\top \in \mathbb{Z}_q^{1 \times m'}$.
- (b) Set $\mathbf{S} = \mathbf{C}$. Also set $\overline{\mathbf{A}}_{\text{circ}} = \overline{\mathbf{A}}_{\text{circ}}' + (1, \text{bits}(\mathbf{S})) \otimes \overline{\mathbf{G}}, \mathbf{A}_{\text{circ}} = \begin{pmatrix} \overline{\mathbf{A}}_{\text{circ}} \\ \mathbf{a}_{\text{circ}}^\top \end{pmatrix}$, where $\mathbf{a}_{\text{circ}} \leftarrow \mathbb{Z}_q^L$, and $\mathbf{c}_{\text{circ}}^\top = \tilde{\mathbf{c}}_{\text{circ}}^\top - (\mathbf{a}_{\text{circ}}^\top - (1, \text{bits}(\mathbf{S})) \otimes \underline{\mathbf{G}})$, where $\overline{\mathbf{G}}$ and $\underline{\mathbf{G}}$ denotes the first n rows and $n+1$ -th row of the gadget matrix $\mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m}$, respectively.
- (c) For all $i \in [Q]$, sample $\mathbf{e}_{\mathbf{P},i} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}, \leq \sigma_{\text{pre}} \sqrt{\lambda}}^\ell$ and set $\mathbf{c}_{\mathbf{P},i}^\top = \mathbf{c}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}},\mathbf{S}}^{f_i} + \mathbf{a}_{\text{circ}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{circ}},\mathbf{S}}^{f_i} + \mathbf{c}_i^\top + \mathbf{e}_{\mathbf{P},i}^\top$.

3. Forward the following to $\mathcal{A}$.

$$\left(1^\lambda, \mathbf{A}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \mathbf{c}_{\text{fhe}}^\top, \mathbf{S}, \mathbf{c}_{\text{circ}}^\top, (\mathbf{c}')^\top, \mathbf{c}_{\mathbf{B}}^\top, \{f_i, \mathbf{c}_{\mathbf{P},i}^\top\}_{i \in [Q]} \right)$$

4. $\mathcal{A}$ outputs a bit β' . $\mathcal{B}$ forwards the bit β' to the Equation (12) challenger.

We observe that if $\beta = 0$, then $\mathcal{B}$ simulates the distribution of Hybrid 1, and if $\beta = 1$, then the distribution of Hybrid 2. The indistinguishability of Hyb_1 and Hyb_2 follows from the indistinguishability of Equation (12). $\square$

Hence, we achieve the following distribution:

$$\left(\begin{array}{l} 1^\lambda, \mathbf{A}_{\text{circ}}, \overline{\mathbf{A}}', \overline{\mathbf{A}}_{\text{fhe}}, \mathbf{B}, \mathbf{c}_{\text{fhe}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \mathbf{S} \leftarrow \mathbb{Z}_q^{(n+1) \times (n+1) \lceil \log q \rceil m}, \\ \mathbf{c}_{\text{circ}}^\top \leftarrow \mathbb{Z}_q^{1 \times (Ls+1)m}, \\ (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times m'}, \mathbf{c}_{\mathbf{B}}^\top \leftarrow \mathbb{Z}_q^{1 \times m}, \{f_i \leftarrow \mathcal{F}_{\text{prm}}, \mathbf{c}_{\mathbf{P},i}^\top \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]} \end{array} \right) \quad (19)$$

where $\mathbf{A}_{\text{fhe}} = \begin{pmatrix} \overline{\mathbf{A}}_{\text{fhe}} \\ \mathbf{c}_{\text{fhe}}^\top \end{pmatrix}$. The distribution as in Equation (19) is the same as $\hat{D}_1^{\text{pre}}$ as in Equation (18). Hence, the proof. $\square$

$\square$

$\square$

3.2 Attacks when Pre-Condition Error is Larger

In this section, we develop new attacks that exploit the case where the error in the pre-condition is set larger than the error in the post-condition.

3.2.1 Attack 1

Theorem 3.13. There exists an efficient and public-coin ELWE sampler Samp as defined in Assumption 2.12 (with $\mathbf{A}$ and LWE sample with respect to it) such that the pre-condition holds under the LWE assumption, but the post-condition does not- i.e., there exists a distinguisher $\mathcal{A}^{\text{post}}$ that distinguishes post-condition with non-negligible probability.

Proof. We first define the sampler.

Defining Evasive LWE Sampler. Let $n, m, \ell, p, q, r, B \in \mathbb{N}$ be parameters and λ be the security parameter. We assume that p and r are primes and $q = pr$. The public coin sampler Samp of the Evasive LWE assumption is defined as follows.

1. Sample $\mathbf{D} \leftarrow [0, r-1]^{n \times \ell}$ and set $\mathbf{P} = p\mathbf{D} \bmod q$.
2. Samples $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$.
3. Output $\mathbf{S} = \mathbf{s}^\top$, $\mathbf{P}$, and $\text{aux} = \mathbf{D}$.

We can see that the sampler is public-coin, since the only randomness that should be kept secret is $\mathbf{s}$.

Parameter Setting. We set the parameters as follows.

$$\begin{aligned} n &= \text{poly}(\lambda), \quad \ell = n^2, \quad m = O(n \log q), \quad r > 2^{9\lambda}, \quad p > 2^{11\lambda}, \quad q = pr, \quad B = 2^{6\lambda}, \\ \sigma_{\text{post}} &= 2^{5\lambda}, \quad \sigma_{\text{pre}} = \sigma'_{\text{pre}} = 2^{15\lambda}, \quad \sigma_s = 2^{2\lambda}, \quad \tau = O\left(\sqrt{(n+1) \log q}\right) \end{aligned}$$

Here r and p are primes greater than $2^{9\lambda}$ and $2^{11\lambda}$ respectively.

Distinguishing Algorithm/Attack Strategy. The adversary $\mathcal{A}^{\text{post}}$ holds the distribution D_β^{post} for $\beta \leftarrow \{0, 1\}$. Thus, for $\beta = 0$, $\mathbf{c}_\mathbf{B}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top$ and for $\beta = 1$, $\mathbf{c}_\mathbf{B}^\top \leftarrow \mathbb{Z}_q^{1 \times m}$. It does the following:

1. Compute a vector $\mathbf{d}^\top \stackrel{\text{def}}{=} \mathbf{c}_\mathbf{B}^\top \cdot \mathbf{B}^{-1}(\mathbf{P}) \bmod q$.
2. Find (the unique) $\mathbf{d}_1 \in [0, r-1]^n$ and $\mathbf{d}_2 \in [0, p-1]^n$ such that $\mathbf{d} = p\mathbf{d}_1 + \mathbf{d}_2 \bmod q$.
3. Find $\mathbf{s}_0 \in [-(r-1)/2, (r-1)/2]^n$ such that $\mathbf{s}_0^\top \mathbf{D} = \mathbf{d}_1^\top \bmod r$.
4. Compute $\gamma = \|\mathbf{c}_\mathbf{B}^\top - \mathbf{s}_0^\top \mathbf{B} \bmod q\|$.
5. Output 1 if $\gamma < B$ and 0 otherwise.

We observe that in the case of $\beta = 0$, we have

$$\mathbf{c}_\mathbf{B}^\top \cdot \mathbf{B}^{-1}(\mathbf{P}) = (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top) \mathbf{B}^{-1}(\mathbf{P}) = p\mathbf{s}^\top \mathbf{D} + \mathbf{e}_\mathbf{B}^\top \mathbf{B}^{-1}(\mathbf{P}).$$

By our parameter setting, $\|\mathbf{e}_\mathbf{B}^\top \mathbf{B}^{-1}(\mathbf{P})\| \leq \lambda \sigma_{\text{post}} \tau m < p$. We therefore have $\mathbf{s}^\top \mathbf{D} = \mathbf{d}_1 \bmod r$. Since $\mathbf{D}$ is full rank over $\mathbb{Z}_r$ with probability $\prod_{i=0}^n (1 - 1/r^{\ell-i}) \geq 1 - n/r^{\ell-n} = 1 - \text{negl}(\lambda)$, we have $\mathbf{s}^\top = \mathbf{s}_0^\top \bmod r$. By our choice of σ_s and B , we have $\|\mathbf{s}\| < r$ with overwhelming probability and thus $\mathbf{s}_0 = \mathbf{s}$ holds over the integer. Therefore, $\gamma = \|\mathbf{e}_\mathbf{B}\| < B$ with overwhelming probability by our choice of σ_{post} . In the case of $\beta = 1$, we argue that such $\mathbf{s}$ does not exist with overwhelming probability. This can be observed by the counting argument. For an arbitrarily fixed $\mathbf{B}$, for each $\mathbf{s}_0 \in \mathbb{Z}_q^n$, there are $(2B+1)^m$ points whose distance from $\mathbf{s}_0^\top \mathbf{B}$ is within B in infinity norm. The probability that random $\mathbf{c}_\mathbf{B}$ is within distance B is thus at most $(2B+1)^m / q^m$. Taking union bound over $\mathbf{s}_0$, the probability that random $\mathbf{c}_\mathbf{B}$ is within distance B from the span of $\mathbf{B}$ is at most $q^n (2B+1)^m / q^m$. By our choice of B , q , and m , this probability is negligible. Summarizing the above discussion, the adversary distinguishes the distributions with high advantage.

Proving the pre-condition. Here, we prove that the pre-condition with respect to the sampler holds. To do so, we prove the pre-condition of Assumption 2.12 by a series of hybrids Hyb_0 to Hyb_3 where Hyb_0 is the D_0^{pre} distribution and Hyb_3 is the D_1^{pre} distribution of (32). We prove that $\text{Hyb}_0 \approx_c \text{Hyb}_1 \approx_c \dots \approx_c \text{Hyb}_3$.

Hyb₀: This is the D_0^{pre} distribution. Namely, the adversary is given $(\mathbf{B}, \mathbf{P}, \mathbf{c}_\mathbf{B}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top, \mathbf{c}_\mathbf{P}^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_\mathbf{P}^\top, \mathbf{D})$.

Hyb₁: This is identical to Hyb_0 except that we replace the term $\mathbf{c}_\mathbf{P}^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_\mathbf{P}^\top$ with $\mathbf{s}^\top \mathbf{P} + \mathbf{s}^\top \mathbf{D}' + \mathbf{e}_\mathbf{P}^\top$, where $\mathbf{D}' \leftarrow [0, p-1]^{n \times \ell}$. By our choice of σ_{pre} , σ_s , and p , we have $\|\mathbf{e}_\mathbf{P}\| > \lambda^{\omega(1)} \|\mathbf{s}^\top \mathbf{D}'\|$ and thus $\mathbf{e}_\mathbf{P}^\top \approx_s \mathbf{s}^\top \mathbf{D}' + \mathbf{e}_\mathbf{P}^\top$. Therefore, this hybrid is statistically indistinguishable from the previous one.

Hyb₂: In this hybrid, we further replace $\mathbf{c}_\mathbf{P}^\top$ and $\mathbf{D}$. Here, we sample $\mathbf{F} \leftarrow \mathbb{Z}_q^{n \times \ell}$ and set $\mathbf{c}_\mathbf{P}^\top = \mathbf{s}^\top \mathbf{F} + \mathbf{e}_\mathbf{P}^\top$ and $\mathbf{D} = \lfloor \mathbf{F}/p \rfloor \bmod r$. This hybrid is essentially the same as previous hybrid. To see this, it suffices to see that the distribution of $(\mathbf{P} + \mathbf{D}' \bmod q, \mathbf{D} \bmod r) = (p\mathbf{D} + \mathbf{D}' \bmod q, \mathbf{D} \bmod r)$ in the previous hybrid is the same as $(\mathbf{F} \bmod q, \lfloor \mathbf{F}/p \rfloor \bmod r)$ in this hybrid.

Hyb₃: This is the D_1^{pre} distribution. Compared with the previous hybrid, here we replace $\mathbf{c}_\mathbf{B}^\top$ and $\mathbf{c}_\mathbf{P}^\top$ with random vectors with the same dimensions. The indistinguishability from the previous hybrid can be shown by a straightforward reduction to LWE.

This completes the proof of the theorem. □

3.2.2 Attack 2.

Our first attack is a bit unnatural in that each entry of $\mathbf{P}$ is in the ideal generated by p in $\mathbb{Z}_q$ (i.e., a multiple of p). Here, we show another example for uniform $\mathbf{P}$ over $\mathbb{Z}_q^{n \times \ell}$.

Theorem 3.14. Let us consider a variant of ELWE as defined in Assumption 2.12. There exists an efficient and public-coin ELWE sampler Samp such that the pre-condition holds under the LWE assumption, but the post-condition does not- i.e., there exists a distinguisher $\mathcal{A}^{\text{post}}$ that distinguishes post-condition with non-negligible probability. In this example, $\mathbf{P}$ is distributed uniformly over $\mathbb{Z}_q^{n \times \ell}$.

Proof. Let $n, m, \ell, p, q, r, B \in \mathbb{N}$ be parameters and λ be the security parameter. We assume that p and r are primes and $q = pr$. The public coin sampler Samp of the Evasive LWE assumption is defined as follows.

1. Sample $\mathbf{A} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{D} \leftarrow [0, r-1]^{n \times \ell}$, and $\mathbf{E} \leftarrow \{0, 1\}^{m \times \ell}$ and set $\mathbf{P} = \mathbf{A}\mathbf{E} + p\mathbf{D} \bmod q$.
2. Samples $\mathbf{s} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$.
3. Output $\mathbf{S} = \mathbf{s}^\top$, $\mathbf{P}$, $\mathbf{A}$, and $\text{aux} = (\mathbf{D}, \mathbf{E})$.

We can see that the sampler is public-coin, since the only randomness that should be kept secret is $\mathbf{s}$.

Parameter Setting. We set the parameter as follows.

$$n = \text{poly}(\lambda), \ell = n^2, m = O(n \log q), r > 2^{9\lambda}, p > 2^{11\lambda}, q = pr, B = 2^{6\lambda}, \\ \sigma_{\text{post}} = 2^{5\lambda}, \sigma_{\text{pre}} = 2^{15\lambda}, \sigma'_{\text{pre}} = 2^{12\lambda}, \sigma_s = 2^{2\lambda}, \tau = O(\sqrt{m \log q})$$

Distinguishing Algorithm/Attack Strategy. The adversary $\mathcal{A}^{\text{post}}$ holds the distribution D_β^{post} for $\beta \leftarrow \{0, 1\}$. Thus, for $\beta = 0$, $\mathbf{c}_\mathbf{A}^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_\mathbf{A}^\top$ and $\mathbf{c}_\mathbf{B}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top$ where $\mathbf{e}_\mathbf{A}, \mathbf{e}_\mathbf{B} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{post}}}^m$, and for $\beta = 1$, $\mathbf{c}_\mathbf{A}^\top, \mathbf{c}_\mathbf{B}^\top \leftarrow \mathbb{Z}_q^{1 \times m}$. It does the following:

1. Compute a vector $\mathbf{d}^\top \stackrel{\text{def}}{=} \mathbf{c}_\mathbf{B}^\top \cdot \mathbf{B}^{-1}(\mathbf{P}) - \mathbf{c}_\mathbf{A}^\top \cdot \mathbf{E} \bmod q$.
2. Find $\mathbf{d}_1 \in [0, r-1]^n$ and $\mathbf{d}_2 \in [0, p-1]^n$ such that $\mathbf{d} = p\mathbf{d}_1 + \mathbf{d}_2 \bmod q$.
3. Find $\mathbf{s}_0 \in [-(r-1)/2, (r-1)/2]$ such that $\mathbf{s}_0^\top \mathbf{D} = \mathbf{d}_1^\top \bmod r$.
4. Compute $\gamma = \|\mathbf{c}_\mathbf{B}^\top - \mathbf{s}_0^\top \mathbf{B} \bmod q\|$.
5. Output 1 if $\gamma < B$ and 0 otherwise.

We observe that in the case of $\beta = 0$, we have

$$\mathbf{c}_\mathbf{B}^\top \cdot \mathbf{B}^{-1}(\mathbf{P}) - \mathbf{c}_\mathbf{A}^\top \cdot \mathbf{E} = (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top) \mathbf{B}^{-1}(\mathbf{P}) - (\mathbf{s}^\top \mathbf{A} + \mathbf{e}_\mathbf{A}^\top) \mathbf{E} = p\mathbf{s}^\top \mathbf{D} + \mathbf{e}_\mathbf{B}^\top \mathbf{B}^{-1}(\mathbf{P}) - \mathbf{e}_\mathbf{A}^\top \mathbf{E}.$$

By our parameter setting, $\|\mathbf{e}_\mathbf{B}^\top \mathbf{B}^{-1}(\mathbf{P}) - \mathbf{e}_\mathbf{A}^\top \mathbf{E}\| \leq m\sigma_{\text{post}}\sqrt{\lambda}(\tau\sqrt{\lambda} + 1) < p$. We therefore have $\mathbf{s}^\top \mathbf{D} = \mathbf{d}_1^\top \bmod r$. Since $\mathbf{D}$ is full rank over $\mathbb{Z}_r$ with probability $\prod_{i=0}^n (1 - 1/r^{\ell-i}) \geq 1 - n/r^{\ell-n} = 1 - \text{negl}(\lambda)$, we have $\mathbf{s}^\top = \mathbf{s}_0^\top \bmod r$. By our choice of σ_s and B , we have $\|\mathbf{s}\| < r$ with overwhelming probability and thus $\mathbf{s}_0 = \mathbf{s}$ holds over the integer. Therefore, $\gamma = \|\mathbf{e}_\mathbf{B}\| < B$ with overwhelming probability by our choice of σ_{post} . Furthermore, in the case of $\beta = 1$, we argue that $\mathbf{s}_0$ such that $\|\mathbf{c}_\mathbf{B}^\top - \mathbf{s}_0^\top \mathbf{B} \bmod q\| < B$ does not exist with overwhelming probability. This is shown by the same argument as that of Theorem 3.13. Therefore, the above constitutes a valid adversary.

Proving the pre-condition. Here, we prove that the pre-condition with respect to the sampler holds. To do so, we prove pre-condition of Assumption 2.12 by a series of hybrids Hyb_0 to Hyb_3 where Hyb_0 is the D_0^{pre} distribution and Hyb_3 is the D_1^{pre} distribution of Equation (32). We prove that $\text{Hyb}_0 \approx_c \text{Hyb}_1 \approx_c \dots \approx_c \text{Hyb}_3$.

Hyb₀: This is the D_0^{pre} distribution. Namely, the adversary is given $(\mathbf{B}, \mathbf{P}, \mathbf{c}_B^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{c}_A^\top = \mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top, \mathbf{c}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_P^\top, \mathbf{A}, \mathbf{D}, \mathbf{E})$, where $\mathbf{e}_A \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}}^m$, $\mathbf{e}_B \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}}^m$ and $\mathbf{e}_P \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{pre}}}^\ell$.

Hyb₁: This is identical to Hyb_0 except that we replace the term $\mathbf{c}_P^\top$ with $\mathbf{c}_A^\top \mathbf{E} + \mathbf{s}^\top (p\mathbf{D} + \mathbf{D}') + \mathbf{e}_P^\top$, where $\mathbf{D}' \leftarrow [0, p-1]^{n \times \ell}$. We have

$$\mathbf{c}_A^\top \mathbf{E} + \mathbf{s}^\top (p\mathbf{D} + \mathbf{D}') + \mathbf{e}_P^\top = (\mathbf{s}^\top \mathbf{A} + \mathbf{e}_A^\top) \mathbf{E} + \mathbf{s}^\top (p\mathbf{D} + \mathbf{D}') + \mathbf{e}_P^\top = \mathbf{s}^\top \mathbf{P} + \mathbf{e}_A^\top \mathbf{E} + \mathbf{s}^\top \mathbf{D}' + \mathbf{e}_P^\top$$

By our choice of σ_{pre} , σ_s , and p , we have $\|\mathbf{e}_A^\top \mathbf{E} + \mathbf{s}^\top \mathbf{D}'\| < \sqrt{\lambda}(m\sigma_{\text{pre}}' + n(p-1)\sigma_s)$ and thus $\|\mathbf{e}_P\| > \lambda^{\omega(1)} \|\mathbf{e}_A^\top \mathbf{E} + \mathbf{s}^\top \mathbf{D}'\|$. This implies $\mathbf{e}_P^\top \approx_s \mathbf{e}_P^\top + \mathbf{e}_A^\top \mathbf{E} + \mathbf{s}^\top \mathbf{D}'$. Therefore, this hybrid is statistically indistinguishable from the previous one.

Hyb₂: In this hybrid, we further replace $\mathbf{c}_P$ and $\mathbf{D}$. Here, we sample $\mathbf{F} \leftarrow \mathbb{Z}_q^{n \times \ell}$ and set $\mathbf{c}_P = \mathbf{s}^\top \mathbf{F} + \mathbf{e}_P^\top$ and $\mathbf{D} = \lfloor \mathbf{F}/p \rfloor \bmod r$. This hybrid is essentially the same as previous hybrid. To see this, it suffices to see that the distribution of $(\mathbf{P} \bmod q, \mathbf{D} \bmod r) = (p\mathbf{D} + \mathbf{D}' \bmod q, \mathbf{D} \bmod r)$ in the previous hybrid is the same as $(\mathbf{F} \bmod q, \lfloor \mathbf{F}/p \rfloor \bmod r)$ in this hybrid.

Hyb₃: This is the D_1^{pre} distribution. Compared with the previous hybrid, here we replace $\mathbf{c}_B$ and $\mathbf{c}_P$ with random vectors with the same dimensions. The indistinguishability from the previous hybrid can be shown by a straightforward reduction to LWE.

Proving that $\mathbf{P}$ is distributed uniformly at random over $\mathbb{Z}_q^{n \times \ell}$. We argue that $\mathbf{P}$ is distributed uniformly at random over $\mathbb{Z}_q^{n \times \ell}$ since so is $\mathbf{A}\mathbf{E}$. The latter can be shown by the leftover hash lemma (Lemma 2.7), which can be applied since the function family $\{\mathbf{e} \mapsto \mathbf{A}\mathbf{e}\}_A$ is 2-universal. $\square$

4 Counter-Examples for Private-Coin Evasive LWE

In this section, we present counter-examples for various versions of private coin Evasive LWE that have appeared in the literature.

4.1 Counter-Example for AKY Assumption

Theorem 4.1. There exists an efficient ELWE sampler Samp as defined in Assumption 2.12 such that the pre-condition holds, but the post-condition does not- i.e., there exists a distinguisher $\mathcal{A}^{\text{post}}$ that distinguishes post-condition with non-negligible probability.

Proof. Our sampler uses a PRF with a contrived circuit implementation as described below.

PRF Circuit Description. Let $\text{PRF} : \{\{1\} \times \{0, 1\}^\lambda\} \times \{0, 1\}^\lambda \rightarrow [-q/4 + B, q/4 - B]^\ell$ where B is chosen to be exponentially smaller than $q/4$ ⁹. Let C_{PRF_r} , with hardwired $\mathbf{r}$, outputting $\text{PRF}(\text{sd}, \mathbf{r})$ (in binary) on input $\text{sd} = (1, \text{sd})$, be any circuit implementing PRF. We construct a modified circuit C'_{PRF_r} implementing PRF as follows.

⁹Note that fixing the first bit of seed to 1 does not affect PRF security, since the actual computation can always ignore the first bit.

1. Let C_0 be a circuit which outputs the first bit of its input $\mathbf{x}$. Further assume that the first bit of $\mathbf{x}$ is always 1.
2. To the output wire of C_{PRF_r} corresponding to the lowest-order bit of $\text{PRF}(\text{sd}, \mathbf{r})[1]$, attach a new gate which performs the correlation-inducing transformation described in Corollary 3.4, using the output wire of C_0 as the special “1” input. Inputs to C_0 can be the bits of sd .
3. To all the other output wires of C_{PRF_r} , attach a gate performing "even randomness" transformation as in Lemma 3.2.

Defining Evasive LWE Sampler. Let $n, m, \ell, Q, q \in \mathbb{N}$ be parameters and λ be the security parameter. The private coin sampler Samp with private coins $\text{coins}_{\text{priv}}^{\text{Samp}} = (\text{sd}, \mathbf{R}, \mathbf{e}_{\text{att}}, \mathbf{e}_{\text{fhe}}, \bar{\mathbf{A}}_{\text{fhe}})$ of the Evasive LWE assumption is defined as follows. The sampler is almost the same as the one used by the previous version of AKY [AKY24a], except that it uses contrived implementation of the PRF circuit. The sampler Samp on input 1^λ does the following:

1. Let $\{\mathcal{F}_{\text{prm}} = \{f : \{0, 1\}^L \rightarrow \{0, 1\}^{1 \times \ell}\}\}_{\text{prm}}$ be a family of functions, where $f \in \mathcal{F}_{\text{prm}}$ can be computed by a circuit of depth $\text{dep}(\lambda) = \text{poly}(\lambda)$ and $\text{prm} = (1^{L(\lambda)}, 1^{\ell(\lambda)}, 1^{\text{dep}(\lambda)})$. Choose $\{f_i \in \mathcal{F}_{\text{prm}}\}_{i \in [Q]}$ and an input vector $\mathbf{x} \in \{0, 1\}^L$ so that

$$(1^\lambda, \{f_i, f_i(\mathbf{x})\}_{i \in [Q]}) \approx_c (1^\lambda, \{f_i, \Delta_i\}_{i \in [Q]}),$$

where $\Delta_i \leftarrow \{0, 1\}^{1 \times \ell}$ for $i \in [Q]$. A concrete example of the choice for such $\mathbf{x}$ and f_i would be to define $\mathbf{x}$ as a random PRF seed and $f_i(\mathbf{x})$ as the PRF value on input i w.r.t the key $\mathbf{x}$.

Define the following circuit C'_{f_i} implementing a scaled version of f_i . That is, C'_{f_i} takes $\mathbf{x} \in \{0, 1\}^L$ as input and outputs $\lfloor q/2 \rfloor f_i(\mathbf{x})$. Start with any circuit C_{f_i} outputting $\lfloor q/2 \rfloor f_i(\mathbf{x})$ (in binary) on input $\mathbf{x}$. To all the output wires of C_{f_i} , attach a gate performing "even randomness" transformation as in Lemma 3.2. We can observe that C'_{f_i} is of depth at most $d + 3$.

2. Samples $\{\mathbf{r}_i \leftarrow \{0, 1\}^\lambda\}_{i \in [Q]}$, $\tilde{\text{sd}} \leftarrow \{0, 1\}^\lambda$ and set $\text{sd} = (1, \tilde{\text{sd}})$.
3. Then it defines a function $F_i = F[f_i, \mathbf{r}_i]$ with $f_i, \mathbf{r}_i$ hardwired as follows: On input $(\mathbf{x}, \text{sd})$, compute and output $f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) \in \mathbb{Z}_q^{1 \times \ell}$. Let C'_{F_i} be a circuit that implements F_i and uses circuits C'_{f_i} and $C'_{\text{PRF}_{\mathbf{r}_i}}$ to compute the respective components, and adds their outputs to get the final output of C'_{F_i} .
4. Samples $\mathbf{A}_{\text{att}} \leftarrow \mathbb{Z}_q^{(n+1) \times (L_X+1)m}$ where $L_X = m(\lambda + L + 1)(n + 1) \lceil \log q \rceil$. Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$, where the homomorphic evaluations of the even randomness gate and the correlation inducing gates are implemented as specified in Lemma 3.2 and Corollary 3.4, respectively. Compute $\mathbf{H}_{\mathbf{A}_{\text{att}}}^{F_i} = \text{MEvalC}(\mathbf{A}_{\text{att}}, \text{VEval}_{C'_{F_i}}) \in \mathbb{Z}_q^{(L_X+1)m \times \ell}$ for all $i \in [Q]$. It then computes $\mathbf{A}_{F_i} = \mathbf{A}_{\text{att}} \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}}^{F_i}$ for all $i \in [Q]$ and sets $\mathbf{P} = (\mathbf{A}_{F_1}, \dots, \mathbf{A}_{F_Q})$.
5. It then samples $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and set $\mathbf{s} = (\bar{\mathbf{s}}^\top, -1)^\top$.
6. Sample $\bar{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{e}_{\text{fhe}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{fhe}}}^m$, $\mathbf{R}_j \leftarrow \{0, 1\}^{m \times m}$ for all $1 \leq j \leq (L + \lambda + 1)$ and compute a GSW encryption as follows.

$$\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe}} \\ \bar{\mathbf{s}}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix}, \quad \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m(\lambda+L+1)}$$

where $\mathbf{R} = (\mathbf{R}_1, \dots, \mathbf{R}_{(\lambda+L+1)})$. Let $L_X = m(\lambda + L + 1)(n + 1) \lceil \log q \rceil$ be the bit length of $\mathbf{X}$.

7. Sample $\mathbf{e}_{\text{att}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{att}}}^{(L_X+1)m}$ and compute $\mathbf{c}_{\text{att}}^\top := \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X}))^{10} \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top$.
8. Output

$$\begin{aligned}\mathbf{S} &= \mathbf{s}^\top \\ \text{aux} &= (\mathbf{X}, \mathbf{c}_{\text{att}}^\top, f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}}) \\ \mathbf{P} &= (\mathbf{A}_{F_1}, \dots, \mathbf{A}_{F_Q}).\end{aligned}$$

Parameter Setting. We set our parameters the same as AKY.

$$\begin{aligned}\tilde{\beta} &= 2^{O(\text{dep} \cdot \log^3 \lambda)}, \quad q \in (2^{8\lambda} \tilde{\beta}, 2^{8\lambda+1} \tilde{\beta}], \quad n = \text{poly}(\lambda, \text{dep}), \quad m = O(n \log q), \\ \tau &= O\left(\sqrt{(n+1) \log q}\right), \quad B = 2^{6\lambda} \tilde{\beta}, \quad \sigma_s = \sigma_{\text{fhe}} = \sigma_{\text{att}} = \sigma = 2^{2\lambda}, \quad \sigma_B = 2^{5\lambda} \tilde{\beta}, \\ \sigma_1 &= 2^{4\lambda+O(1)} \tilde{\beta} / \text{poly}(\lambda), \quad Q = \lambda + mw + (L_X + 1)m\end{aligned}$$

Counter-example Roadmap. Now, we show a counterexample to Evasive LWE (Assumption 2.12) with respect to the sampler specified above. This is proven by showing that the precondition distributions, as defined in Assumption 2.12 are computationally indistinguishable, while the postcondition distributions as defined in Assumption 2.12 are distinguishable.

Concretely, for our counter-example, we need to show that for $(\mathbf{s}^\top, \text{aux}, \mathbf{P}) \leftarrow \text{Samp}(1^\lambda)$, where $\text{aux} = (\mathbf{X}, \mathbf{c}_{\text{att}}^\top, f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}})$ and $(\mathbf{B}, \mathbf{B}_\tau^{-1}) \leftarrow \text{TrapGen}(1^{n+1}, 1^{m \log q}, q)$,

1. (Precondition) For all PPT adversary $\mathcal{A}^{\text{pre}}$, the following two distributions, D_0^{pre} and D_1^{pre} are indistinguishable.

$$D_0^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{s}^\top \mathbf{P} + (\mathbf{e}')^\top, \\ \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right) \approx D_1^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times \ell Q}, \\ \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right),$$

where, $\mathbf{e}_B \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_B}^{m \log q}$, $\mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_1}^{\ell Q}$. We prove this in Lemma 4.4.

2. (Attack against postcondition) There exists an adversary $\mathcal{A}^{\text{post}}$ who distinguishes the following distributions, D_0^{post} and D_1^{post} with non-negligible probability:

$$\begin{aligned}D_0^{\text{post}} &:= \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \mathbf{K}, \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right) \\ D_1^{\text{post}} &:= \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, \mathbf{K}, \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right)\end{aligned}$$

where $\mathbf{e}_B \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_B}^{m \log q}$, $\mathbf{K}_i \leftarrow \mathbf{B}_\tau^{-1}(\mathbf{A}_{F_i})$ and $\mathbf{K} = (\mathbf{K}_1, \dots, \mathbf{K}_Q)$. We prove this in Lemma 4.2.

¹⁰Note that AKY uses $\text{bits}(1, \mathbf{X})$ in place of $(1, \text{bits}(\mathbf{X}))$ to represent the same thing.

Distinguishing Algorithm/Attack Strategy. The adversary $\mathcal{A}^{\text{post}}$ holds the distribution D_β^{post} for $\beta \leftarrow \{0, 1\}$. Thus, for $\beta = 0$, $\mathbf{c}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top$, and for $\beta = 1$, $\mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}$. It does the following:

1. For each $i \in [Q]$, compute $\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} = \text{MEvalCX}(\mathbf{A}_{\text{att}}, \text{VEval}_{C'_{F_i}}(1, \text{bits}(\mathbf{X})))$ and

$$\mathbf{z}_i = \mathbf{c}^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \pmod{q}$$

2. For all $i \in [Q]$ and $j \in [\ell]$, set $y_{i,j} = 0$, if $z_{i,j} \in [-q/4, q/4]$ and $y_{i,j} = 1$ otherwise, where $z_{i,j}$ is the j -th element of $\mathbf{z}_i$. Now it subtracts $\mathbf{y}_i^\top \lfloor q/2 \rfloor$ ¹¹ from $\mathbf{z}_i$ to get

$$\begin{aligned} \forall i \in [Q], \tilde{\mathbf{z}}_i &= \mathbf{z}_i - \mathbf{y}_i^\top \lfloor q/2 \rfloor \pmod{q} \\ &= \mathbf{c}^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} - \mathbf{y}_i^\top \lfloor q/2 \rfloor \pmod{q} \end{aligned}$$

where $\mathbf{y}_i^\top = (y_{i,1}, \dots, y_{i,\ell})$.

3. Let $w = \lceil \log_2 q \rceil$. $\mathcal{A}^{\text{post}}$ defines the following set of linear equations (Equation (20)) in variables $\mathbf{e}_\mathbf{B}[1], \dots, \mathbf{e}_\mathbf{B}[mw]$ and $\mathbf{e}_{\text{att}}[1], \dots, \mathbf{e}_{\text{att}}[(L_X + 1)m]$ and outputs $\beta' = 0$ if (20) is solvable; otherwise, output $\beta' = 1$.

$$\begin{aligned} &\mathbf{e}_\mathbf{B}[1]\mathbf{K}_1[1,1] + \dots + \mathbf{e}_\mathbf{B}[mw]\mathbf{K}_1[mw,1] - \mathbf{e}_{\text{att}}[1]\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_1}[1,1] - \dots \\ &\quad - \mathbf{e}_{\text{att}}[(L_X + 1)m]\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_1}[(L_X + 1)m,1] = \tilde{\mathbf{z}}_1[1] \pmod{2} \\ &\mathbf{e}_\mathbf{B}[1]\mathbf{K}_2[1,1] + \dots + \mathbf{e}_\mathbf{B}[mw]\mathbf{K}_2[mw,1] - \mathbf{e}_{\text{att}}[1]\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_2}[1,1] - \dots \\ &\quad - \mathbf{e}_{\text{att}}[(L_X + 1)m]\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_2}[(L_X + 1)m,1] = \tilde{\mathbf{z}}_2[1] \pmod{2} \quad (20) \\ &\quad \vdots \\ &\mathbf{e}_\mathbf{B}[1]\mathbf{K}_Q[1,1] + \dots + \mathbf{e}_\mathbf{B}[mw]\mathbf{K}_Q[mw,1] - \mathbf{e}_{\text{att}}[1]\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_Q}[1,1] - \dots \\ &\quad - \mathbf{e}_{\text{att}}[(L_X + 1)m]\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_Q}[(L_X + 1)m,1] = \tilde{\mathbf{z}}_Q[1] \pmod{2} \end{aligned}$$

Analyzing success probability of $\mathcal{A}^{\text{post}}$.

Lemma 4.2. $\mathcal{A}^{\text{post}}$ wins (i.e., $\beta = \beta'$) with $3/4 - \text{negl}(\lambda)$ probability when $Q \geq \lambda + mw + (L_X + 1)m$.

Proof. Let us first analyze the solvability of equations defined in (20) for the two cases: $\beta = 0$ and $\beta = 1$.

When $\beta = 0$,

$$\begin{aligned} \forall i \in [Q], \mathbf{z}_i &= \mathbf{c}^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \\ &= (\mathbf{s}^\top \mathbf{B} + \mathbf{e}_\mathbf{B}^\top) \cdot \mathbf{K}_i - (\mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top) \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \\ &= \mathbf{s}^\top \mathbf{B} \mathbf{K}_i + \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i - \mathbf{s}^\top \mathbf{A}_{\text{att}} \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} + \mathbf{s}^\top \text{VEval}_{C'_{F_i}}(\text{bits}(\mathbf{X})) - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \quad (21) \\ &= \mathbf{s}^\top \mathbf{A}_{F_i} + \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i - \mathbf{s}^\top \mathbf{A}_{F_i} + F_i(\mathbf{x}, \text{sd}) - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \\ &= \mathbf{e}_\mathbf{B}^\top \mathbf{K}_i + f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \end{aligned}$$

where $\mathbf{R}_{C'_{F_i}}$ is the randomness in the homomorphically evaluated encryption of $F_i(\mathbf{x}, \text{sd})$. Let the depth of C_{F_i} be $d = \text{poly}(\lambda)$ and from Lemma 2.14, the depth of $\text{VEval}_{C_{F_i}}$ is

$$d_{\text{VEval}_{C_{F_i}}} = (dO(\log m \log \log q) + O(\log^2 \log q)).$$

¹¹observe that $\mathbf{y}_i^\top = f_i(\mathbf{x})$ in case of $\beta = 0$.

The depth of circuit C'_{F_i} increases at most by constant $O(1)$ from d due to the addition of even randomness and correlation-inducing gate, and hence, it is $d + O(1)$. The depth of $\text{VEval}_{C'_{F_i}}$ is

$$d_{\text{VEval}_{C'_{F_i}}} = ((d + O(1))O(\log m \log \log q) + O(\log^2 \log q)).$$

From Lemma 2.14 and parameter setting, we have

$$\begin{aligned} \|\mathbf{R}_{C'_{F_i}}\| &\leq (m+2)^{d+O(1)} \lceil \log q \rceil \max_{\ell \in [L+\lambda+1]} \|\mathbf{R}_\ell^\top\| \\ &\leq (m+2)^{d+O(1)} \lceil \log q \rceil m \\ &\leq (m+2)^{d+O(1)} \lceil \log q \rceil \leq \tilde{\beta} \end{aligned}$$

and using the depth bound as in Section 2.5, we have

$$\|\mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}\| \leq (m+2)^{d_{\text{VEval}_{C'_{F_i}}}} \lceil \log q \rceil \leq \tilde{\beta}$$

Now, from parameter setting, we have

$$\|\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} + \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}\| \leq 2^{2\lambda+1} \sqrt{\lambda} \tilde{\beta} \leq 2^{3\lambda} \tilde{\beta}$$

and

$$\|\mathbf{e}_{\mathbf{B}}^\top \mathbf{K}_i - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}\| \leq 2^{5\lambda+1} \sqrt{\lambda} \tilde{\beta} + 2^{3\lambda} \tilde{\beta} < 2^{6\lambda} \tilde{\beta} < B$$

Hence, $\|\text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{\mathbf{B}}^\top \mathbf{K}_i - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}\| < (q/4 - B) + B = q/4$, $\mathcal{A}^{\text{post}}$ will correctly recover $f_i(\mathbf{x})$ i.e. $\mathbf{y}_i^\top = f_i(\mathbf{x})$ and hence, we will have

$$\begin{aligned} \forall i \in [Q], \\ \tilde{\mathbf{z}}_i &= f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{\mathbf{B}}^\top \mathbf{K}_i - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} - f_i(\mathbf{x}) \lfloor q/2 \rfloor \\ &= \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{\mathbf{B}}^\top \mathbf{K}_i - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}} - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \in [-q/4, q/4] \quad \text{over integer} \end{aligned}$$

We prove in Claim 4.3 that $\forall i \in [Q]$, $\text{PRF}(\text{sd}, \mathbf{r}_i)[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}}[1] = 0 \pmod{2}$ with probability $\geq 1/2$. This gives us that with probability $\geq 1/2$,

$$\forall i \in [Q], \tilde{\mathbf{z}}_i[1] = \mathbf{e}_{\mathbf{B}}^\top \mathbf{K}_i[1] - \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}[1] \pmod{2}.$$

Observe that this is the same as the set of equations defined in (20). Hence, in case of $\beta = 0$, (20) is solvable with probability $1/2 + \text{negl}(\lambda)$. This gives us

$$\Pr(\beta' = 0 \mid \beta = 0) = 1/2 + \text{negl}(\lambda). \quad (22)$$

When $\beta = 1$,

$$\forall i \in [Q], \mathbf{z}_i = \mathbf{c}^\top \cdot \mathbf{K}_i - \mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}$$

where $\mathbf{c} \leftarrow \mathbb{Z}_q^{mw}$.

Here, we argue that the distribution of $\{\mathbf{c}_{\mathbf{B}}^\top \cdot \mathbf{K}_i \pmod{q}\}_{i \in [Q]}$ is statistically close to uniform distribution over $(\mathbb{Z}_q^{1 \times \ell})^Q$, even given $\{\mathbf{K}_i \pmod{2}\}_{i \in [Q]}$. To see this, we first observe that by Lemma 2.6, each column of $\mathbf{K}_i[\cdot, j]$ has min-entropy at least $mw \log(\sigma_{-1}/\lambda) > 2mw$ except for negligible probability over $\mathbf{B}$ and $(\mathbf{K}_i[\cdot, j] \pmod{2})$ can be represented by a string of mw -bits. We then apply the generalized Leftover Hash Lemma Lemma 2.7, which implies that the two distributions are within statistical distance $Q\sqrt{2^{mw} \cdot 2^{-2mw} q} = \text{negl}(\lambda)$ by our choice of parameters. We therefore have that $\{\mathbf{z}_i \pmod{q}\}_{i \in [Q]}$

is distributed uniformly at random over $(\mathbb{Z}_q^{1 \times \ell})^Q$, even given $\{\mathbf{K}_i \bmod 2\}_i$. This, in turn, implies $\{\mathbf{z}_i \bmod 2\}_{i \in [Q]}$ (which forms the RHS of equations in Equation (20)) is also statistically close to a uniform distribution independent of $\{\mathbf{K}_i \bmod 2\}_i$ when $\beta = 1$. Hence, Equation (20), having $mw + (\mathbf{L}_X + 1)m$ number of variables, is unsolvable with overwhelming probability if $Q > \lambda + mw + (\mathbf{L}_X + 1)m$, since the RHS of the equation falls into the space for which there exists a solution with probability at most $2^{mw + (\mathbf{L}_X + 1)m} / 2^{\lambda + mw + (\mathbf{L}_X + 1)m} = 2^{-\lambda}$.

This gives us

$$\Pr(\beta' = 1 \mid \beta = 1) \geq 1 - \text{negl}(\lambda). \quad (23)$$

Thus, from Equation (22) and (23), we get $\Pr(\beta' = \beta) \geq 3/4 - \text{negl}(\lambda)$. $\square$

Claim 4.3. $\forall i \in [Q]$, $\text{PRF}(\text{sd}, \mathbf{r}_i)[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}}[1] = 0 \bmod 2$ with probability $1/2 + \text{negl}(\lambda)$.

Proof. Let us start by analyzing the $\tilde{\mathbf{e}}_{\text{fhe}}^\top[1] = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}}[1]$ error. Recall that $\mathbf{R}_{C'_{F_i}}$ is the randomness in the homomorphically evaluated encryption of $F_i(\mathbf{x}, \text{sd})$ (of different form: $\mathbf{A}_{\text{fhe}} \mathbf{R}_{C'_{F_i}} - \begin{pmatrix} 0^{n \times \ell} \\ F_i(\mathbf{x}, \text{sd}) \end{pmatrix}$), output by $\text{VEval}_{C'_{F_i}}$ circuit. Since $F_i(\mathbf{x}, \text{sd}) = f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i)$, we can write $\tilde{\mathbf{e}}_{\text{fhe}}^\top[1] = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{F_i}}[1] = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{f_i}}[1] + \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{\mathbf{r}_i}}}[1]$ because the additive property is preserved after homomorphic addition.

We recall from Section 2.4 that for any circuit $C : \{0, 1\}^L \rightarrow \mathbb{Z}_q^{1 \times \ell}$, VEval_C on input $\mathbf{X}$ - a GSW encryption of some $\mathbf{x} \in \{0, 1\}^L$ - outputs $\mathbf{A}_{\text{fhe}} \mathbf{R}_C - \begin{pmatrix} 0^{n \times \ell} \\ C(\mathbf{x}) \end{pmatrix}$. Let $C(\mathbf{x})[u, v] \in \{-1, 0, 1\}$ be (the signed version of) the v -th bit of $C(\mathbf{x})[u] \in \mathbb{Z}_q$, for $u \in [\ell]$ and $v \in [0, w]$, $w = \lceil \log_2 q \rceil$, then VEval_C computation firstly homomorphically computes GSW encryption of $C(\mathbf{x})[u, v]$, which is of the form $\mathbf{C}_{u,v} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C,u,v} - C(\mathbf{x})[u, v] \mathbf{G}$. Then for each $u \in [\ell]$, it linearly aggregates $\{\mathbf{C}_{u,v}\}_{v \in [0, w-1]}$ to get $\mathbf{C}_u = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C,u} - \begin{pmatrix} 0^{n \times 1} \\ C(\mathbf{x})[u] \end{pmatrix}$ as $\mathbf{C}_u = \sum_v \mathbf{C}_{u,v} \mathbf{G}^{-1}(2^v \iota_{n+1})$. In particular,

$$\mathbf{R}_{C,u} = \sum_v \mathbf{R}_{C,u,v} \mathbf{G}^{-1}(2^v \iota_{n+1}), \text{ and } \mathbf{R}_C = (\mathbf{R}_{C,1}, \dots, \mathbf{R}_{C,\ell}).$$

Coming back to our analysis, $(\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{f_i}})[1] = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{f_i},1} = \mathbf{e}_{\text{fhe}}^\top \sum_{v=0}^{w-1} \mathbf{R}_{C'_{f_i},1,v} \mathbf{G}^{-1}(2^v \iota_{n+1})$, where $\mathbf{R}_{C'_{f_i},1}$ is the first column of $\mathbf{R}_{C'_{f_i}}$ and $\mathbf{R}_{C'_{f_i},1,v}$ is the randomness in homomorphically computed GSW encryption of the v -th bit of $f_i(\mathbf{x}) \lfloor q/2 \rfloor [1]$. Since $f_i(\mathbf{x}) \lfloor q/2 \rfloor$ is implemented by C'_{f_i} , which involves the even randomness transformation (Lemma 3.2) on all its output bits, $\mathbf{R}_{C'_{f_i},1,v}$ has all the entries as even due to Lemma 3.2. This implies $\mathbf{R}_{C'_{f_i},1}$ has even entries, which in turn implies that $(\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{f_i}})[1]$ is even.

So, now our goal is to show that

$$\text{PRF}(\text{sd}, \mathbf{r}_i)[1] - (\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{\mathbf{r}_i}}})[1] = 0 \bmod 2$$

Similar to above, we have $(\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{\mathbf{r}_i}}})[1] = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{\mathbf{r}_i}},1} = \mathbf{e}_{\text{fhe}}^\top \sum_{v=0}^{w-1} \mathbf{R}_{C'_{\text{PRF}_{\mathbf{r}_i}},1,v} \mathbf{G}^{-1}(2^v \iota_{n+1})$.

Again, since in $C'_{\text{PRF}_{\mathbf{r}_i}}$ all the output bits pass through even randomness, except the lowest order bit of

the first entry, which passes through the correlation inducing gate, we have

$$\begin{aligned}
& \text{PRF}(\text{sd}, \mathbf{r}_i)[1] - (\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{r_i}}})[1] \bmod 2 \\
&= \text{PRF}(\text{sd}, \mathbf{r}_i)[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{r_i}}, 1, 0} \mathbf{G}^{-1}(\iota_{n+1}) - \mathbf{e}_{\text{fhe}}^\top \sum_{v=1}^{w-1} \mathbf{R}_{C'_{\text{PRF}_{r_i}}, 1, v} \mathbf{G}^{-1}(2^v \iota_{n+1}) \bmod 2 \\
&= \text{PRF}(\text{sd}, \mathbf{r}_i)[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{r_i}}, 1, 0} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \quad (\text{due to Lemma 3.2}) \\
&= \text{PRF}(\text{sd}, \mathbf{r}_i)[1, 0] + \sum_{v=1}^{w-2} 2^v \text{PRF}(\text{sd}, \mathbf{r}_i)[1, v] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{r_i}}, 1, 0} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\
&= \text{PRF}(\text{sd}, \mathbf{r}_i)[1, 0] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{r_i}}, 1, 0} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\
&= \text{PRF}(\text{sd}, \mathbf{r}_i)[1, 0] - \text{PRF}(\text{sd}, \mathbf{r}_i)[1, 0] \mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\
&\quad (\text{due to Corollary 3.4}) \\
&= 0 \bmod 2, \text{ if } \mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \text{ is odd.}
\end{aligned}$$

In the above, $\text{PRF}(\text{sd}, \mathbf{r}_i)[u, v]$ represents the v -th bit in the binary representation of $\text{PRF}(\text{sd}, \mathbf{r}_i)[u]$ and $\mathbf{R}^*$ is the randomness in the ciphertext of special "1" input used in Corollary 3.4. We then argue that $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1})$ is odd with probability negligibly close to $1/2$. To see this, we first observe that each entry of $\mathbf{R}^*$ is distributed over $\{0, 1\}$ uniformly at random and thus so is each entry of $\mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2$. In particular, this implies that $\mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2$ is a non-zero vector with overwhelming probability. We then observe that each entry of $\mathbf{e}_{\text{fhe}} \bmod 2$ is statistically close to the uniform distribution over $\{0, 1\}$ by Lemma 2.9. Combining these facts, it follows that the distribution of $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2$ is statistically close to the uniform distribution over $\{0, 1\}$, as desired. Note that the term $\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1})$ does not depend on i and hence, for all $i \in [Q]$, the probability that $\text{PRF}(\text{sd}, \mathbf{r}_i)[1] - \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C'_{\text{PRF}_{r_i}}}[1] = 0 \bmod 2$ is $1/2 + \text{negl}(\lambda)$. Hence, the proof. $\square$

Proving the pre-condition. We now show that the pre-condition of ELWE holds with respect to the above sampler defined in Section 4.1.

Lemma 4.4. For $(\mathbf{s}^\top, \text{aux}, \mathbf{P}) \leftarrow \text{Samp}(1^\lambda)$, where $\text{aux} = (\mathbf{X}, \mathbf{c}_{\text{att}}^\top, f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}})$,

$$D_0^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_{\mathbf{B}}^\top, (\mathbf{c}')^\top = \mathbf{s}^\top \mathbf{P} + (\mathbf{e}')^\top, \\ \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right) \approx D_1^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times \ell Q}, \\ \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right)$$

where Samp is as defined in Section 4.1, $\mathbf{e}_{\mathbf{B}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\mathbf{B}}}^{m \log q}$ and $\mathbf{e}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_1}^{\ell Q}$.

Proof. To prove this, we introduce one intermediate distribution $D_{\text{inter}}^{\text{pre}}$ as

$$D_{\text{inter}}^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \mathbf{P}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, (\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times \ell Q}, \\ \mathbf{X} \leftarrow \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}, \\ \mathbf{c}_{\text{att}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_X+1)m}, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right)$$

and then prove that $D_0^{\text{pre}} \approx D_{\text{inter}}^{\text{pre}}$ and $D_{\text{inter}}^{\text{pre}} \approx D_1^{\text{pre}}$. This implies, $D_0^{\text{pre}} \approx D_1^{\text{pre}}$.

Proof of $D_0^{\text{pre}} \approx D_{\text{inter}}^{\text{pre}}$

Note that the distribution D_0^{pre} and $D_{\text{inter}}^{\text{pre}}$ is exactly the same as the L.H.S and R.H.S distribution

(respectively) in equation (4) of the security proof of Functional Encryption for pseudorandom functionality (PRFE) construction as in the previous version of [AKY24a]. Hence, almost the same security proof of PRFE applies to the proof of Lemma 4.4. However, for completeness, we provide the full proof here. Since, $(\mathbf{c}')^\top = \mathbf{s}^\top \mathbf{P} + (\mathbf{e}')^\top = (\mathbf{s}^\top \mathbf{A}_{F_1} + (\mathbf{e}_{P,1})^\top, \dots, \mathbf{s}^\top \mathbf{A}_{F_Q} + (\mathbf{e}_{P,Q})^\top) = \{\mathbf{s}^\top \mathbf{A}_{F_i} + (\mathbf{e}_{P,i})^\top\}_{i \in [Q]} = \{\mathbf{c}_{P,i}^\top\}_{i \in [Q]}$, it suffices to prove the following Equation (24).

$$D_0^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \{\mathbf{A}_{F_i}\}_{i \in [Q]}, \mathbf{c}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \{\mathbf{c}_{P,i} = \mathbf{s}^\top \mathbf{A}_{F_i} + (\mathbf{e}'_i)^\top\}_{i \in [Q]}, \\ \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \\ \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right) \quad (24)$$

$$\approx D_{\text{inter}}^{\text{pre}} := \left(\begin{array}{l} \mathbf{B}, \{\mathbf{A}_{F_i}\}_{i \in [Q]}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, \{\mathbf{c}_{P,i} \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]}, \\ \mathbf{X} \leftarrow \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}, \\ \mathbf{c}_{\text{att}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_X+1)m}, \\ f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \end{array} \right)$$

where $\mathbf{e}'_i \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_1}^\ell$ for all $i \in [Q]$. We prove Equation (24) via the following sequence of hybrids.

Hyb₀. This is L.H.S distribution of Equation (24).

Hyb₁. This hybrid is same as Hyb₀, except we compute $\mathbf{c}_{P,i}^\top$ as

$$\mathbf{c}_{P,i}^\top = \mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} + f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{P,i}^\top$$

where $\mathbf{e}_{P,i} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_1}^\ell$. We claim that Hyb₁ and Hyb₂ are statistically indistinguishable. To see this, note that from Equation (21), we have :

$$\mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} = \mathbf{s}^\top \mathbf{A}_{F_i} - f_i(\mathbf{x}) \lfloor q/2 \rfloor - \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{F_i}^\top$$

where $\mathbf{e}_{F_i}^\top = \mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C_{F_i}'} + \mathbf{e}_{\text{att}}^\top \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i}$. This implies,

$$\mathbf{c}_{P,i}^\top = \mathbf{c}_{\text{att}}^\top \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} + f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{P,i}^\top = \mathbf{s}^\top \mathbf{A}_{F_i} + \mathbf{e}_{F_i}^\top + \mathbf{e}_{P,i}^\top$$

where $\|\mathbf{e}_{F_i}^\top\| \leq 2^{3\lambda} \tilde{\beta}$ as proved in Lemma 4.2.

Next, we note that $\|\mathbf{e}_{F_i}^\top\| \leq 2^{4\lambda+O(1)} \beta / \text{poly}(\lambda) = \|\mathbf{e}_{P,i}\|$. Thus by noise flooding (Lemma 2.8) we have $\mathbf{e}_{F_i}^\top + \mathbf{e}_{P,i}^\top \approx_s \mathbf{e}_{P,i}^\top$ with a statistical distance of $\text{negl}(\lambda)$. Thus, it suffices to show the pseudorandomness of the following distribution given $f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}}$.

$$\left(\begin{array}{l} \mathbf{B}, \{\mathbf{A}_{F_i}\}_{i \in [Q]}, \mathbf{c}^\top = \mathbf{s}^\top \mathbf{B} + \mathbf{e}_B^\top, \{\tilde{F}_i = f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{P,i}^\top\}_{i \in [Q]}, \\ \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top \end{array} \right)$$

Hyb₂. This hybrid is same as Hyb₁ except that, we sample $\mathbf{c} \leftarrow \mathbb{Z}_q^{mw}$, $\mathbf{c}_{\text{att}} \leftarrow \mathbb{Z}_q^m$ and $\mathbf{A}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{(n+1) \times m}$, where $\mathbf{A}_{\text{fhe}}$ is the fhe public key used to compute $\mathbf{X}$. We have $\text{Hyb}_1 \approx_c \text{Hyb}_2$ using LWE.

To prove this, we show that if there exists an adversary $\mathcal{A}$ who can distinguish between the two hybrids with non-negligible advantage, then there is a reduction $\mathcal{B}$ that breaks LWE security with non-negligible advantage. The reduction is as follows.

1. On receiving security parameter from $\mathcal{A}$, forward it to LWE challenger.

2. The LWE challenger samples $\beta \leftarrow \{0, 1\}$. It also samples $\mathbf{A}_{\text{LWE}} \leftarrow \mathbb{Z}_q^{n \times (mw+m+(L_X+1)m)}$, $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and sets $\mathbf{b}^\top = \bar{\mathbf{s}}^\top \mathbf{A}_{\text{LWE}} + \mathbf{e}_{\text{fhe}}^\top \in \mathbb{Z}_q^{1 \times (mw+m+(L_X+1)m)}$ if $\beta = 0$, else it samples $\mathbf{b}^\top \leftarrow \mathbb{Z}_q^{1 \times (mw+m+(L_X+1)m)}$ if $\beta = 1$. Sends $(\mathbf{A}_{\text{LWE}}, \mathbf{b})$ to $\mathcal{B}$.
3. $\mathcal{B}$ parses $\mathbf{A}_{\text{LWE}} = (\mathbf{B}', \hat{\mathbf{A}}_{\text{fhe}}, \mathbf{A}'_{\text{att}})$, where $\mathbf{B}' \in \mathbb{Z}_q^{n \times mw}$, $\hat{\mathbf{A}}_{\text{fhe}} \in \mathbb{Z}_q^{n \times m}$, $\mathbf{A}'_{\text{att}} \in \mathbb{Z}_q^{n \times ((L_X+1)m)}$ and $\mathbf{b}^\top = (\mathbf{b}_{\mathbf{B}}^\top, \mathbf{b}_{\text{fhe}}^\top, \mathbf{b}_{\text{att}}^\top)$.
4. Choose $\{f_i \in \mathcal{F}_{\text{prm}}\}_{i \in [Q]}$ and an input vector $\mathbf{x} \in \{0, 1\}^L$.
5. Samples $\{\mathbf{r}_i \leftarrow \{0, 1\}^\lambda\}_{i \in [Q]}$, $\tilde{\text{sd}} \leftarrow \{0, 1\}^\lambda$ and set $\text{sd} = (1, \tilde{\text{sd}})$.
6. Samples $\hat{\mathbf{b}} \leftarrow \mathbb{Z}_q^{mw}$ and sets $\mathbf{B} = \begin{pmatrix} \mathbf{B}' \\ \hat{\mathbf{b}}^\top \end{pmatrix}$ and $\mathbf{c}^\top := \mathbf{b}_{\mathbf{B}}^\top - \hat{\mathbf{b}}^\top$.
7. Sets $\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \hat{\mathbf{A}}_{\text{fhe}} \\ \mathbf{b}_{\text{fhe}}^\top \end{pmatrix}$ and computes $\mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}$ as in the construction.
8. Sets $\bar{\mathbf{A}}_{\text{att}} = \mathbf{A}'_{\text{att}} + (1, \text{bits}(\mathbf{X})) \otimes \bar{\mathbf{G}}$, $\mathbf{A}_{\text{att}} = \begin{pmatrix} \bar{\mathbf{A}}_{\text{att}} \\ \mathbf{a}_{\text{att}}^\top \end{pmatrix}$, where $\mathbf{a}_{\text{att}} \leftarrow \mathbb{Z}_q^{(L_X+1)m}$, and $\mathbf{c}_{\text{att}}^\top = \mathbf{b}_{\text{att}}^\top - (\mathbf{a}_{\text{att}}^\top - (1, \text{bits}(\mathbf{X})) \otimes \underline{\mathbf{G}})$, where $\bar{\mathbf{G}}$ and $\underline{\mathbf{G}}$ denotes the first n rows and $n+1$ -th row of the gadget matrix $\mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m}$, respectively.
9. Compute $\tilde{\mathbf{F}}_i$ and $\{\mathbf{A}_{\mathbf{F}_i}\}_{i \in [Q]}$ as in hybrid 1.
10. Sends $(\mathbf{B}, \{\mathbf{A}_{\mathbf{F}_i}\}_{i \in [Q]}, \mathbf{c}^\top, \tilde{\mathbf{F}}_i, \mathbf{X}, \mathbf{c}_{\text{att}}^\top, f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}})$ to $\mathcal{A}$.
11. $\mathcal{A}$ outputs a bit β' . $\mathcal{B}$ forwards the bit β' to the LWE challenger.

It is easy to see that if LWE challenger sent $\mathbf{b}^\top = \bar{\mathbf{s}}^\top \mathbf{A}_{\text{LWE}} + \mathbf{e}_{\text{LWE}}^\top$, then $\mathcal{B}$ simulated Hyb_1 with $\mathcal{A}$ else if LWE challenger sent random $\mathbf{b} \leftarrow \mathbb{Z}_q^{mw+m+(L_X+1)m}$ then $\mathcal{B}$ simulated Hyb_2 with $\mathcal{A}$.

Thus, it suffices to show the pseudorandomness of the following distribution given $f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}}$.

$$\left(\mathbf{B}, \{\mathbf{A}_{\mathbf{F}_i}\}_{i \in [Q]}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, \{\tilde{\mathbf{F}}_i = f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{\mathbf{P},i}^\top\}_{i \in [Q]}, \right. \\ \left. \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G}, \mathbf{c}_{\text{att}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_X+1)m} \right)$$

Hyb_3 . This hybrid is same as Hyb_2 except that we sample $\mathbf{X} \leftarrow \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}$. We have $\text{Hyb}_2 \approx_s \text{Hyb}_3$ using leftover hash lemma. We now argue that $\mathbf{A}_{\text{fhe}} \mathbf{R}$ is statistically close to a uniform distribution over $\mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}$. This can be seen by noting that, for a uniformly sampled vector of length m where the entries are sampled from $\{0, 1\}$, the min-entropy is m . This implies the min-entropy of each column of $\mathbf{R}$ is m . Furthermore, by the Leftover Hash Lemma (Lemma 2.7), each entry of $\mathbf{A}_{\text{fhe}} \mathbf{R}$ is statistically close to being uniformly distributed over $\mathbb{Z}_q$, with statistical distance $\text{negl}(\lambda)$. This implies that the same holds for $\mathbf{X}$. Thus, it suffices to show the pseudorandomness of the following distribution given $f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}}$.

$$\left(\mathbf{B}, \{\mathbf{A}_{\mathbf{F}_i}\}_{i \in [Q]}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, \{\tilde{\mathbf{F}}_i = f_i(\mathbf{x}) \lfloor q/2 \rfloor + \text{PRF}(\text{sd}, \mathbf{r}_i) + \mathbf{e}_{\mathbf{P},i}^\top\}_{i \in [Q]}, \right. \\ \left. \mathbf{X} \leftarrow \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}, \mathbf{c}_{\text{att}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_X+1)m} \right)$$

Hyb_4 . This hybrid is the same as the previous one except that we replace $\text{PRF}(\text{sd}, \cdot)$ with the real random function $\mathbf{R}(\cdot)$. Since sd is not used anywhere else, we can use the security of PRF to conclude that this hybrid is computationally indistinguishable from the previous one.

Hyb₅. This hybrid is same as the previous one except that we output a failure symbol if the set $\{\mathbf{r}_i\}_{i \in [Q]}$, in aux, contains a collision. We prove that the probability with which there occurs a collision is negligible in λ . To prove this it suffices to show that there is no $i, i' \in [Q]$ such that $i \neq i'$ and $\mathbf{r}_i = \mathbf{r}_{i'}$. The probability of this happening can be bounded by $Q^2/2^\lambda$ by taking the union bound with respect to all the combinations of i, i' . Thus, the probability of outputting the failure symbol is $Q^2/2^\lambda$, which is $\text{negl}(\lambda)$.

Hyb₆. In this hybrid we compute $\tilde{F}_i$ as

$$\tilde{F}_i = f_i(\mathbf{x}) \lfloor q/2 \rfloor + R_i + \mathbf{e}_{\mathbf{P},i}^\top$$

for all $i \in [Q]$. Namely, we use fresh randomness $R_i \leftarrow [-q/4 + B, q/4 - B]^{1 \times \ell}$ instead of deriving the randomness by $R(\mathbf{r}_i)$. We claim that this change is only conceptual. To see this, we observe that unless the failure condition introduced in Hyb₅ is satisfied, every invocation of the function R is with respect to a fresh input, and thus, the output can be replaced with a fresh randomness.

Thus, it suffices to show the pseudorandomness of the following distribution given $f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}}$.

$$\left(\mathbf{B}, \{\mathbf{A}_{F_i}\}_{i \in [Q]}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, \{\tilde{F}_i = f_i(\mathbf{x}) \lfloor q/2 \rfloor + \mathbf{R}_i + \mathbf{e}_{\mathbf{P},i}^\top\}_{i \in [Q]}, \right. \\ \left. \mathbf{X} \leftarrow \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}, \mathbf{c}_{\text{att}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_X+1)m} \right)$$

Hyb₇. This hybrid is same as the previous one except we sample $\tilde{F}_i \leftarrow \mathbb{Z}_q^\ell$ for $i \in [Q]$. This follows from the pseudorandomness of $\{f_i(\mathbf{x})\}_i$ and the fact that B is exponentially smaller than $q/4$. To see this note that we have

$$(1^\lambda, \{f_i, f_i(\mathbf{x})\}_{i \in [Q]}) \approx_c (1^\lambda, \{f_i, \Delta_i \leftarrow \{0,1\}^{1 \times \ell}\}_{i \in [Q]})$$

which implies

$$(1^\lambda, \{f_i, \tilde{F}_i = f_i(\mathbf{x}) \lfloor q/2 \rfloor + R_i + \mathbf{e}_{\mathbf{P},i}^\top\}_{i \in [Q]}) \approx_c (1^\lambda, \{f_i, \tilde{F}_i \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]})$$

Hence, we achieve the following distribution

$$\left(\mathbf{B}, \{\mathbf{A}_{F_i}\}_{i \in [Q]}, \mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}, \{\tilde{F}_i \leftarrow \mathbb{Z}_q^{1 \times \ell}\}_{i \in [Q]}, \mathbf{X} \leftarrow \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}, \right. \\ \left. \mathbf{c}_{\text{att}}^\top \leftarrow \mathbb{Z}_q^{1 \times (L_X+1)m}, f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}} \right)$$

which is the R.H.S distribution of Equation (24), hence the proof.

Proof of $D_{\text{inter}}^{\text{pre}} \approx D_1^{\text{pre}}$

We prove $D_{\text{inter}}^{\text{pre}} \approx D_1^{\text{pre}}$ via the following sequence of hybrids.

Hyb₀. This is L.H.S distribution i.e. $D_{\text{inter}}^{\text{pre}}$.

Hyb₁. This hybrid is same as Hyb₀ except that we compute $\mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \text{sd}) \otimes \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}$, where $\mathbf{A}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{(n+1) \times m}$. We have $\text{Hyb}_0 \approx_s \text{Hyb}_1$ using leftover hash lemma. We can argue that the uniform distribution $\mathbb{Z}_q^{(n+1) \times (m(\lambda+L+1))}$ is statistically close to $\mathbf{A}_{\text{fhe}} \mathbf{R}$ using the same argument as used to prove $\text{Hyb}_2 \approx \text{Hyb}_3$ in proof of $D_0^{\text{pre}} \approx D_{\text{inter}}^{\text{pre}}$. Hence, we skip the indistinguishability argument.

Hyb₂. This hybrid is same as Hyb₁ except that we compute $\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \hat{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m} \\ \mathbf{b}_{\text{fhe}}^\top = \bar{\mathbf{s}}^\top \mathbf{A}_{\text{LWE}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix}$ and $\mathbf{c}_{\text{att}}^\top = \mathbf{s}^\top (\mathbf{A}_{\text{att}} - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G}) + \mathbf{e}_{\text{att}}^\top$. To prove this, we show that if there exists an adversary $\mathcal{A}$ who can distinguish between the two hybrids with non-negligible advantage, then there is a reduction $\mathcal{B}$ that breaks LWE security with non-negligible advantage. The reduction is as follows.

1. On receiving security parameter from $\mathcal{A}$, forward it to LWE challenger.
2. The LWE challenger samples $\beta \leftarrow \{0, 1\}$. It also samples $\mathbf{A}_{\text{LWE}} \leftarrow \mathbb{Z}_q^{n \times (m + (L_X + 1)m)}$, $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and sets $\mathbf{b}^\top = \bar{\mathbf{s}}^\top \mathbf{A}_{\text{LWE}} + \mathbf{e}_{\text{fhe}}^\top \in \mathbb{Z}_q^{1 \times (m + (L_X + 1)m)}$ if $\beta = 1$, else it samples $\mathbf{b}^\top \leftarrow \mathbb{Z}_q^{1 \times (m + (L_X + 1)m)}$ if $\beta = 0$. Sends $(\mathbf{A}_{\text{LWE}}, \mathbf{b})$ to $\mathcal{B}$.
3. $\mathcal{B}$ parses $\mathbf{A}_{\text{LWE}} = (\hat{\mathbf{A}}_{\text{fhe}}, \mathbf{A}'_{\text{att}})$, where $\hat{\mathbf{A}}_{\text{fhe}} \in \mathbb{Z}_q^{n \times m}$, $\mathbf{A}'_{\text{att}} \in \mathbb{Z}_q^{n \times (L_X + 1)m}$ and $\mathbf{b}^\top = (\mathbf{b}_{\text{fhe}}^\top, \mathbf{b}_{\text{att}}^\top)$.
4. Choose $\{f_i \in \mathcal{F}_{\text{prm}}\}_{i \in [Q]}$ and an input vector $\mathbf{x} \in \{0, 1\}^L$.
5. Samples $\{\mathbf{r}_i \leftarrow \{0, 1\}^\lambda\}_{i \in [Q]}$, $\tilde{\mathbf{s}} \leftarrow \{0, 1\}^\lambda$ and set $\mathbf{sd} = (1, \tilde{\mathbf{s}})$.
6. Sets $\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \hat{\mathbf{A}}_{\text{fhe}} \\ \mathbf{b}_{\text{fhe}}^\top \end{pmatrix}$ and computes $\mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R} - (\mathbf{x}, \mathbf{sd}) \otimes \mathbf{G}$ as in the construction.
7. Sets $\bar{\mathbf{A}}_{\text{att}} = \mathbf{A}'_{\text{att}} + (1, \text{bits}(\mathbf{X})) \otimes \bar{\mathbf{G}}$, $\mathbf{A}_{\text{att}} = \begin{pmatrix} \bar{\mathbf{A}}_{\text{att}} \\ \mathbf{a}_{\text{att}}^\top \end{pmatrix}$, where $\mathbf{a}_{\text{att}} \leftarrow \mathbb{Z}_q^{(L_X + 1)m}$, and $\mathbf{c}_{\text{att}}^\top = \mathbf{b}_{\text{att}}^\top - (\mathbf{a}_{\text{att}}^\top - (1, \text{bits}(\mathbf{X})) \otimes \mathbf{G})$, where $\bar{\mathbf{G}}$ and $\mathbf{G}$ denotes the first n rows and $n + 1$ -th row of the gadget matrix $\mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m}$, respectively.
8. Sample $\mathbf{B} \leftarrow \mathbb{Z}_q^{(n+1) \times m \log q}$, $\mathbf{c}^\top \leftarrow \mathbb{Z}_q^{1 \times m \log q}$, $(\mathbf{c}')^\top \leftarrow \mathbb{Z}_q^{1 \times \ell_Q}$.
9. Compute $\{\mathbf{A}_{F_i}\}_{i \in [Q]}$ as in hybrid 1.
10. Sends $(\mathbf{B}, \{\mathbf{A}_{F_i}\}_{i \in [Q]}, \mathbf{c}^\top, (\mathbf{c}')^\top, \mathbf{X}, \mathbf{c}_{\text{att}}^\top, f_1, \dots, f_Q, \mathbf{r}_1, \dots, \mathbf{r}_Q, \mathbf{A}_{\text{att}})$ to $\mathcal{A}$.
11. $\mathcal{A}$ outputs a bit β' . $\mathcal{B}$ forwards the bit β' to the LWE challenger.

It is easy to see that if LWE challenger sent random $\mathbf{b} \leftarrow \mathbb{Z}_q^{m + (L_X + 1)m}$ then $\mathcal{B}$ simulated Hyb₁ with $\mathcal{A}$, else if LWE challenger sent $\mathbf{b}^\top = \bar{\mathbf{s}}^\top \mathbf{A}_{\text{LWE}} + \mathbf{e}_{\text{LWE}}^\top$, then $\mathcal{B}$ simulated Hyb₂ with $\mathcal{A}$.

Hence, $D_{\text{inter}}^{\text{pre}} \approx D_1^{\text{pre}}$. Hence, the proof. □

□

□

4.2 Counter-Example for BDJMPV Assumption

Assumption 4.5 (Evasive LWE [BDJ⁺25]). We recall the version of evasive LWE assumption considered in BDJMPV, which is adapted from [MPV24b] and [BUW24]. Let $m, n, k, \kappa > 0$ be integers and let q be a modulus. Let $\tau, \sigma, \sigma' > 0$. Let Samp be an algorithm which takes 1^λ and a matrix $\mathbf{P} \in \mathbb{Z}_q^{k \times n}$ and outputs a matrix $\mathbf{S} \in \mathbb{Z}_q^{n \times \kappa}$ and auxiliary information aux. Let

$$\begin{aligned} \mathbf{D} &\leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}, \quad \mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}, \quad \mathbf{P} = \mathbf{D} \cdot \mathbf{B}, \quad (\mathbf{S}, \text{aux}) \leftarrow \text{Samp}(1^\lambda, \mathbf{P}), \\ \mathbf{E} &\leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}, \quad \mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}, \quad \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \quad \mathbf{C}' \leftarrow \mathbb{Z}_q^{k \times \kappa} \end{aligned}$$

For PPT distinguishers, $\mathcal{A}^{\text{pre}}$ and $\mathcal{A}^{\text{post}}$ define the following functions:

$$\text{Adv}_{\mathcal{A}_0}^{\text{pre}}(\lambda) = |\Pr[\mathcal{A}_0(\mathbf{B}, \mathbf{P}, \mathbf{B}\mathbf{S} + \mathbf{E}, \mathbf{P}\mathbf{S} + \mathbf{E}', \text{aux}) = 1] - \Pr[\mathcal{A}_0(\mathbf{B}, \mathbf{P}, \mathbf{C}, \mathbf{C}', \text{aux}) = 1]|$$

$$\text{Adv}_{\mathcal{A}_1}^{\text{post}}(\lambda) = |\Pr[\mathcal{A}_1(\mathbf{B}, \mathbf{P}, \mathbf{B}\mathbf{S} + \mathbf{E}, \mathbf{D}, \text{aux}) = 1] - \Pr[\mathcal{A}_1(\mathbf{B}, \mathbf{P}, \mathbf{C}, \mathbf{D}, \text{aux}) = 1]|$$

We say that the evasive LWE assumption $\text{evLWE}(q, m, n, k, \kappa, \text{Samp}, \tau, \sigma, \sigma')$ holds, if for every PPT distinguisher $\mathcal{A}_1$, there exists a PPT distinguisher $\mathcal{A}_0$ and polynomial $Q(\lambda)$ such that

$$\text{Adv}_{\mathcal{A}_0}^{\text{pre}}(\lambda) \geq \text{Adv}_{\mathcal{A}_1}^{\text{post}}(\lambda)/Q(\lambda) - \text{negl}(\lambda)$$

and $\text{time}(\mathcal{A}_0) \leq \text{time}(\mathcal{A}_1) \cdot Q(\lambda)$.

Remark 4.6. In this section, we stick to the notation used in [BDJ⁺25], where $\mathbf{B}$ is of size $m \times n$ and thus “tall” matrix, whereas $\mathbf{B}$ is a “wide” matrix of size $n \times m$ in other sections.

Remark 4.7. We remark that this version of evasive LWE is different from that in [AKY24a] (considered in Section 4.1). In this version the matrix $\mathbf{P}$ is a random matrix and cannot be controlled by the sampler as it is provided to the sampler as an input. However, aux may contain contrived terms which are also related to $\mathbf{P}$ and thus can still leak some information making it vulnerable to attacks. Indeed, we exploit such a term in aux in our counterexample.

Theorem 4.8. There exists an efficient ELWE sampler Samp as defined in Assumption 4.5 such that the pre-condition holds, but the post-condition does not- i.e., there exists a distinguisher $\mathcal{A}^{\text{post}}$ that distinguishes the post-condition with non-negligible probability.

The sampler used for proving the theorem is almost the same as the one used in the construction of exponentially efficient doubly pseudorandom obfuscation scheme (xdPRO) of [BDJ⁺25], except that it uses a contrived implementation of the PRF and f (as defined below) circuit and set q to be an odd number rather than power of two.

Proof. Our sampler uses a PRF with a contrived circuit implementation, as described below.

PRF Circuit Description. Let $\text{PRF} : \{\{1\} \times \{0, 1\}^\lambda\} \times \{0, 1\}^{\log(\kappa)} \rightarrow [-q/4 + \hat{B}, q/4 - \hat{B}]^k$ where $\hat{B}$ is chosen to be exponentially smaller than $q/4$. Let C_{PRF_i} , with hardwired i , outputting $\text{PRF}(K', i)$ (in binary) on input $K' = (1, K'')$, be any circuit implementing PRF. We construct a modified circuit C'_{PRF_i} implementing PRF as follows.

1. Let C_0 be a circuit that outputs the first bit of its input $\mathbf{x}$. Further, assume that the first bit of $\mathbf{x}$ is always 1.
2. To the output wire of C_{PRF_i} corresponding to the lowest-order bit of $\text{PRF}(K', i)[j]$ for all $j \in [k]$, attach a new gate which performs the correlation-inducing transformation described in Corollary 3.4, using the output wire of C_0 as the special “1” input. Inputs to C_0 can be the bits of K' .
3. To all the other output wires of C_{PRF_i} , attach a gate performing “even randomness” transformation as in Lemma 3.2.

Defining Evasive LWE Sampler. Let $n, m, k, \kappa, \eta \in \mathbb{N}$ be parameters, q be an odd number, and λ be the security parameter. They will be set after the description of the sampler below. The private coin sampler Samp with private coins $\text{coins}_{\text{priv}}^{\text{Samp}} = (K'', \mathbf{R}, \bar{\mathbf{E}}, \mathbf{S}, \bar{\mathbf{s}}, \{\mathbf{R}_{\text{fhe}, \ell}\}_{\ell \in [\eta+1+\lambda]}, \bar{\mathbf{A}}_{\text{fhe}}, \mathbf{e}_{\text{fhe}})$ of the Evasive LWE assumption as in Assumption 4.5 is defined as follows. The sampler Samp on input $(1^\lambda, \mathbf{P})$, where $\mathbf{P} \in \mathbb{Z}_q^{k \times n}$ does the following:

1. Let $\{\mathcal{F}_{\text{prm}} = \{f : \{0, 1\}^\eta \times \{0, 1\}^{\log(\kappa)} \rightarrow \{0, 1\}^k\}\}_{\text{prm}}$ be a family of functions, where $f \in \mathcal{F}_{\text{prm}}$ can be computed by a circuit of depth $d(\lambda) = \text{poly}(\lambda)$ and $\text{prm} = (1^{\eta(\lambda)}, 1^{k(\lambda)}, 1^{\kappa(\lambda)}, 1^{d(\lambda)})$. Let KeySamp be a sampling algorithm that, on input 1^λ , outputs key $K \in \{0, 1\}^\eta$ and auxiliary information aux_K dependent on K . Choose $f \in \mathcal{F}_{\text{prm}}$ so that

$$(\{f(K, i)\}_{i \in [k]}, \text{aux}_K) \approx_c (\{\mathbf{u}_i : \mathbf{u}_i \leftarrow \{0, 1\}^k\}_{i \in [k]}, \text{aux}_K),$$

over the random choice of $(K, \text{aux}_K) \leftarrow \text{KeySamp}(1^\lambda)$. A concrete example of such KeySamp is an algorithm that chooses random PRF key K and sets $\text{aux}_K = \perp$. The function $f(K, i)$ is defined to be a PRF value with respect to the key K and input i .

For all $i \in [\kappa]$, let C_{f_i} , with hardwired i , outputting scaled version of $f_i(K) = f(K, i)$, i.e. outputting $\lfloor q/2 \rfloor f_i(K)$ (in binary) on input K , be any circuit implementing f_i . We construct a modified circuit C'_{f_i} implementing f_i as follows. To all the output wires of C_{f_i} , attach a gate performing "even randomness" transformation as in Lemma 3.2. We can observe that C'_{f_i} is of depth at most $d + 3$.

2. Sample $K'' \leftarrow \{0, 1\}^\lambda$ and set $K' = (1, K'')$ and $\bar{K} = (K, K')$.
3. Then it defines a function F_i with i hardwired as follows: On input $\bar{K}$, compute and output $f(K, i) \lfloor q/2 \rfloor + \text{PRF}(K', i) \in \mathbb{Z}_q^k$. Let C'_{F_i} be a circuit that implements F_i and uses circuits C'_{f_i} and C'_{PRF_i} to compute the respective components and adds their outputs to get the final output of C'_{F_i} .
4. Sample $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and set $\mathbf{s} = (\bar{\mathbf{s}}^\top, -1)^\top$.
5. Sample $\bar{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{e}_{\text{fhe}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{fhe}}}^m$, $\mathbf{R}_{\text{fhe}, \ell} \leftarrow \{0, 1\}^{m \times m}$ for all $\ell \in [\eta + \lambda + 1]$ and compute a GSW encryption as follows.

$$\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe}} \\ \bar{\mathbf{s}}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix}, \quad \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (\bar{K}) \otimes \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m(\lambda + \eta + 1)}$$

where $\mathbf{R}_{\text{fhe}} = (\mathbf{R}_{\text{fhe}, 1}, \dots, \mathbf{R}_{\text{fhe}, (\lambda + \eta + 1)})$.

6. Parse $\mathbf{s}^\top = (s_1, \dots, s_n, -1) \in \mathbb{Z}_q^{1 \times n+1}$, sample $\mathbf{R} \leftarrow \mathbb{Z}_q^{n \times (k+1)m}$ and $\bar{\mathbf{E}} \leftarrow \mathcal{D}_{\mathbb{Z}, \rho}^{k \times (k+1)m}$ and compute $\mathbf{A} = \mathbf{P}\mathbf{R} + \bar{\mathbf{E}} + (\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{k \times (k+1)m}$ where $\mathbf{0} \in \mathbb{Z}_q^k$ is all zero column vector and $\mathbf{I}_k$ is the identity matrix with k rows and columns.
7. For $i \in [\kappa]$

- Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$, where the homomorphic evaluations of the even randomness gate and the correlation inducing gates are implemented as specified in Lemma 3.2 and Corollary 3.4, respectively. Furthermore, the homomorphic evaluation corresponding to the addition of the output of C'_{f_i} and C'_{PRF_i} in Step 3 is done by simple addition of the GSW ciphertexts.

- Compute $\mathbf{X}_{C'_{F_i}} = \text{VEval}_{C'_{F_i}}(\mathbf{X})$. Thus, $\mathbf{X}_{C'_{F_i}} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C'_{F_i}} - \begin{pmatrix} 0^{n \times k} \\ C'_{F_i}(\bar{K}) \end{pmatrix} \in \mathbb{Z}_q^{(n+1) \times k}$, where $\mathbf{R}_{C'_{F_i}}$ is the randomness after homomorphic evaluation.

- Let $\mathbf{f}_i \in \mathbb{Z}_q^{k(n+1)}$ be the vectorization of $\mathbf{X}_{C'_{F_i}}$ such that $(\mathbf{I}_k \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top) \mathbf{G}^{-1}(\mathbf{f}_i) \approx C'_{F_i}(\bar{K})$. That is,

$$\mathbf{f}_i = \left(\mathbf{X}_{C'_{F_i}}[1, 1], \dots, \mathbf{X}_{C'_{F_i}}[n+1, 1], \mathbf{X}_{C'_{F_i}}[1, 2], \dots, \mathbf{X}_{C'_{F_i}}[n+1, 2], \dots, \mathbf{X}_{C'_{F_i}}[1, k], \dots, \mathbf{X}_{C'_{F_i}}[n+1, k] \right)^\top.$$

- Sample $\mathbf{r}_i \leftarrow \{0, 1\}^m$.

8. Set $\mathbf{F} = (\mathbf{G}_{\mathbf{r}_1}^{-1}(\mathbf{f}_1), \dots, \mathbf{G}_{\mathbf{r}_\kappa}^{-1}(\mathbf{f}_\kappa)) \in \{0, 1\}^{(k+1)m \times \kappa}$. Here $\mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) = \begin{pmatrix} \tilde{\mathbf{f}}_i \\ \mathbf{r}_i \end{pmatrix}$ where $\tilde{\mathbf{f}}_i \in \mathbb{Z}_q^{k(n+1) \log q}$ is the unique vector with $(\mathbf{I}_k \otimes \mathbf{G}) \tilde{\mathbf{f}}_i = \mathbf{f}_i$.

9. Sample $\mathbf{S} \leftarrow \mathbb{Z}_q^{n \times \kappa}$ and set $\mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}$.
10. Output $(\mathbf{S}, \text{aux})$ where $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$.

Parameter Setting. We set our parameters the same as [BDJ⁺25], except that we explicitly define $\tilde{\beta}$, $\tilde{B}$ and explicitly set κ and k .

$$\begin{aligned} \tilde{\beta} &= 2^{O(d \cdot \log^3 \lambda)}, \quad \tilde{B} = \lambda \sigma_{\text{fhe}} \tilde{\beta} m, \quad \hat{B} = \tilde{B} + \lambda m \tau \sigma + \lambda m(k+1)\rho, \\ \sigma &= \sigma' = 2^\lambda (\tilde{B} + \lambda(k+1)m\rho), \quad q \in (2^\lambda \kappa k \hat{B}, 2^{\lambda+1} \kappa k \hat{B}], \quad n = \text{poly}(\lambda), \\ m &= (n+1) \lceil \log_2 q \rceil, \quad k = m^2, \quad \kappa = m^4 \end{aligned}$$

Counter-example Roadmap. Now, for our counter-example, we need to show that for $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$, $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$, $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$, $(\mathbf{S}, \text{aux}) \leftarrow \text{Samp}(1^\lambda, \mathbf{P})$ where $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$,

1. (Attack against postcondition) There exists an adversary who distinguishes the following distributions, D_0^{post} and D_1^{post} with non-negligible probability.

$$D_0^{\text{post}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}, \mathbf{D}, \text{aux}) \text{ and } D_1^{\text{post}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \mathbf{D}, \text{aux}),$$

where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$.

We begin with Step 1. To prove this step, we introduce one extra distribution D_2^{post} as

$$D_2^{\text{post}} := (\mathbf{B}, \mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}, \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \mathbf{D}, \text{aux} = (\mathbf{H}, \text{aux}'))$$

where $(\mathbf{B}, \mathbf{B}_\tau^{-1}) \leftarrow \text{TrapGen}(1^m, 1^n, q)$, $\mathbf{D} \leftarrow \mathbf{B}_\tau^{-1}(\mathbf{P})$,¹² $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$ and $\text{aux}' = (\mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$, and then prove that,

- (a) for all PPT adversary, the two distributions, D_1^{post} and D_2^{post} are statistically indistinguishable. We prove this in Claim 4.9.
- (b) there exists an adversary $\mathcal{A}^{\text{post}}$ who distinguishes the distributions, D_0^{post} and D_2^{post} with non-negligible probability. We prove this in Lemma 4.10.

This implies the adversary $\mathcal{A}^{\text{post}}$ distinguishes D_0^{post} and D_1^{post} with non-negligible probability.

2. (Precondition) For all PPT adversary $\mathcal{A}^{\text{pre}}$, the following two distributions, D_0^{pre} and D_1^{pre} are indistinguishable.

$$D_0^{\text{pre}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}, \mathbf{C}' = \mathbf{P}\mathbf{S} + \mathbf{E}', \text{aux}) \approx D_1^{\text{pre}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \mathbf{C}' \leftarrow \mathbb{Z}_q^{k \times \kappa}, \text{aux}),$$

where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$, $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$. We prove this in Lemma 4.12.

We start by showing the following useful claim.

¹²Here, we overload the notation and $\mathbf{B}_\tau^{-1}(\mathbf{P})$ denotes a distribution over short $\mathbf{D}$ such that $\mathbf{D}\mathbf{B} = \mathbf{P}$, rather than $\mathbf{B}\mathbf{D} = \mathbf{P}$. This change is introduced due to our choice of the size of $\mathbf{B}$ in this section, where we choose $\mathbf{B}$ to be a tall matrix rather than the wide matrix.

Proving $D_1^{\text{post}} \approx_s D_2^{\text{post}}$. We show below that $D_1^{\text{post}} \approx_s D_2^{\text{post}}$ holds with respect to the above sampler defined in Section 4.2.

Claim 4.9. For $(\mathbf{S}, \text{aux}) \leftarrow \text{Samp}(1^\lambda, \mathbf{P})$ where $\text{aux} = (\mathbf{H}, \text{aux}')$, $\text{aux}' = (\mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$,

$$D_1^{\text{post}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \mathbf{D}, \mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}, \text{aux}')$$

$$\approx_s D_2^{\text{post}} := (\mathbf{B}, \mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}, \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \mathbf{D}, \mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}, \text{aux}'),$$

where in D_1^{post} , $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$, $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$, $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$, $\mathbf{R} \leftarrow \mathbb{Z}_q^{n \times (k+1)m}$ and in D_2^{post} , $(\mathbf{B}, \mathbf{B}_\tau^{-1}) \leftarrow \text{TrapGen}(1^m, 1^n, q)$, $\mathbf{D} \leftarrow \mathbf{B}_\tau^{-1}(\mathbf{P})$.

Proof. We prove Claim 4.9 using a series of hybrids $\text{Hyb}_0, \dots, \text{Hyb}_3$ and proving that $\text{Hyb}_0 \approx \text{Hyb}_1 \approx \dots \approx \text{Hyb}_3$, where Hyb_0 is the D_1^{post} distribution and Hyb_3 is the D_2^{post} distribution.

Hyb₁: This is identical to Hyb_0 except that we sample matrix $\mathbf{H}$ as $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$. In Hyb_0 , $\mathbf{H}$ is computed by $\mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}$ for a uniformly random $\mathbf{S} \leftarrow \mathbb{Z}_q^{n \times \kappa}$ that is not used anywhere else. Therefore, Hyb_0 and Hyb_1 are identically distributed.

Hyb₂: This is identical to Hyb_1 except that instead of uniformly sampling $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$, we compute it as $(\mathbf{B}, \mathbf{B}_\tau^{-1}) \leftarrow \text{TrapGen}(1^m, 1^n, q)$. Therefore, the statistical indistinguishability between Hyb_1 and Hyb_2 follows from Lemma 2.3.

Hyb₃: This is identical to Hyb_2 except that instead of first sampling $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$ and setting $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$, we uniformly sample $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$ and set $\mathbf{D} \leftarrow \mathbf{B}_\tau^{-1}(\mathbf{P})$. Therefore, the statistical indistinguishability between Hyb_2 and Hyb_3 follows from the trapdoor sampling property (Section 2.3). Note that this distribution is the same as D_2^{post} .

□

Distinguishing Algorithm/Attack Strategy for D_0^{post} and D_2^{post} . The adversary $\mathcal{A}^{\text{post}}$ holds the distribution D_0^{post} if $\beta = 0$ and D_2^{post} if $\beta = 1$ for $\beta \leftarrow \{0, 1\}$. Thus, for $\beta = 0$, $\mathbf{P} = \mathbf{D}\mathbf{B}$, $\mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}$, $\mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}$ and for $\beta = 1$, $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$, $\mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}$, $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$. It does the following:

1. For $i \in [\kappa]$

- Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$.
- Compute $\mathbf{X}_{C'_{F_i}} = \text{VEval}_{C'_{F_i}}(\mathbf{X}) = \mathbf{A}_{\text{fhe}}\mathbf{R}_{C'_{F_i}} - \begin{pmatrix} 0^{n \times k} \\ C'_{F_i}(\bar{K}) \end{pmatrix} \in \mathbb{Z}_q^{n+1 \times k}$.
- Compute $\mathbf{f}_i \in \mathbb{Z}_q^{k(n+1)}$ as the vectorization of $\mathbf{X}_{C'_{F_i}}$ i.e.

$$\mathbf{f}_i = \left(\mathbf{X}_{C'_{F_i}}[1, 1], \dots, \mathbf{X}_{C'_{F_i}}[n+1, 1], \mathbf{X}_{C'_{F_i}}[1, 2], \dots, \mathbf{X}_{C'_{F_i}}[n+1, 2], \dots, \mathbf{X}_{C'_{F_i}}[1, k], \dots, \mathbf{X}_{C'_{F_i}}[n+1, k] \right)^T$$

2. Compute $\mathbf{F} = (\mathbf{G}_{\mathbf{r}_1}^{-1}(\mathbf{f}_1), \dots, \mathbf{G}_{\mathbf{r}_\kappa}^{-1}(\mathbf{f}_\kappa))$ where $\{\mathbf{r}_i\}_{i \in [\kappa]} \in \text{aux}$.

3. Compute $\mathbf{Y} = \mathbf{A}\mathbf{F} + \mathbf{D}\mathbf{C} - \mathbf{P}\mathbf{H} \in \mathbb{Z}_q^{k \times \kappa}$ and parse $\mathbf{Y} = (\mathbf{y}_1, \dots, \mathbf{y}_\kappa)$ column-wise.

4. For $i \in [\kappa]$, compute $\mathbf{z}_i = \mathbf{y}_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}_i)^{13}$.

¹³Observe that $\text{MSB}(\mathbf{y}_i) = f(K, i)$ when $\beta = 0$

5. $\mathcal{A}^{\text{post}}$ defines the following set of linear equations (Equation (25)), where $\bar{\mathbf{E}}, \mathbf{E}$ are unknown variables (i.e., there are $k(k+1)m + m\kappa$ variables). It outputs $\beta' = 0$ if Equation (25) is solvable; otherwise, output $\beta' = 1$.

$$\left\{ \begin{aligned} &\bar{\mathbf{E}}[j, 1]\mathbf{F}[1, i] + \dots + \bar{\mathbf{E}}[j, (k+1)m]\mathbf{F}[(k+1)m, i] + \mathbf{D}[j, 1]\mathbf{E}[1, i] + \dots + \mathbf{D}[j, m]\mathbf{E}[m, i] \\ &= \mathbf{z}_i[j] \bmod 2 \end{aligned} \right\}_{i \in [\kappa], j \in [k]} \quad (25)$$

We then analyze the success probability of $\mathcal{A}^{\text{post}}$.

Lemma 4.10. $\mathcal{A}^{\text{post}}$ wins (i.e., $\beta = \beta'$) with probability $3/4 - \text{negl}(\lambda)$ when $k(k+1)m + m\kappa + \lambda < k\kappa$.

Proof. Let us first analyze the solvability of equations defined in Equation (25) for the two cases: $\beta = 0$ and $\beta = 1$.

The Case of $\beta = 0$. When $\beta = 0$, for all $i \in [\kappa]$, it holds that

$$\begin{aligned} &((\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top) \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) \\ = &\begin{pmatrix} (s_1 \mathbf{g}^\top, \dots, s_n \mathbf{g}^\top, -\mathbf{g}^\top) & 0^m & \dots & 0^m & 0^m \\ 0^m & (s_1 \mathbf{g}^\top, \dots, s_n \mathbf{g}^\top, -\mathbf{g}^\top) & \dots & 0^m & 0^m \\ \vdots & \vdots & \dots & \vdots & \vdots \\ 0^m & 0^m & \vdots & (s_1 \mathbf{g}^\top, \dots, s_n \mathbf{g}^\top, -\mathbf{g}^\top) & 0^m \end{pmatrix}_{k \times (k+1)m} \begin{pmatrix} \tilde{\mathbf{f}}_i \\ \mathbf{r}_i \end{pmatrix}_{(k+1)m \times 1} \\ = &\begin{pmatrix} (s_1 \mathbf{g}^\top \tilde{\mathbf{f}}_i[1; \log q] + \dots + s_n \mathbf{g}^\top \tilde{\mathbf{f}}_i[(n-1) \log q + 1; n \log q] - \mathbf{g}^\top \tilde{\mathbf{f}}_i[n \log q + 1; m]) \\ \vdots \\ (s_1 \mathbf{g}^\top \tilde{\mathbf{f}}_i[(k-1)m + 1; (k-1)m + 1 + \log q] + \dots \\ + s_n \mathbf{g}^\top \tilde{\mathbf{f}}_i[km - 2 \log q + 01; km - \log q] - \mathbf{g}^\top \tilde{\mathbf{f}}_i[km - \log q + 1; km]) \end{pmatrix}_{k \times 1} \\ = &\begin{pmatrix} (s_1 \mathbf{f}_i[1] + \dots + s_n \mathbf{f}_i[n] - \mathbf{f}_i[n+1]) \\ \vdots \\ (s_1 \mathbf{f}_i[(k-1)(n+1) + 1] + \dots \\ + s_n \mathbf{f}_i[k(n+1) - 1] - \mathbf{f}_i[k(n+1)]) \end{pmatrix}_{k \times 1} \quad (\text{the equality follows from the property of } \mathbf{G}_{\mathbf{r}_i}^{-1}) \\ = &\begin{pmatrix} \mathbf{s}^\top \mathbf{X}_{C_{\mathbf{F}_i}}'[\cdot, 1] \\ \vdots \\ \mathbf{s}^\top \mathbf{X}_{C_{\mathbf{F}_i}}'[\cdot, k] \end{pmatrix}_{k \times 1} = \begin{pmatrix} C_{\mathbf{F}_i}'(\bar{K})[1] - \mathbf{e}_{\text{fhe}}^\top(\mathbf{R}_{C_{\mathbf{F}_i}'}[\cdot, 1]) \\ \vdots \\ C_{\mathbf{F}_i}'(\bar{K})[k] - \mathbf{e}_{\text{fhe}}^\top(\mathbf{R}_{C_{\mathbf{F}_i}'}[\cdot, k]) \end{pmatrix}_{k \times 1} = \mathbf{F}(\bar{K}, i) - \tilde{\mathbf{e}}_i \quad (\text{by correctness of GSW}) \end{aligned}$$

where $\tilde{\mathbf{f}}_i[\ell; \ell']$ is the part of vector $\tilde{\mathbf{f}}_i$ starting from ℓ -th element and ending at ℓ' -th element and

$\tilde{\mathbf{e}}_i = \begin{pmatrix} \mathbf{e}_{\text{fhe}}^\top(\mathbf{R}_{C_{\mathbf{F}_i}'}[\cdot, 1]) \\ \vdots \\ \mathbf{e}_{\text{fhe}}^\top(\mathbf{R}_{C_{\mathbf{F}_i}'}[\cdot, k]) \end{pmatrix}$ is the GSW decryption noise terms. Here, $\mathbf{s}^\top \mathbf{X}_{C_{\mathbf{F}_i}'}[\cdot, j]$ denotes the j -th column

of $\mathbf{s}^\top \mathbf{X}_{C_{\mathbf{F}_i}'}'$ with $n+1$ rows. Therefore, for all $i \in [\kappa]$,

$$\begin{aligned} \mathbf{A} \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) &= (\mathbf{P} \mathbf{R} + \bar{\mathbf{E}} + ((\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top)) \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) \\ &= \mathbf{P} \mathbf{R} \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) + \bar{\mathbf{E}} \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) + ((\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top) \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) \\ &= \mathbf{P} \mathbf{R} \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) + \bar{\mathbf{E}} \mathbf{G}_{\mathbf{r}_i}^{-1}(\mathbf{f}_i) + \mathbf{F}(\bar{K}, i) - \tilde{\mathbf{e}}_i \end{aligned}$$

Hence, it holds that

$$\mathbf{AF} = \mathbf{PRF} + \bar{\mathbf{E}}\mathbf{F} + (F(\bar{K}, 1), \dots, F(\bar{K}, \kappa)) - \tilde{\mathbf{E}}$$

where $\tilde{\mathbf{E}} = (\tilde{\mathbf{e}}_1, \dots, \tilde{\mathbf{e}}_\kappa)$. Also, $\mathbf{DC} = \mathbf{D}(\mathbf{BS} + \mathbf{E}) = \mathbf{DBS} + \mathbf{DE} = \mathbf{PS} + \mathbf{DE}$ and $\mathbf{PH} = \mathbf{P}(\mathbf{S} + \mathbf{RF}) = \mathbf{PS} + \mathbf{PRF}$. Hence, we have

$$\mathbf{Y} = \mathbf{AF} + \mathbf{DC} - \mathbf{PH} = (F(\bar{K}, 1), \dots, F(\bar{K}, \kappa)) - \tilde{\mathbf{E}} + \bar{\mathbf{E}}\mathbf{F} + \mathbf{DE} \quad (26)$$

Let $\hat{\mathbf{E}} = -\tilde{\mathbf{E}} + \bar{\mathbf{E}}\mathbf{F} + \mathbf{DE} = (\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_\kappa)$. Note that since $\bar{\mathbf{E}} \leftarrow \mathcal{D}_{\mathbb{Z}, \rho}^{k \times (k+1)m}$, for each column $\bar{\mathbf{E}}[\cdot, i]$ of $\bar{\mathbf{E}}$, we have $\|\bar{\mathbf{E}}[\cdot, i]\| \leq \sqrt{\lambda}\rho$. Therefore, $\|\bar{\mathbf{E}} \cdot \mathbf{f}_i\| \leq \sqrt{\lambda}(k+1)m\rho$. Similarly, since $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$, for each column $\mathbf{E}[\cdot, i]$ of $\mathbf{E}$, $\|\mathbf{E}[\cdot, i]\| \leq \sqrt{\lambda}\sigma$ and every row $\mathbf{D}[j, \cdot]$ of $\mathbf{D}$, $\|\mathbf{D}[j, \cdot]\| \leq \sqrt{\lambda}\tau$. Therefore, $\|\mathbf{D} \cdot \mathbf{E}[\cdot, i]\| \leq \lambda m \tau \sigma$.

Now for the error bound $\tilde{\mathbf{e}}_i$, we have the following. From Lemma 2.14 and parameter setting, we have,

$$\begin{aligned} \|\mathbf{R}_{C_{F_i}}\| &\leq (m+2)^{d+O(1)} \lceil \log q \rceil \max_{\ell \in [\eta+1+\lambda]} \|\mathbf{R}_\ell^\top\| \\ &\leq (m+2)^{d+O(1)} \lceil \log q \rceil m \\ &\leq (m+2)^{d+O(1)} \lceil \log q \rceil \leq \tilde{\beta} \end{aligned}$$

Now, from the parameter setting, we have

$$\begin{aligned} \|\tilde{\mathbf{e}}_i\| &\leq \max_{j \in [k]} \|\mathbf{e}_{\text{fhe}}^\top(\mathbf{R}_{C_{F_i}}[\cdot, j])\| \\ &\leq \|\mathbf{e}_{\text{fhe}}^\top \mathbf{R}_{C_{F_i}}\| \\ &\leq \sqrt{\lambda} \sigma_{\text{fhe}} \tilde{\beta} m \leq \tilde{B}. \end{aligned}$$

Therefore, by setting

$$\hat{B} = \tilde{B} + \lambda m \tau \sigma + \lambda m (k+1) \rho, \text{ we get } \|\hat{\mathbf{e}}_i\| \leq \hat{B}.$$

Now, from Equation (26), we have for $i \in [\kappa]$, $\mathbf{y}_i = \lfloor q/2 \rfloor f(K, i) + \mathbf{PRF}(K', i) + \hat{\mathbf{e}}_i$. Since, $\mathbf{PRF}(K', i) \in [-q/4 + \hat{B}, q/4 - \hat{B}]^k$, we have $\mathbf{PRF}(K', i) + \hat{\mathbf{e}}_i \in [-q/4, q/4]^k$. Hence, we have the following.

$$\text{MSB}(\mathbf{y}_i) = \text{MSB}(\lfloor q/2 \rfloor f(K, i) + \mathbf{PRF}(K', i) + \hat{\mathbf{e}}_i) = f(K, i)$$

Now, for $i \in [\kappa]$, $\mathbf{z}_i = \mathbf{y}_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}_i) = \mathbf{PRF}(K', i) + \hat{\mathbf{e}}_i = \mathbf{PRF}(K', i) - \tilde{\mathbf{E}}[\cdot, i] + \bar{\mathbf{E}}\mathbf{F}[\cdot, i] + \mathbf{DE}[\cdot, i]$. where $\tilde{\mathbf{E}}[\cdot, i], \bar{\mathbf{E}}\mathbf{F}[\cdot, i], \mathbf{DE}[\cdot, i]$ denotes the i th column of $\tilde{\mathbf{E}}, \bar{\mathbf{E}}\mathbf{F}, \mathbf{DE}$ respectively. We prove in Claim 4.11 that $\mathbf{PRF}(K', i)[j] - \tilde{\mathbf{E}}[j, i] = 0 \bmod 2$ for all $i \in [\kappa], j \in [k]$ with probability $1/2 + \text{negl}(\lambda)$. This gives us that with probability $1/2 + \text{negl}(\lambda)$,

$$\forall i \in [\kappa], j \in [k], \bar{\mathbf{E}}\mathbf{F}[j, i] + \mathbf{DE}[j, i] = \mathbf{z}_i[j] \bmod 2 \quad (27)$$

We can rewrite Equation (27) as Equation (28) below for all $i \in [\kappa], j \in [k]$.

$$\bar{\mathbf{E}}[j, 1]\mathbf{F}[1, i] + \dots + \bar{\mathbf{E}}[j, (k+1)m]\mathbf{F}[(k+1)m, i] + \mathbf{D}[j, 1]\mathbf{E}[1, i] + \dots + \mathbf{D}[j, m]\mathbf{E}[m, i] = \mathbf{z}_i[j] \bmod 2 \quad (28)$$

Observe that this is the same as the set of equations defined in Equation (25). Hence, in case of $\beta = 0$, Equation (25) is solvable with probability $1/2 + \text{negl}(\lambda)$. This gives us

$$\Pr(\beta' = 0 \mid \beta = 0) = 1/2 + \text{negl}(\lambda). \quad (29)$$

The Case of $\beta = 1$. When $\beta = 1$, for all $i \in [\kappa]$, we have,

$$\mathbf{z}_i = \mathbf{y}_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}_i) = \mathbf{AF}[\cdot, i] + \mathbf{DC}[\cdot, i] - \mathbf{PH}[\cdot, i] - \lfloor q/2 \rfloor \text{MSB}(\mathbf{AF}[\cdot, i] + \mathbf{DC}[\cdot, i] - \mathbf{PH}[\cdot, i]) \quad (30)$$

where $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$, $\mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}$, and $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$. We argue that Equation (25) is not solvable with overwhelming probability in this case. It would have been easy to show this, if $\{\mathbf{z}_i \bmod 2\}_{i \in [\kappa]}$ was distributed uniformly at random independent from $\mathbf{D}$ and $\mathbf{F}$. However, we are not able to prove that $\{\mathbf{z}_i \bmod 2\}_{i \in [\kappa]}$ is random. Instead, we prove that $\{\mathbf{z}_i \bmod 2\}_i$ is pseudorandom given $\mathbf{D}$ and $\mathbf{F}$, which is sufficient for our purpose. Concretely, we consider the following sequence of hybrids. We denote the probability that Equation (25) is not solvable in Hyb_{xx} by $\Pr[\mathbf{E}_{\text{xx}}]$.

Hyb₁: This is the same as D_2^{post} except that we only consider $(\mathbf{D}, \mathbf{F}, \{\mathbf{z}_i \bmod 2\}_i)$, where $\{\mathbf{z}_i\}_i$ is defined as in Equation (30) and other elements such as $\mathbf{A}$ are discarded. Clearly, discarding other elements does not change the probability of Equation (25) being not solvable. We want to prove $\Pr[\mathbf{E}_1] = 1 - \text{negl}(\lambda)$.

Hyb₂: In this hybrid, we change the definition of $\mathbf{z}_i$ as

$$\mathbf{z}_i = \mathbf{y}_i - 2\mathbf{e}'_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}_i - 2\mathbf{e}'_i) \bmod q$$

for freshly chosen $\mathbf{e}'_i \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^k$. We argue that the distribution of $\{\mathbf{z}_i \bmod 2\}_i$ is statistically close to that of the previous hybrid. To see this, we first observe that for any $y \in \mathbb{Z}_q$ outside of the “boundary” $[-q/4 - B, -q/4 + B] \cup [q/4 - B, q/4 + B]$, we have $\text{MSB}(y) = \text{MSB}(y - 2e')$ for any e' such that $|e'| < B/2$. For such y and e' , we have

$$y - \lfloor q/2 \rfloor \text{MSB}(y) = y - \lfloor q/2 \rfloor \text{MSB}(y - 2e') = y - 2e' - \lfloor q/2 \rfloor \text{MSB}(y - 2e') \bmod 2.$$

We then set $B = 2\sigma' \sqrt{\lambda}$. Then, we have $|e'_i| < B$ with overwhelming probability. Furthermore, each $\mathbf{y}_i$ is distributed uniformly at random over $\mathbb{Z}_q^k$, since $\mathbf{P}$ and $\mathbf{H}$ are independently and uniformly sampled. Therefore, by the union bound, the probability that there exists i such that some of $\mathbf{y}_i$'s entry falls into the boundary is negligible. Hence, we have $\mathbf{y}_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}_i) = \mathbf{y}_i - 2\mathbf{e}'_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}_i - 2\mathbf{e}'_i) \bmod 2$ with overwhelming probability, as desired. This implies $|\Pr[\mathbf{E}_1] - \Pr[\mathbf{E}_2]| = \text{negl}(\lambda)$.

Hyb₃: In this hybrid, we further change the definition of $\mathbf{z}_i$ as

$$\mathbf{z}_i = \mathbf{y}'_i - \lfloor q/2 \rfloor \text{MSB}(\mathbf{y}'_i)$$

for $\mathbf{y}'_i \leftarrow \mathbb{Z}_q^k$. We argue that under the LWE assumption, we have $|\Pr[\mathbf{E}_2] - \Pr[\mathbf{E}_3]| = \text{negl}(\lambda)$.

To show this, we consider the following reduction. The reduction algorithm is given an LWE challenge $(\mathbf{P}, \mathbf{U})$, where $\mathbf{U} = \mathbf{P}\mathbf{H} + 2\mathbf{E}'$ if $\beta = 0$ and $\mathbf{U} \leftarrow \mathbb{Z}_q^{k \times \kappa}$.¹⁴ Then, the reduction algorithm chooses $(\mathbf{B}, \mathbf{B}_\tau^{-1}) \leftarrow \text{TrapGen}(1^m, 1^n, q)$, $\mathbf{D} \leftarrow \mathbf{B}_\tau^{-1}(\mathbf{P})$, and $\text{aux}' = (\mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$ by itself. Then, it defines

$$\mathbf{y}'_i = \mathbf{A}\mathbf{F}[\cdot, i] + \mathbf{D}\mathbf{C}[\cdot, i] - \mathbf{U}[\cdot, i] - \lfloor q/2 \rfloor \text{MSB}(\mathbf{A}\mathbf{F}[\cdot, i] + \mathbf{D}\mathbf{C}[\cdot, i] - \mathbf{U}[\cdot, i])$$

and $\mathbf{z}_i$ based on the value. Then it outputs 0 if Equation (25) is solvable and 1 otherwise.

We can see that the advantage of the adversary above is $|\Pr[\mathbf{E}_2] - \Pr[\mathbf{E}_3]|$. Therefore, unless the LWE assumption is broken, we have that $|\Pr[\mathbf{E}_2] - \Pr[\mathbf{E}_3]| \leq \text{negl}(\lambda)$.

Hyb₄: In this hybrid, $\mathbf{z}_i$ is uniformly random in $[-q/4, q/4]^k$ for all $i \in [\kappa]$. It is easy to see that this game is statistically close to the previous hybrid, since $\mathbf{y}'_i$ is uniformly random.

¹⁴Here, we consider a variant of the LWE problem where the noise is always even. This variant is known to be equivalent to the standard one when q is odd.

We finally observe that $\{\mathbf{z}_i \bmod 2\}_i$ is statistically close to uniform distribution over $\{0, 1\}^{k \times \kappa}$.

Therefore, Equation (25), consisting of $k(k+1)m + m\kappa$ variables, is unsolvable with overwhelming probability if $k(k+1)m + m\kappa + \lambda < k\kappa$ in Hyb₄. Note that for our setting of parameters, this relation is indeed satisfied. Therefore, we have $\Pr[E_1] = 1 - \text{negl}(\lambda)$ from the above discussion.

This gives us

$$\Pr(\beta' = 1 \mid \beta = 1) = 1 - \text{negl}(\lambda). \quad (31)$$

Thus, from Equation (29) and (31), we get $\Pr(\beta' = \beta) \approx 3/4$. $\square$

Claim 4.11. $\forall i \in [\kappa], j \in [k], \text{PRF}(K', i)[j] - \tilde{\mathbf{E}}[j, i] = 0 \bmod 2$ with probability $1/2 + \text{negl}(\lambda)$.

Proof. The proof is similar to that of Claim 4.3. Let us start by analyzing the error $\tilde{\mathbf{E}}[j, i]$. Note that $\forall i \in [\kappa], j \in [k], \tilde{\mathbf{E}}[j, i] = \mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{F_i}}[\cdot, j])$ where $\mathbf{R}_{C'_{F_i}}[\cdot, j]$ is the randomness in the homomorphically evaluated encryption of $F(\bar{K}, i)[j]$ (of different form: $\mathbf{A}_{\text{fhe}}(\mathbf{R}_{C'_{F_i}}[\cdot, j]) - \begin{pmatrix} 0^{n \times 1} \\ F(\bar{K}, i)[j] \end{pmatrix}$), output by $\text{VEval}_{C'_{F_i}}$ circuit. Here, $\mathbf{A}_{\text{fhe}}$ is the FHE public key used to compute $\mathbf{X}$, i.e. the encryption of key $\bar{K}$.

Recall that $F(\bar{K}, i)[j] = f(K, i) \lfloor q/2 \rfloor [j] + \text{PRF}(K', i)[j]$ and we can write $\tilde{\mathbf{E}}[j, i] = \mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{F_i}}[\cdot, j]) = \mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{f_i}}[\cdot, j]) + \mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{\text{PRF}_i}}[\cdot, j])$ follows by the way we implemented the homomorphic addition.

Now, $\mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{f_i}}[\cdot, j]) = \mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{f_i}, j} = \mathbf{e}_{\text{fhe}}^T \sum_{v=0}^{\log_2 q - 1} \mathbf{R}_{C'_{f_i}, j, v} \mathbf{G}^{-1}(2^v \iota_{n+1})$, where $\mathbf{R}_{C'_{f_i}, j} = \mathbf{R}_{C'_{f_i}}[\cdot, j]$ is the j th column of $\mathbf{R}_{C'_{f_i}}$ and $\mathbf{R}_{C'_{f_i}, j, v}$ is the randomness in homomorphically computed GSW encryption of the v -th bit of $f(K, i) \lfloor q/2 \rfloor [j]$. Here $v \in [0, \log_2 q)$. Since $f(K, i) \lfloor q/2 \rfloor$ is implemented by C'_{f_i} , which involves the even randomness transformation (Lemma 3.2) on all its output bits, $\mathbf{R}_{C'_{f_i}, j, v}$ has all the entries as even due to Lemma 3.2. This implies $\mathbf{R}_{C'_{f_i}, j}$ has even entries, which in turn implies that $\mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{f_i}, j}$ is even.

So, now our goal is to show that

$$\text{PRF}(K', i)[j] - \mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{\text{PRF}_i}}[\cdot, j]) = 0 \bmod 2$$

Similar to above, we have $\mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{\text{PRF}_i}}[\cdot, j]) = \mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{\text{PRF}_i}, j} = \mathbf{e}_{\text{fhe}}^T \sum_{v=0}^{w-1} \mathbf{R}_{C'_{\text{PRF}_i}, j, v} \mathbf{G}^{-1}(2^v \iota_{n+1})$.

Again, since in C'_{PRF_i} all the output bits pass through even randomness, except the lowest order bit of the j th entry of the output, for all $j \in [k]$, which passes through the correlation inducing gate, we have

$$\begin{aligned} & \text{PRF}(K', i)[j] - \mathbf{e}_{\text{fhe}}^T(\mathbf{R}_{C'_{\text{PRF}_i}}[\cdot, j]) \bmod 2 \\ &= \text{PRF}(K', i)[j] - \mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{\text{PRF}_i}, j, 0} \mathbf{G}^{-1}(\iota_{n+1}) - \mathbf{e}_{\text{fhe}}^T \sum_{v=1}^{\log_2 q - 1} \mathbf{R}_{C'_{\text{PRF}_i}, j, v} \mathbf{G}^{-1}(2^v \iota_{n+1}) \bmod 2 \\ &= \text{PRF}(K', i)[j] - \mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{\text{PRF}_i}, j, 0} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \quad (\text{due to Lemma 3.2}) \\ &= \text{PRF}(K', i)[j, 0] + \sum_{v=1}^{\log_2 q - 2} 2^v \text{PRF}(K', i)[j, v] - \mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{\text{PRF}_i}, j, 0} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\ &= \text{PRF}(K', i)[j, 0] - \mathbf{e}_{\text{fhe}}^T \mathbf{R}_{C'_{\text{PRF}_i}, j, 0} \mathbf{G}^{-1}(\iota_{n+1}) \bmod 2 \\ &= \text{PRF}(K', i)[j, 0] - \text{PRF}(K', i)[j, 0] (\mathbf{e}_{\text{fhe}}^T \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1})) \bmod 2 \\ & \quad (\text{due to Corollary 3.4}) \\ &= 0 \bmod 2, \text{ if } (\mathbf{e}_{\text{fhe}}^T \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1})) \text{ is odd.} \end{aligned}$$

In the above, $\text{PRF}(K', i)[j, v] \in \{-1, 0, 1\}$ represents the v -th (signed) bit in the binary representation of $\text{PRF}(K', i)[j]$ and $\mathbf{R}^*$ is the randomness in the ciphertext of special "1" input used in Corollary 3.4.

Note that $(\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}))$ is odd with probability negligibly close to $1/2$ using the same argument as in Claim 4.3. Also, note that the term $(\mathbf{e}_{\text{fhe}}^\top \mathbf{R}^* \mathbf{G}^{-1}(\iota_{n+1}))$ does not depend on i, j and hence, for all $i \in [\kappa]$, $j \in [k]$, the probability that $\text{PRF}(K', i)[j] - \mathbf{e}_{\text{fhe}}^\top (\mathbf{R}_{C'_{\text{PRF}_i}}[\cdot, j]) = 0 \bmod 2$ is $1/2 + \text{negl}(\lambda)$. Hence, the proof. $\square$

Proving the pre-condition. We now show that the pre-condition of Evasive LWE as in Assumption 4.5 holds with respect to the above sampler defined in Section 4.2.

Lemma 4.12. For $(\mathbf{S}, \text{aux}) \leftarrow \text{Samp}(1^\lambda, \mathbf{P})$ where $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$,

$$D_0^{\text{pre}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}, \mathbf{C}' = \mathbf{P}\mathbf{S} + \mathbf{E}', \text{aux}) \approx D_1^{\text{pre}} := (\mathbf{B}, \mathbf{P}, \mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}, \mathbf{C}' \leftarrow \mathbb{Z}_q^{k \times \kappa}, \text{aux}) \quad (32)$$

where $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$, $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$, $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$, $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$, $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$.

Proof. The proof of this claim is similar to the proof of pre-condition of evasive LWE used in the security proof of exponentially efficient doubly pseudorandom obfuscation scheme (xdPRO) of [BDJ⁺25]. However, for completeness, we provide the proof here. We prove Lemma 4.12 by a series of hybrids Hyb_0 to Hyb_{11} where Hyb_0 is the D_0^{pre} distribution and Hyb_{11} is the D_1^{pre} distribution of (32). We prove that $\text{Hyb}_0 \approx \text{Hyb}_1 \approx \dots \approx \text{Hyb}_{11}$.

Hyb₀: This is the D_0^{pre} distribution. We recall the distribution here.

1. Sample $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$, $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$ and set $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$.
2. Choose $(K, \text{aux}_K) \leftarrow \text{KeySamp}(1^\lambda)$, sample $K'' \leftarrow \{0, 1\}^\lambda$ and set $K' = (1, K'')$ and $\bar{K} = (K, K')$.
3. Samples $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and set $\mathbf{s} = (\bar{\mathbf{s}}^\top, -1)^\top$.
4. Sample $\bar{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{e}_{\text{fhe}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{fhe}}}^m$, $\mathbf{R}_{\text{fhe}, \ell} \leftarrow \{0, 1\}^{m \times m}$ for all $\ell \in [\eta + \lambda + 1]$ and compute a GSW encryption as follows.

$$\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \bar{\mathbf{s}}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix}, \quad \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (\bar{K}) \otimes \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m(\lambda + \eta + 1)}$$

where $\mathbf{R}_{\text{fhe}} = (\mathbf{R}_{\text{fhe}, 1}, \dots, \mathbf{R}_{\text{fhe}, (\lambda + \eta + 1)})$.

5. Parse $\mathbf{s}^\top = (s_1, \dots, s_n, -1) \in \mathbb{Z}_q^{1 \times (n+1)}$, sample $\mathbf{R} \leftarrow \mathbb{Z}_q^{n \times (k+1)m}$ and $\bar{\mathbf{E}} \leftarrow \mathcal{D}_{\mathbb{Z}, \rho}^{k \times (k+1)m}$ and compute $\mathbf{A} = \mathbf{P}\mathbf{R} + \bar{\mathbf{E}} + (\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{k \times (k+1)m}$ where $\mathbf{0} \in \mathbb{Z}_q^k$ is all zero column vector and $\mathbf{I}_k$ is the identity matrix with k rows and columns.
6. For $i \in [\kappa]$
 - Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$.
 - Compute $\mathbf{X}_{C'_{F_i}} = \text{VEval}_{C'_{F_i}}(\mathbf{X})$. Hence, $\mathbf{X}_{C'_{F_i}} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{C'_{F_i}} - \begin{pmatrix} 0^{n \times k} \\ C'_{F_i}(\bar{K}) \end{pmatrix} \in \mathbb{Z}_q^{n+1 \times k}$.
 - Let $\mathbf{f}_i \in \mathbb{Z}_q^{k(n+1)}$ be the vectorization of $\mathbf{X}_{C'_{F_i}}$ i.e.

$$\mathbf{f}_i = \begin{pmatrix} \mathbf{X}_{C'_{F_i}}[1, 1], \dots, \mathbf{X}_{C'_{F_i}}[n+1, 1], \mathbf{X}_{C'_{F_i}}[1, 2], \dots, \mathbf{X}_{C'_{F_i}}[n+1, 2], \dots, \mathbf{X}_{C'_{F_i}}[1, k], \\ \dots, \mathbf{X}_{C'_{F_i}}[n+1, k] \end{pmatrix}^\top$$

- Sample $\mathbf{r}_i \leftarrow \{0, 1\}^m$.

7. Set $\mathbf{F} = (\mathbf{G}_{\mathbf{r}_1}^{-1}(\mathbf{f}_1), \dots, \mathbf{G}_{\mathbf{r}_\kappa}^{-1}(\mathbf{f}_\kappa))$.
8. Sample $\mathbf{S} \leftarrow \mathbb{Z}_q^{n \times \kappa}$ and set $\mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}$.
9. Set $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$.
10. Compute $\mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E}$ where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$.
11. Compute $\mathbf{C}' = \mathbf{P}\mathbf{S} + \mathbf{E}'$ where $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$.
12. Output $(\mathbf{B}, \mathbf{P}, \mathbf{C}, \mathbf{C}', \text{aux})$.

Hyb₁: This is same as Hyb₀ except the way we compute $\mathbf{P}$. Instead of sampling $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$ and setting $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$, we sample $\mathbf{P}$ uniformly at random i.e. $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$. We now prove that $\text{Hyb}_0 \approx_s \text{Hyb}_1$. We prove that $(\mathbf{B}, \mathbf{D}\mathbf{B}) \approx_s (\mathbf{B}, \mathbf{P})$ where $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$, $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$, $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$. This can be seen by observing that, by Lemma 2.5, the min-entropy of each row of $\mathbf{D}$ is at least m . Hence, by the Leftover Hash Lemma (Lemma 2.7), each entry of $\mathbf{D}\mathbf{B}$, is statistically close to being uniformly distributed over $\mathbb{Z}_q$.

Hyb₂: This is same as Hyb₁ except the way we compute $\mathbf{H}, \mathbf{C}, \mathbf{C}'$. Instead of choosing $\mathbf{S}$ uniformly at random, sample $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$ and set $\mathbf{S} = \mathbf{H} - \mathbf{R}\mathbf{F}$. Now,

- compute $\mathbf{U} = \mathbf{B}\mathbf{R} + \mathbf{E}^*$ where $\mathbf{E}^* \leftarrow \mathcal{D}_{\mathbb{Z}, \rho}^{m \times (k+1)m}$.
- compute $\mathbf{C} = \mathbf{B}\mathbf{H} - \mathbf{U}\mathbf{F} + \mathbf{E}^*\mathbf{F} + \mathbf{E}$ where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$.
- compute $\mathbf{C}' = \mathbf{P}\mathbf{H} - \mathbf{A}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) + \bar{\mathbf{E}}\mathbf{F} + \tilde{\mathbf{E}} + \mathbf{E}'$ where $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$ where $\tilde{\mathbf{E}}$ is such that $((\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top)\mathbf{F} = (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) + \tilde{\mathbf{E}}$.

It is easy to see that Hyb₁ and Hyb₂ are identically distributed. When we substitute $\mathbf{S} = \mathbf{H} - \mathbf{R}\mathbf{F}$ in $\mathbf{C}$ and $\mathbf{C}'$, we get $\mathbf{C} = \mathbf{B}\mathbf{S} + \mathbf{E} = \mathbf{B}(\mathbf{H} - \mathbf{R}\mathbf{F}) + \mathbf{E} = \mathbf{B}\mathbf{H} - \mathbf{U}\mathbf{F} + \mathbf{E}^*\mathbf{F} + \mathbf{E}$ and $\mathbf{C}' = \mathbf{P}\mathbf{S} + \mathbf{E}' = \mathbf{P}(\mathbf{H} - \mathbf{R}\mathbf{F}) + \mathbf{E}' = \mathbf{P}\mathbf{H} - \mathbf{P}\mathbf{R}\mathbf{F} + \mathbf{E}' = \mathbf{P}\mathbf{H} - (\mathbf{A}\mathbf{F} - (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) - \tilde{\mathbf{E}} - \bar{\mathbf{E}}\mathbf{F}) + \mathbf{E}' = \mathbf{P}\mathbf{H} - \mathbf{A}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) + \bar{\mathbf{E}}\mathbf{F} + \tilde{\mathbf{E}} + \mathbf{E}'$. Therefore, $\text{Hyb}_1 \approx_s \text{Hyb}_2$.

Hyb₃: This is same as Hyb₂ except the way we compute $\mathbf{C}, \mathbf{C}'$. In this hybrid, we compute $\mathbf{C} = \mathbf{B}\mathbf{H} - \mathbf{U}\mathbf{F} + \mathbf{E}$ where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$ and $\mathbf{C}' = \mathbf{P}\mathbf{H} - \mathbf{A}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) + \mathbf{E}'$ where $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$. Note that by GSW noise bound setting, we have $\|\tilde{\mathbf{E}}\| \leq \tilde{B}$, from the analysis of $\beta = 0$ case in Lemma 4.10, we have $\|\bar{\mathbf{E}}\mathbf{F}\| \leq \lambda(k+1)m\rho$ and $\|\mathbf{E}^*\mathbf{F}\| \leq \lambda(k+1)m\rho$ and by parameter setting we have $\sigma = \sigma' \geq 2^\lambda(\tilde{B} + \lambda(k+1)m\rho)$, we have $\mathbf{E}^*\mathbf{F} + \mathbf{E} \approx_s \mathbf{E}$ and $\bar{\mathbf{E}}\mathbf{F} + \tilde{\mathbf{E}} + \mathbf{E}' \approx_s \mathbf{E}'$. Therefore, $\text{Hyb}_2 \approx_s \text{Hyb}_3$.

Hyb₄: This is same as Hyb₃ except the way we compute $\mathbf{A}, \mathbf{U}$. In this hybrid, we sample $\mathbf{A} \leftarrow \mathbb{Z}_q^{k \times (k+1)m}$ and $\mathbf{U} \leftarrow \mathbb{Z}_q^{m \times (k+1)m}$. We now prove that $\text{Hyb}_3 \approx_c \text{Hyb}_4$. Let $\mathcal{A}$ be the adversary that distinguishes between Hyb₃ and Hyb₄. Then we construct a reduction $\mathcal{B}$ that acts as an adversary against LWE. $\mathcal{B}$ does the following.

- Challenger of LWE samples $\beta \leftarrow \{0, 1\}$ and sends $(\mathbf{V}, \mathbf{Z} = \mathbf{V}\mathbf{R} + \mathbf{E})$, where $\mathbf{E} = \begin{pmatrix} \mathbf{E}' \\ \mathbf{E}^* \end{pmatrix}$ if $\beta = 0$ and $(\mathbf{V}, \mathbf{Z} \leftarrow \mathbb{Z}_q^{k+m \times (k+1)m})$ if $\beta = 1$. Here $\mathbf{V} \leftarrow \mathbb{Z}_q^{k+m \times n}$.
- Parse $\mathbf{V} = \begin{pmatrix} \mathbf{P} \\ \mathbf{B} \end{pmatrix}$ and $\mathbf{Z} = \begin{pmatrix} \mathbf{A}' \\ \mathbf{U} \end{pmatrix}$.
- Choose $(K, \text{aux}_K) \leftarrow \text{KeySamp}(1^\lambda)$, sample $K'' \leftarrow \{0, 1\}^\lambda$ and set $K' = (1, K'')$ and $\bar{K} = (K, K')$.

- Samples $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and set $\mathbf{s} = (\bar{\mathbf{s}}^\top, -1)^\top$.
- Sample $\bar{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{e}_{\text{fhe}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{fhe}}}^m$, $\mathbf{R}_{\text{fhe}, \ell} \leftarrow \{0, 1\}^{m \times m}$ for all $\ell \in [\eta + \lambda + 1]$ and compute a GSW encryption as follows.

$$\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe}} \\ \bar{\mathbf{s}}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix}, \quad \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (\bar{K}) \otimes \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m(\lambda + \eta + 1)}$$

where $\mathbf{R}_{\text{fhe}} = (\mathbf{R}_{\text{fhe}, 1}, \dots, \mathbf{R}_{\text{fhe}, (\lambda + \eta + 1)})$.

- Set $\mathbf{A} = \mathbf{A}' + (\mathbf{I}_k, \mathbf{0}) \otimes \mathbf{s}^\top \otimes \mathbf{g}^\top$.
- For $i \in [\kappa]$
 - Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$.
 - Compute $\mathbf{X}_{C'_{F_i}} = \text{VEval}_{C'_{F_i}}(\mathbf{X})$.
 - Let $\mathbf{f}_i \in \mathbb{Z}_q^{k(n+1)}$ be the vectorization of $\mathbf{X}_{C'_{F_i}}$.
 - Sample $\mathbf{r}_i \leftarrow \{0, 1\}^m$.
- Set $\mathbf{F} = (\mathbf{G}_{\mathbf{r}_1}^{-1}(\mathbf{f}_1), \dots, \mathbf{G}_{\mathbf{r}_\kappa}^{-1}(\mathbf{f}_\kappa))$.
- Sample $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$.
- Set $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$.
- Compute $\mathbf{C} = \mathbf{B}\mathbf{H} - \mathbf{U}\mathbf{F} + \mathbf{E}$ where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$.
- Compute $\mathbf{C}' = \mathbf{P}\mathbf{H} - \mathbf{A}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) + \mathbf{E}'$ where $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$.
- Output $(\mathbf{B}, \mathbf{P}, \mathbf{C}, \mathbf{C}', \text{aux})$ to $\mathcal{A}$.
- On receiving β' from $\mathcal{A}$, outputs β' .

It is straightforward to see that if $\beta = 0$, then $\mathcal{B}$ simulated Hyb_3 as $\mathbf{A} = \mathbf{A}' + (\mathbf{I}_k, \mathbf{0}) \otimes \mathbf{s}^\top \otimes \mathbf{g}^\top = \mathbf{P}\mathbf{R} + \mathbf{E}' + (\mathbf{I}_k, \mathbf{0}) \otimes \mathbf{s}^\top \otimes \mathbf{g}^\top$ and $\mathbf{U} = \mathbf{B}\mathbf{R} + \mathbf{E}^*$, else it simulated Hyb_4 as $\mathbf{A} = \mathbf{A}' + (\mathbf{I}_k, \mathbf{0}) \otimes \mathbf{s}^\top \otimes \mathbf{g}^\top$ is uniformly random and $\mathbf{U}$ is uniformly random. Hence, $\text{Hyb}_3 \approx_c \text{Hyb}_4$ by the hardness of LWE.

Hyb₅: This is same as Hyb_4 except that we compute $\mathbf{X}$ as $\mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (0^{\eta + \lambda + 1}) \otimes \mathbf{G}$. We now prove that $\text{Hyb}_4 \approx_c \text{Hyb}_5$. Let $\mathcal{A}$ be the adversary that distinguishes between Hyb_4 and Hyb_5 . Then we construct a reduction $\mathcal{B}$ that breaks the IND-CPA security of the GSW FHE scheme. $\mathcal{B}$ does the following.

- Sample $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$, $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$.
- Choose $(K, \text{aux}_K) \leftarrow \text{KeySamp}(1^\lambda)$, sample $K'' \leftarrow \{0, 1\}^\lambda$ and set $K' = (1, K'')$ and $\bar{K} = (K, K')$.
- $\mathcal{B}$ sends the messages, $\bar{K}$ and $0^{\eta + \lambda + 1}$ to the challenger of GSW.
- Challenger of GSW FHE scheme samples $\beta \leftarrow \{0, 1\}$ and sends $(\mathbf{A}_{\text{fhe}}, \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (\bar{K}) \otimes \mathbf{G})$ if $\beta = 0$ and $(\mathbf{A}_{\text{fhe}}, \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (0^{\eta + \lambda + 1}) \otimes \mathbf{G})$ if $\beta = 1$.
- Sample $\mathbf{A} \leftarrow \mathbb{Z}_q^{k \times (k+1)m}$, $\mathbf{U} \leftarrow \mathbb{Z}_q^{m \times (k+1)m}$.
- For $i \in [\kappa]$
 - Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$.
 - Compute $\mathbf{X}_{C'_{F_i}} = \text{VEval}_{C'_{F_i}}(\mathbf{X})$.
 - Let $\mathbf{f}_i \in \mathbb{Z}_q^{k(n+1)}$ be the vectorization of $\mathbf{X}_{C'_{F_i}}$.

- Sample $\mathbf{r}_i \leftarrow \{0, 1\}^m$.
- Set $\mathbf{F} = (\mathbf{G}_{\mathbf{r}_1}^{-1}(\mathbf{f}_1), \dots, \mathbf{G}_{\mathbf{r}_\kappa}^{-1}(\mathbf{f}_\kappa))$.
- Sample $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$.
- Set $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$.
- Compute $\mathbf{C} = \mathbf{B}\mathbf{H} - \mathbf{U}\mathbf{F} + \mathbf{E}$ where $\mathbf{E} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma}^{m \times \kappa}$.
- Compute $\mathbf{C}' = \mathbf{P}\mathbf{H} - \mathbf{A}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) + \mathbf{E}'$ where $\mathbf{E}' \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma'}^{k \times \kappa}$.
- Output $(\mathbf{B}, \mathbf{P}, \mathbf{C}, \mathbf{C}', \text{aux})$ to $\mathcal{A}$.
- On receiving β' from $\mathcal{A}$, outputs β' .

It is straightforward to see that if $\beta = 0$, then $\mathcal{B}$ simulated Hyb_4 , else it simulated Hyb_5 . Hence, $\text{Hyb}_4 \approx_c \text{Hyb}_5$ by the IND-CPA security of the GSW FHE scheme.

Hyb₆: This is same as Hyb_5 except the we sample $\mathbf{C}'$ as $\mathbf{C}' \leftarrow \mathbb{Z}_q^{k \times \kappa}$. Note that for $f \in \mathcal{F}_{\text{prm}}$,

$$(\{f(K, i)\}_{i \in [\kappa]}, \text{aux}_K) \approx_c (\{\mathbf{u}_i : \mathbf{u}_i \leftarrow \{0, 1\}^k\}_{i \in [\kappa]}, \text{aux}_K),$$

which implies

$$(\{\lfloor q/2 \rfloor f(K, i) + \text{PRF}(K', i) + \mathbf{E}'[\cdot, i]\}_{i \in [\kappa]}, \text{aux}_K) \approx_c (\{\tilde{\mathbf{c}}_i : \tilde{\mathbf{c}}_i \leftarrow \mathbb{Z}_q^k\}_{i \in [\kappa]}, \text{aux}_K),$$

where $\|\mathbf{E}'[\cdot, i]\| \leq \sqrt{\lambda} \sigma' \leq \hat{B}$ and $\hat{B}$ is exponentially smaller than q . Therefore, $\text{Hyb}_5 \approx_c \text{Hyb}_6$.

Hyb₇: This is same as Hyb_6 except that in this hybrid, we sample $\mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}$. We prove that $\text{Hyb}_6 \approx_c \text{Hyb}_7$ using the hardness of LWE. Let $\mathcal{A}$ be the adversary that distinguishes between Hyb_6 and Hyb_7 . Then we construct a reduction $\mathcal{B}$ that acts as an adversary against LWE. $\mathcal{B}$ does the following.

- Challenger of LWE samples $\beta \leftarrow \{0, 1\}$ and sends $(\mathbf{V}, \mathbf{Z} = \mathbf{T}\mathbf{V} + \mathbf{E} \in \mathbb{Z}_q^{m \times \kappa})$, if $\beta = 0$ and $(\mathbf{V}, \mathbf{Z} \leftarrow \mathbb{Z}_q^{m \times \kappa})$ if $\beta = 1$. Here $\mathbf{V} \leftarrow \mathbb{Z}_q^{(n+1) \times \kappa}$.
- Sample $\mathbf{U}' \leftarrow \mathbb{Z}_q^{m \times (k+1)m}$.
- Sample $\mathbf{P} \leftarrow \mathbb{Z}_q^{k \times n}$, $\mathbf{B} \leftarrow \mathbb{Z}_q^{m \times n}$.
- Samples $\bar{\mathbf{s}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_s}^n$ and set $\mathbf{s} = (\bar{\mathbf{s}}^\top, -1)^\top$.
- Sample $\bar{\mathbf{A}}_{\text{fhe}} \leftarrow \mathbb{Z}_q^{n \times m}$, $\mathbf{e}_{\text{fhe}} \leftarrow \mathcal{D}_{\mathbb{Z}, \sigma_{\text{fhe}}}^m$, $\mathbf{R}_{\text{fhe}, \ell} \leftarrow \{0, 1\}^{m \times m}$ for all $\ell \in [\eta + \lambda + 1]$ and compute a GSW encryption as follows.

$$\mathbf{A}_{\text{fhe}} := \begin{pmatrix} \bar{\mathbf{A}}_{\text{fhe}} \\ \bar{\mathbf{s}}^\top \bar{\mathbf{A}}_{\text{fhe}} + \mathbf{e}_{\text{fhe}}^\top \end{pmatrix}, \quad \mathbf{X} = \mathbf{A}_{\text{fhe}} \mathbf{R}_{\text{fhe}} - (0^{\eta+\lambda+1}) \otimes \mathbf{G} \in \mathbb{Z}_q^{(n+1) \times m(\lambda+\eta+1)}$$

where $\mathbf{R}_{\text{fhe}} = (\mathbf{R}_{\text{fhe}, 1}, \dots, \mathbf{R}_{\text{fhe}, (\lambda+\eta+1)})$.

- Sample $\mathbf{A} \leftarrow \mathbb{Z}_q^{k \times (k+1)m}$
- For $i \in [\kappa]$
 - Define the homomorphic evaluation circuit $\text{VEval}_{C'_{F_i}} = \text{MakeVEvalCkt}(n, m, q, C'_{F_i})$.
 - Compute $\mathbf{X}_{C'_{F_i}} = \text{VEval}_{C'_{F_i}}(\mathbf{X})$.
 - Let $\mathbf{f}_i \in \mathbb{Z}_q^{k(n+1)}$ be the vectorization of $\mathbf{X}_{C'_{F_i}}$.
- Set $\text{aux} = (\mathbf{H}, \mathbf{A}_{\text{fhe}}, \mathbf{X}, \mathbf{A}, \{\mathbf{r}_i\}_{i \in [\kappa]}, \text{aux}_K)$.

- Compute $\mathbf{C}' \leftarrow \mathbb{Z}_q^{k \times \kappa}$.
- For $i \in [k]$, set $\mathbf{r}_i = \mathbf{G}^{-1}(\mathbf{V}[\cdot, i])$ where $\mathbf{V}[\cdot, i]$ is the i th column of $\mathbf{V}$.
- Set $\mathbf{F} = (\mathbf{G}_{\mathbf{r}_1}^{-1}(\mathbf{f}_1), \dots, \mathbf{G}_{\mathbf{r}_\kappa}^{-1}(\mathbf{f}_\kappa))$.
- Sample $\mathbf{H} \leftarrow \mathbb{Z}_q^{n \times \kappa}$.
- Compute $\mathbf{C} = \mathbf{B}\mathbf{H} + \mathbf{Z} + \mathbf{U}'\mathbf{F}$.
- Output $(\mathbf{B}, \mathbf{P}, \mathbf{C}, \mathbf{C}', \text{aux})$ to $\mathcal{A}$.
- On receiving β' from $\mathcal{A}$, outputs β' .

Since, $\mathbf{V}$ is uniformly random, $\mathbf{r}_i$ are i.i.d uniformly random. Now when $\mathbf{Z} = \mathbf{T}\mathbf{V} + \mathbf{E}$, we have $\mathbf{Z} + \mathbf{U}'\mathbf{F} = \mathbf{T}\mathbf{V} + \mathbf{E} + \mathbf{U}'\mathbf{F} = (0^{m \times km}, \mathbf{T}\mathbf{G})\mathbf{F} + \mathbf{E} + \mathbf{U}'\mathbf{F} = ((0^{m \times km}, \mathbf{T}\mathbf{G}) + \mathbf{U}')\mathbf{F} + \mathbf{E} = \mathbf{U}\mathbf{F} + \mathbf{E}$. Here $\mathbf{U} = (0^{m \times km}, \mathbf{T}\mathbf{G}) + \mathbf{U}'$ is uniformly random. Hence, in this case, we have $\mathbf{C} = \mathbf{B}\mathbf{H} + \mathbf{Z} + \mathbf{U}'\mathbf{F} = \mathbf{B}\mathbf{H} + \mathbf{U}\mathbf{F} + \mathbf{E}$ which is identical to distribution of $\mathbf{C}$ in Hyb_6 .

When $\mathbf{Z} \leftarrow \mathbb{Z}_q^{m \times \kappa}$, we have $\mathbf{C} \leftarrow \mathbb{Z}_q^{m \times \kappa}$. Therefore, $\text{Hyb}_6 \approx_c \text{Hyb}_7$ by the hardness of LWE.

Hyb₈: This is same as Hyb_7 except that we undo the changes made in Hyb_5 . In this hybrid, we compute $c \leftarrow \text{Enc}(\mathbf{A}_{\text{fhe}}, \bar{K})$. $\text{Hyb}_7 \approx_c \text{Hyb}_8$ by the IND-CPA security of the GSW FHE scheme. We skip the indistinguishability argument as it is the same as that of $\text{Hyb}_4 \approx \text{Hyb}_5$.

Hyb₉: This is same as Hyb_8 except that we undo the changes made in Hyb_4 . We compute $\mathbf{A}$ as: Parse $\mathbf{s} = (s_1, \dots, s_n, -1) \in \mathbb{Z}_q^{n+1}$, sample $\mathbf{R} \leftarrow \mathbb{Z}_q^{n \times (k+1)m}$ and $\bar{\mathbf{E}} \leftarrow \mathcal{D}_{\mathbb{Z}, \rho}^{k \times (k+1)m}$ and compute $\mathbf{A} = \mathbf{P}\mathbf{R} + \bar{\mathbf{E}} + (\mathbf{I}_k, \mathbf{0}) \otimes (s_1, \dots, s_n, -1) \otimes \mathbf{g}^\top \in \mathbb{Z}_q^{k \times (k+1)m}$. $\text{Hyb}_8 \approx_c \text{Hyb}_9$ by the hardness of LWE. We skip the indistinguishability argument as it is the same as that of $\text{Hyb}_3 \approx \text{Hyb}_4$.

Hyb₁₀: This is same as Hyb_9 except that we undo the changes made in Hyb_2 . In this hybrid, we sample $\mathbf{S} \leftarrow \mathbb{Z}_q^{n \times \kappa}$ and set $\mathbf{H} = \mathbf{S} + \mathbf{R}\mathbf{F}$. Hyb_9 and Hyb_{10} are identically distributed. We skip the indistinguishability argument as it is the same as that of $\text{Hyb}_1 \approx \text{Hyb}_2$.

Hyb₁₁: This is same as Hyb_{10} except that we undo the changes made in Hyb_1 . We sample $\mathbf{D} \leftarrow \mathcal{D}_{\mathbb{Z}, \tau}^{k \times m}$ and set $\mathbf{P} = \mathbf{D} \cdot \mathbf{B}$. We have $\text{Hyb}_{10} \approx \text{Hyb}_{11}$ from Lemma 2.5 and Lemma 2.7. We skip the indistinguishability argument as it is the same as that of $\text{Hyb}_0 \approx \text{Hyb}_1$.

The distribution as in Hyb_{11} is the same as $\mathbf{D}_1^{\text{pre}}$. Hence, the proof. □

□

Remark 4.13 (On giving out $\mathbf{B}$ and $\mathbf{P}$). We make the following observations: (i) the indistinguishability in the pre-condition holds in the presence of both $\mathbf{B}$ and $\mathbf{P}$ as we have shown above, which implies that the indistinguishability will hold even in the absence of either or both of them, (ii) the distinguisher in the post-condition does not use the matrix $\mathbf{B}$. It uses $\mathbf{P}$ to compute $\mathbf{Y} = \mathbf{A}\mathbf{F} + \mathbf{D}\mathbf{C} - \mathbf{P}\mathbf{H}$, where $\mathbf{A}$, $\mathbf{H}$ and $\mathbf{F}$ are either provided directly in aux or can be computed from aux. If we modify the sampler to replace all the terms in aux with just $\mathbf{V} = \mathbf{A}\mathbf{F} - \mathbf{P}\mathbf{H}$ then it is easy to see that it won't affect the indistinguishability in pre-condition. On the other hand, the distinguisher in the post condition will not need $\mathbf{P}$ to compute $\mathbf{Y}$, which can now be computed as $\mathbf{V} + \mathbf{D}\mathbf{C}$. We note that

$$\begin{aligned} \mathbf{A}\mathbf{F} - \mathbf{P}\mathbf{H} &= \mathbf{P}\mathbf{R}\mathbf{F} + \bar{\mathbf{E}}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) - \tilde{\mathbf{E}} - \mathbf{P}\mathbf{S} - \mathbf{P}\mathbf{R}\mathbf{F} \\ &= \bar{\mathbf{E}}\mathbf{F} + (\mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa)) - \tilde{\mathbf{E}} - \mathbf{P}\mathbf{S} \\ &\approx \mathbf{F}(\bar{K}, 1), \dots, \mathbf{F}(\bar{K}, \kappa) - \mathbf{P}\mathbf{S} \end{aligned}$$

hides $\mathbf{P}$ since $\mathbf{F}(\bar{K}, \cdot)$ is pseudorandom. The indistinguishability in the third line follows from noise flooding since $\|\bar{\mathbf{E}}\mathbf{F} - \tilde{\mathbf{E}}\|$ is exponentially smaller than the output distribution of $\mathbf{F}(\bar{K}, \cdot)$.

Thus, similar to [DJM⁺25b] (where aux contains $\mathbf{SP} - 2\mathbf{T}$ with each entry of $\mathbf{T}$ sampled uniformly randomly from half the space - $[0, 1, \dots, \lfloor q/2 \rfloor]$), our counterexample also applies to the strongest setting, where both $\mathbf{B}$ and $\mathbf{P}$ are given out in the pre-condition and are hidden in the post-condition.

Same holds true for counterexample in Section 4.1, where we can replace aux with $\mathbf{c}_{\text{att}}^T \cdot \mathbf{H}_{\mathbf{A}_{\text{att}}, \mathbf{X}}^{F_i} \approx \mathbf{s}^T \mathbf{P} + (F_1(\mathbf{x}, \text{sd}), \dots, F_Q(\mathbf{x}, \text{sd}))$.

4.3 Extending BDJMMPV PRO Counterexample to Multi-Challenge PRFE

In this section, we adapt the counterexamples from [BDJ⁺25] against the existence of PRO for all function families to show that there exist pseudorandom function families for which multi-challenge PRFE is not possible. Similar to [BDJ⁺25] we construct the counterexample in two steps:

1. **Counterexample in the presence of auxiliary input.** In this section, we show that for all pseudorandom function families there exists auxiliary information, with respect to which, a multi-challenge PRFE cannot be constructed.

Theorem 4.14. Assuming the existence of sub-exponentially secure, instance hiding witness encryption for NP, for every function family $\mathcal{F}_{\text{prm}} = \{f : \mathcal{X}_{\text{prm}} \rightarrow \mathcal{Y}_{\text{prm}}\}$, where prm consist of polynomially bounded functions of λ ¹⁵, including bound $s = s(\lambda)$ on the size of the circuits computing any function in the family, satisfying

$$\{f_i\}_{i \in [Q_k]}, \{f_i(x_j)\}_{i \in [Q_k], j \in [Q_c]} \approx_c \{f_i\}_{i \in [Q_k]}, \{u_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}, \quad (33)$$

where x_j 's are chosen in such a way that they have enough entropy, there exists a dependent auxiliary input aux , such that there does not exist any PRFE scheme satisfying prCT security.

Proof. Consider any pseudorandom function family $\mathcal{F}$ - for example, a PRF family, with the input of the form $x_j = (K, j)$, and function $f_i(x_j) = \text{PRF}(K, (i, j))$. Suppose $\text{PRFE} = (\text{PRFE.Setup}, \text{PRFE.Enc}, \text{PRFE.KeyGen}, \text{PRFE.Dec})$ be any functional encryption scheme for $\mathcal{F}$, with ciphertext of size at most $p_c(\lambda, s)$ and secret key of length at most $p_k(\lambda, s)$, where p_c, p_k and p are some polynomials. Let $\text{prm}' = (p_c, p_k, Q_c, Q_k)$.

Then, to prove the above theorem, we need to define an efficient sampler that outputs $(\text{aux}, \{f_i\}_{i \in [Q_k]}, \{x_i\}_{i \in [Q_k]})$, such that

- (a) (Pre-condition holds) For all PPT adversary $\mathcal{A}$

$$\begin{aligned} D_0^{\text{pre}} &:= (\text{aux}, \{f_i\}_{i \in [Q_k]}, \{f_i(x_j)\}_{i \in [Q_k], j \in [Q_c]}) \\ &\approx_c D_1^{\text{pre}} := (\text{aux}, \{f_i\}_{i \in [Q_k]}, \{u_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}) \end{aligned} \quad (34)$$

- (b) (Distinguisher for post condition) Define a distinguisher for

$$\begin{aligned} D_0^{\text{post}} &:= (\text{aux}, \{f_i\}_{i \in [Q_k]}, \{\text{ct}_j\}_{j \in [Q_c]}, \{\text{sk}_i\}_{i \in [Q_k]}), \\ D_1^{\text{post}} &:= (\text{aux}, \{f_i\}_{i \in [Q_k]}, \{u_j \leftarrow \mathcal{CT}\}_{j \in [Q_c]}, \{\text{sk}_i\}_{i \in [Q_k]}), \end{aligned} \quad (35)$$

where $(\text{fmpk}, \text{fmsk}) \leftarrow \text{PRFE.Setup}(1^\lambda, \text{prm})$, $\text{ct}_j = \text{PRFE.Enc}(\text{fmpk}, x_j)$ and $\text{sk}_i = \text{PRFE.KeyGen}(\text{fmsk}, f_i)$, for $i \in [Q_k], j \in [Q_c]$, and $\mathcal{CT}$ is the ciphertext space of PRFE.

¹⁵We drop the subscript prm when not explicitly needed, to keep the notations simple.

Defining the sampler Samp. $\text{Samp}(1^\lambda)$ does the following:

- (a) Let Q_k and Q_c are such that $(\log |\mathcal{Y}|)Q_kQ_c > (Q_kp_k + Q_cp_c + 1) + \lambda$.

Define the following NP relation

$$\mathcal{L}_{\text{prm}'} = \{(y_{1,1}, \dots, y_{1,Q_c}, \dots, y_{Q_k,1}, \dots, y_{Q_k,Q_c}) \mid \exists \text{ct}_1, \dots, \text{ct}_{Q_c}, \text{sk}_1, \dots, \text{sk}_{Q_k} \text{ s.t.} \quad (36)$$

$$\forall i \in [Q_k], j \in [Q_c], y_{i,j} = \text{PRFE.Dec}(\text{sk}_i, \text{ct}_j) \text{ and } |\text{ct}_j| \leq p_c \text{ and } |\text{sk}_i| \leq p_k \}.$$

Thus, the length of any witness can be at most $Q_kp_k + Q_cp_c$. Here, we assume that PRFE master public key, fmpk , is also a part of the ciphertexts.

- (b) Choose $f_1, \dots, f_{Q_k} \in \mathcal{F}_{\text{prm}}$, and $\mathbf{x}_1, \dots, \mathbf{x}_{Q_c} \leftarrow \mathcal{X}$, such that the indistinguishability in (33) holds. For example, for the PRF example above, f_i is the PRF function with i hardwired and $\mathbf{x}_j$ can be (K, j) , where K is chosen uniformly from the PRF key space.
- (c) Let $\text{stmt} = (f_1(\mathbf{x}_1), \dots, f_1(\mathbf{x}_{Q_c}), \dots, f_{Q_k}(\mathbf{x}_1), \dots, f_{Q_k}(\mathbf{x}_{Q_c}))$.
- (d) Define $\text{aux} = (\text{WE.Enc}(\text{stmt}, 0), \text{WE.Enc}(\text{stmt}, 1))$.
- (e) Output $(\text{aux}, f_1, \dots, f_{Q_k}, \mathbf{x}_1, \dots, \mathbf{x}_{Q_c})$.

Analyzing the post condition. We first prove the following claim that we will need to motivate/analyze the distinguisher's strategy.

Claim 4.15. In the absence of $\text{ct}_1, \dots, \text{ct}_{Q_c}$ or any other terms dependent on $\{\mathbf{x}_j\}_{j \in [Q_c]}$, for all PPT adversary $\mathcal{A}$,

$$(\{f_i\}_{i \in [Q_k]}, \{\text{sk}_i\}_{i \in [Q_k]}, \text{WE.Enc}(\text{stmt}, 0)) \approx_c (\{f_i\}_{i \in [Q_k]}, \{\text{sk}_i\}_{i \in [Q_k]}, \text{WE.Enc}(\text{stmt}, 1))$$

Proof. We prove the above claim via the following hybrids.

Hyb₀: In this hybrid, $\mathcal{A}$ gets $\text{WE.Enc}(\text{stmt}, 0)$.

Hyb₁: In this hybrid, stmt is replaced with stmt' , where $\text{stmt}' = (u_{1,1}, \dots, u_{1,Q_c}, \dots, u_{Q_k,1}, \dots, u_{Q_k,Q_c})$, where $u_{i,j} \leftarrow \{0, 1\}^\ell$ for all $i \in [Q_k], j \in [Q_c]$. That is, $\mathcal{A}$ gets $\text{WE.Enc}(\text{stmt}', 0)$.

Indistinguishability follows from pseudorandomness of $\mathcal{F}_{\text{prm}}$. In particular, if $\mathcal{A}$ can distinguish between the two hybrids, then we can construct a reduction $\mathcal{B}$ against pseudorandomness of $\mathcal{F}$ as follows.

- (a) Upon input $(\{f_i\}_{i \in [Q_k]}, \{c_{i,j}\}_{i \in [Q_k], j \in [Q_c]})$, where $c_{i,j} = f_i(x_j)$, for some $x_j \in \mathcal{X}$, if $\beta = 0$, else $c_{i,j} \leftarrow \mathcal{Y}$, sample $(\text{fmpk}, \text{fmsk}) \leftarrow \text{PRFE.Setup}(1^\lambda, \text{prm})$. Compute $\text{sk}_i = \text{PRFE.KeyGen}(\text{fmsk}, f_i)$, for $i \in [Q_k]$. Set $\text{stmt} = (c_{1,1}, \dots, c_{Q_k, Q_c})$.
- (b) Send $(\{f_i\}_{i \in [Q_k]}, \{\text{sk}_i\}_{i \in [Q_k]}, \text{WE.Enc}(\text{stmt}, 0))$ to $\mathcal{A}$.
- (c) In the end, $\mathcal{A}$ outputs its guess bit β' . $\mathcal{B}$ also outputs β' .

Clearly, if $\beta = 0$, $\mathcal{B}$ simulates Hyb_0 , else Hyb_1 with $\mathcal{A}$. Hence, $\mathcal{B}$ wins with the same advantage as that of $\mathcal{A}$.

Hyb₂: In this hybrid, $\mathcal{A}$ is given $\text{WE.Enc}(\text{stmt}', 1)$.

Indistinguishability follows from the soundness of WE scheme, since for a randomly chosen set of $\{c_{i,j}\}_{i \in [Q_k], j \in [Q_c]}$, the probability of this being in $\mathcal{L}_{\text{prm}'}$ is negligible. This follows from the standard counting argument as follows: since any witness can be of length at most $Q_cp_c + Q_kp_k$, total number of possible witness is at most $2^{Q_cp_c + Q_kp_k + 1}$. Further, since PRFE decryption is a deterministic operation, there can be at most $2^{Q_cp_c + Q_kp_k + 1}$ members in the language $\mathcal{L}_{\text{prm}'}$. On the other hand, total number of choices for $\{c_{i,j} \in \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}$ is $2^{(\log |\mathcal{Y}|)Q_kQ_c}$. Hence, by the choice of the parameters, for a randomly chosen set of $\{c_{i,j}\}_{i,j}$, this being inside the language is $\leq 2^{-\lambda}$.

Hyb₄: In this hybrid, $\mathcal{A}$ gets $\text{WE.Enc}(\text{stmt}, 1)$.

Indistinguishability, again follows from pseudorandomness of $\mathcal{F}$, similar to the indistinguishability between Hyb₀ and Hyb₁. □

Based on the above claim, the strategy for the distinguisher, $\mathcal{A}^{\text{post}}$, for the post condition is:

- (a) Given $(\text{aux} = (\text{wct}_0, \text{wct}_1), \{f_i\}_{i \in [Q_k]}, \{\text{ct}_j\}_{j \in [Q_c]}, \{\text{sk}_i\}_{i \in [Q_k]})$, where $\text{ct}_j = \text{PRFE.Enc}(\text{fmpk}, x_j)$, if challenge bit $\beta = 0$, and $\text{ct}_j \leftarrow \mathcal{CT}$, if $\beta = 1$.
- (b) Define $\text{wt} = (\text{ct}_1, \dots, \text{ct}_{Q_c}, \text{sk}_1, \dots, \text{sk}_{Q_k})$.
- (c) Compute (μ_0, μ_1) , where $\mu_b = \text{WE.Dec}(\text{wt}, \text{wct}_b)$, for $b \in \{0, 1\}$.
- (d) If $(\mu_0, \mu_1) = (0, 1)$, then output $\beta' = 0$, else return $\beta' = 1$.

With the above strategy, the adversary outputs $\beta' = \beta$ with probability $\approx 7/8$. This is because if $\beta = 0$, then $\Pr(\beta' = 0) = 1$ due to the correctness of WE. When $\beta = 1$, then the probability that $(\mu_0, \mu_1) = (0, 1) \leq 1/4$ due to the above claim. In more detail, Claim 4.15 implies that for a random $\text{ct}_1, \dots, \text{ct}_{Q_c}$, $\text{wt} = (\text{ct}_1, \dots, \text{ct}_{Q_c}, \text{sk}_1, \dots, \text{sk}_{Q_k})$, $\Pr[\text{WE.Dec}(\text{wt}, \text{wct}_0) = 0] = \Pr[\text{WE.Dec}(\text{wt}, \text{wct}_1) = 0] = (p \leq 1, \text{ say})$. Because, otherwise $\mathcal{A}^{\text{post}}$ could distinguish the L.H.S. and R.H.S in the claim, by sampling the random WE ciphertexts itself. Thus, $\Pr[\text{WE.Dec}(\text{wt}, \text{wct}_0) = 1] = \Pr[\text{WE.Dec}(\text{wt}, \text{wct}_1) = 1] = (1 - p)$. This implies that $\Pr[(\mu_0, \mu_1) = (0, 1)] = p(1 - p) \leq 1/4$. Hence $\Pr(\beta' = 1) \geq 3/4$. This gives us the above probability of winning.

Analyzing the pre-condition

Claim 4.16. For all PPT adversary $\mathcal{A}$, $D_0^{\text{pre}} \approx D_1^{\text{pre}}$, where D_0^{pre} and D_1^{pre} are as defined in equation (34).

Proof. We prove the above claim via the following hybrids.

Hyb₀: This is D_0^{pre} . That is, $\mathcal{A}$ gets $((\text{wct}_0, \text{wct}_1), \{f_i\}_{i \in [Q_k]}, \{y_{i,j} = f_i(x_j)\}_{i \in [Q_k], j \in [Q_c]})$, where $\text{wct}_b = \text{WE.Enc}(\text{stmt}, b)$ for $\text{stmt} = (y_{1,1}, \dots, y_{Q_k, Q_c})$ and $b \in \{0, 1\}$.

Hyb₁: In this hybrid, $y_{i,j} \leftarrow \mathcal{Y}$ for all $i \in [Q_k], j \in [Q_c]$. stmt and hence, wct_0 and wct_1 are also computed using random $y_{i,j}$'s.

The indistinguishability between the two hybrids follows from the pseudorandomness of $\mathcal{F}$.

Hyb₂: In this hybrid, the challenger samples $\{u_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}$ (independent of $y_{i,j}$'s which are also sampled randomly) and sets $\text{stmt} = (u_{1,1}, \dots, u_{Q_k, Q_c})$, and computes $\text{wct}_b = \text{WE.Enc}(\text{stmt}, b)$ for $b \in \{0, 1\}$.

Indistinguishability between the two hybrids follows from the instance hiding property of WE because both $(y_{1,1}, \dots, y_{Q_k, Q_c})$ and $(u_{1,1}, \dots, u_{Q_k, Q_c})$ are not in $\mathcal{L}_{\text{prm'}}$ with $1 - \text{negl}(\lambda)$ probability by the same argument as in the proof of claim 4.15 (in the step Hyb₁ $\approx$ Hyb₂).

Hyb₃: In this hybrid, stmt is again set to $(f_1(x_1), \dots, f_{Q_k}(x_{Q_c}))$ and $\text{wct}_b = \text{WE.Enc}(\text{stmt}, b)$ for $b \in \{0, 1\}$.

Since $\{u_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}$ are independent of $\{y_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}$, the indistinguishability between the two hybrids again follow from the pseudorandomness of $\mathcal{F}$.

Finally, the proof completes by observing that Hyb₃ corresponds to the distribution D_1^{pre} . □

This completes the proof of theorem 4.14. $\square$

2. **Counterexample without auxiliary input:** To remove the auxiliary input aux, the idea is to output the WE ciphertexts as FE decryptions. However, since WE ciphertext can be polynomial in its witness size, we cannot perform its computation inside any circuit for which PRFE key is provided. This is because then PRFE ciphertext size and key size (p_c and p_k in above discussions) will be polynomial in Q_k and Q_c and we will not be able to apply incompressibility arguments as before. Instead, the sampler itself computes the WE ciphertext, wct, itself, as before, but now instead of providing it as aux, it divides wct into blocks of fixed lengths and encrypts each block into different PRFE ciphertexts, and provides an additional PRFE key for a function which simply outputs a part of its input. For this, to be able to argue the indistinguishability in the pre-condition, we need pseudorandomness of WE ciphertexts. This can be achieved by combining (plain) WE with compute-and-compare/lockable obfuscation [WZ17, GKW17]. Note that even though this increases the total number of PRFE ciphertexts, not all of them will be a part of the witness. Hence, we will still be able to apply the incompressibility argument as before.

We prove the following theorem.

Theorem 4.17. Assuming the existence of instance hiding WE with pseudorandom ciphertexts, there exists for all λ , a pseudorandom function family $\mathcal{F}_{\text{prm}} = \{f : \mathcal{X}_{\text{prm}} \rightarrow \mathcal{Y}_{\text{prm}}\}$ as defined in equation (33), for which there does not exist a PRFE scheme satisfying prCT security. Here prm consists of some fixed polynomials in λ .

Proof. To prove the above theorem, we need to define a function class $\mathcal{F}_{\text{prm}}$ such that for any PRFE scheme PRFE for $\mathcal{F}_{\text{prm}}$ there exists an efficient sampler Samp that on input (1^λ) outputs $(\{f_i\}_{i \in [Q'_k]}, \{x_j\}_{j \in [Q'_c]})$, for some Q'_k and Q'_c , we will set later, such that

- (a) (Pre-condition holds) For all PPT adversary $\mathcal{A}$,

$$D_0^{\text{pre}} := (\{f_i\}_{i \in [Q'_k]}, \{f_i(x_j)\}_{i \in [Q'_k], j \in [Q'_c]}) \approx_c D_1^{\text{pre}} := (\{f_i\}_{i \in [Q'_k]}, \{u_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q'_k], j \in [Q'_c]}) \quad (37)$$

- (b) (Distinguisher for post condition) Define a distinguisher for

$$D_0^{\text{post}} := (\{f_i\}_{i \in [Q'_k]}, \{\text{ct}_j\}_{j \in [Q'_c]}, \{\text{sk}_i\}_{i \in [Q'_k]}), \quad D_1^{\text{post}} := (\{f_i\}_{i \in [Q'_k]}, \{u_j \leftarrow \mathcal{CT}\}_{j \in [Q'_c]}, \{\text{sk}_i\}_{i \in [Q'_k]}),$$

where $(\text{fmpk}, \text{fmsk}) \leftarrow \text{PRFE.Setup}(1^\lambda)$, $\text{ct}_j = \text{PRFE.Enc}(\text{fmpk}, x_j)$ and $\text{sk}_i = \text{PRFE.KeyGen}(\text{fmsk}, f_i)$, for $i \in [Q'_k], j \in [Q'_c]$, and $\mathcal{CT}$ is the ciphertext space of PRFE.

Defining the sampler. We first describe the ingredients needed to define our sampler.

- Let $\text{PRF} : \mathcal{K} \times (\mathbb{Z}_q \times \mathbb{Z}_q) \rightarrow \mathcal{Y}$ be a pseudorandom function that can be computed by circuits of size s_F , where $\mathcal{K}$ is the PRF key space.
- Set $s = s_F$.
- Let $\mathcal{F}_s$ be a family of pseudorandom functions computable by circuits of size at most s . Let PRFE be any PRFE scheme for $\mathcal{F}_s$. Let $p_c = p_c(\lambda, s)$ and $p_k = p_k(\lambda, s)$, for some polynomials p_k and p_c , be the bound on the sizes of the ciphertexts and the secret keys, respectively, of PRFE scheme.
- Let WE be a witness encryption scheme for the language defined in equation (36), wrt the above PRFE. Thus, witness is of size at most $Q_c p_c + Q_k p_k$. Let $p_w = p_w(Q_c p_c + Q_k p_k)$ be the bound on the size of the ciphertexts of WE. Thus, $p_w > Q_c, Q_k$.

We are now ready to define our sampler, Samp .

$\text{Samp}(1^\lambda)$:

- Let Q_k and Q_c be such that

$$(\log |\mathcal{Y}|)Q_k Q_c > p_c Q_c + p_k Q_k + \lambda + 1. \quad (38)$$

- Let $Q_w = 2p_w / \log |\mathcal{Y}|$. To keep the notations simple, we assume that $\log |\mathcal{Y}|$ divides $2p_w$, else we can use padding. Also, since $p_w = \text{poly}(\lambda, \log |\mathcal{Y}|, Q_c, Q_k)$, we assume $Q_w > Q_c$. In the otherwise case, we could set Q_w to be the maximum of Q_c and $2p_w / \log |\mathcal{Y}|$.
- Set $Q'_k = Q_k + 1$ and $Q'_c = Q_w$.
- For $i \in [Q_k]$, define function $f_i : \mathcal{K} \times \mathbb{Z}_q \times \mathcal{Y} \rightarrow \mathcal{Y}$, such that $f_i(K, j, c) = \text{PRF}(K, (i, j))$.
- Define f_{Q_k+1} to be a selector function, g such that $g(K, j, c) = c$. Note that g is a simple function whose size is smaller than the PRF function and hence, lies in $\mathcal{F}_s$.
- Sample $K \leftarrow \mathcal{K}$ and do the following:
 - (a) For all $i \in [Q_k]$ and $j \in [Q_c]$, compute $y_{i,j} = \text{PRF}(K, (i, j))$.
 - (b) Set $\text{stmt} = (y_{1,1}, \dots, y_{Q_k, Q_c})$.
 - (c) Compute $\text{wct}_0 = \text{WE.Enc}(\text{stmt}, 0)$ and $\text{wct}_1 = \text{WE.Enc}(\text{stmt}, 1)$. Set $\text{wct} = (\text{wct}_0, \text{wct}_1)$. Let w_l be the l -th block of wct , where each block is of length $\log |\mathcal{Y}|$ bits.
 - (d) For $j \in [Q_w]$, set $x_j = (K, j, w_j)$.
- Output $(\{f_1, \dots, f_{Q_k}, g\}, \{x_1, \dots, x_{Q_w}\})$.

Distinguisher's strategy for post condition. Let $\mathcal{A}^{\text{post}}$ be the distinguisher. It is given $(\{f_i\}_{i \in [Q_k]} \cup \{g\}, \{\text{ct}_j\}_{j \in [Q_w]}, \{\text{sk}_i\}_{i \in [Q_k]} \cup \{\text{sk}_g\})$, where $\{\text{sk}_i = \text{PRFE.KeyGen}(\text{fmsk}, f_i)\}_{i \in [Q_k]}$, $\text{sk}_g = \text{PRFE.KeyGen}(\text{fmsk}, g)$ and $\{\text{ct}_j = \text{PRFE.Enc}(\text{fmpk}, x_j)\}_{j \in [Q_w]}$, if the challenge bit $\beta = 0$, else $\{\text{ct}_j \leftarrow \mathcal{CT}\}_{j \in [Q_w]}$, where $(\text{fmpk}, \text{fmsk}) \leftarrow \text{PRFE.Setup}(1^\lambda)$, $\mathcal{CT}$ is the ciphertext space of PRFE. $\mathcal{A}^{\text{post}}$ does the following:

- (a) For $l \in [Q_w]$, compute $\tilde{w}_l = \text{PRFE.Dec}(\text{sk}_g, \text{ct}_l)$.
- (b) Set $\tilde{\text{wct}} = (\tilde{w}_1, \dots, \tilde{w}_{Q_w})$ and parse it as $\tilde{\text{wct}} = (\tilde{\text{wct}}_0, \tilde{\text{wct}}_1)$.
- (c) Set witness $\tilde{\text{wit}} = (\text{ct}_1, \dots, \text{ct}_{Q_c}, \text{sk}_1, \dots, \text{sk}_{Q_k})$.
- (d) Compute $b_0 = \text{WE.Dec}(\tilde{\text{wit}}, \tilde{\text{wct}}_0)$ and $b_1 = \text{WE.Dec}(\tilde{\text{wit}}, \tilde{\text{wct}}_1)$.
- (e) If $(\mu_0, \mu_1) = (0, 1)$, output $\beta' = 0$,
Else, if any of the above steps fails, or if $(\mu_0, \mu_1) \neq (0, 1)$, output $\beta' = 1$.

Analyzing the distinguisher $\mathcal{A}^{\text{post}}$. When $\beta = 0$, it is easy to see by inspection, that $\mathcal{A}^{\text{post}}$ gets $(\mu_0, \mu_1) = (0, 1)$ in Step 2d. In particular, firstly, we observe that $\mathcal{A}^{\text{post}}$ correctly recovers WE encryptions of 0 and 1 as $\tilde{\text{wct}} = (\tilde{\text{wct}}_0, \tilde{\text{wct}}_1)$ due to the correctness of PRFE. Next, we observe that $\tilde{\text{wit}}$ formed by first Q_c ciphertexts and Q_k keys indeed form a valid witness since for all $i \in [Q_k]$, $j \in [Q_c]$, $\text{PRFE.Dec}(\text{sk}_i, \text{ct}_j) = \text{PRF}(K, (i, j)) = y_{i,j}$ due to the correctness of PRFE. Hence, by correctness of WE, $\text{WE.Dec}(\tilde{\text{wit}}, \tilde{\text{wct}}_0) = 0$ and $\text{WE.Dec}(\tilde{\text{wit}}, \tilde{\text{wct}}_1) = 1$. Hence, it outputs $\beta' = 0 = \beta$ with probability close to 1.

In case of $\beta = 1$, since all the Q_w ciphertexts are uniformly random, firstly it is unlikely that $\tilde{\text{wit}}$ formed by (random) c_j 's (along with Q_k keys) will form a valid witness for WE ciphertext computed by $\mathcal{A}^{\text{post}}$. Even if it does, bits 0 and 1 are information theoretically hidden by these ciphertexts.

Hence, $\Pr[(\mu_0, \mu_1) = (0, 1)] \leq 1/4$. Thus, $\mathcal{A}^{\text{post}}$ outputs $\beta' = 0$ with probability at most $1/4$. That is, $\Pr(\beta' = 1 | \beta = 1) \geq 3/4$. This gives overall success probability of adversary close to at least $7/8$.

Analyzing the feasibility of equation (38). We note that the both input and output length of functions in $\mathcal{F}_s$ are independent of either Q_c and Q_k . Thus, $s = s_F$ is independent of Q_k and Q_c since it simply computes a PRF. This implies that both p_k and p_c , which depend on s , are independent of Q_c and Q_k as well. Hence, it is possible to set the values of Q_c and Q_k such that the equation (38) is satisfied.

Analyzing the pre-condition We prove the following claim.

Claim 4.18. For all PPT adversary $\mathcal{A}^{\text{pre}}$, $D_0^{\text{pre}} \approx_c D_1^{\text{pre}}$, where D_0^{pre} and D_1^{pre} are as defined in equation (37)

Proof. We prove the claim via the following hybrids.

Hyb₀: This is D_0^{pre} . That is, $\mathcal{A}^{\text{pre}}$ gets $(\{f_i\}_{i \in [Q_k]} \cup \{g\}, \{y_{i,j}\}_{i \in [Q_k], j \in [Q_w]}, \{y'_l\}_{l \in [Q_w]})$, where $y_{i,j} = \text{PRF}(K, (i, j))$ and $y'_l = g(K, l, w_l) = w_l$ which is the l -th block of $\text{wct} = (\text{wct}_0, \text{wct}_1) = (\text{WE.Enc}(\text{stmt}, 0), \text{WE.Enc}(\text{stmt}, 0))$ for $\text{stmt} = (y_{1,1}, \dots, y_{Q_k, Q_c})$.

Hyb₁: In this hybrid, $y_{i,j} \leftarrow \mathcal{Y}$ for all $i \in [Q_k], j \in [Q_w]$. Note that this also changes the computation of wct_0 and wct_1 , which are now computed with respect to the statement formed by these (random) $y_{i,j}$'s for $i \in [Q_k], j \in [Q_c]$.

Indistinguishability from the previous hybrid follows directly from the pseudorandomness of PRF.

Hyb₂: In this hybrid, y'_j , for all $j \in [Q_w]$ is sampled uniformly randomly. In particular, the challenger samples $\text{wct} = (\text{wct}_0, \text{wct}_1)$ randomly, which implies y'_j 's are random. Observe that this is the same hybrid as D_1^{pre} .

Indistinguishability from the previous hybrid follows by firstly observing that $\text{stmt} = (y_{1,1}, \dots, y_{Q_k, Q_c})$, for $\{y_{i,j} \leftarrow \mathcal{Y}\}_{i \in [Q_k], j \in [Q_c]}$, used in Hyb₁, is not in the language with probability $(1 - \text{negl}(\lambda))$, due to the same arguments as in the proof of claim 4.15. Hence, the indistinguishability follows due to pseudorandomness of WE ciphertexts.

□

□

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